

Cross-Border Spillovers of Bank Regulations: Evidence of a Trade Channel*

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Abstract

We study whether domestic banking regulations transmit internationally through trade linkages. We develop a model in which exporters extend trade credit to bank-rationed importers to illustrate the effects of a credit-driven import demand contraction. Foreign exporters absorb part of the credit cut, less under high trade-credit frictions, and redirect lost sales only for goods that are substitutable enough. Exploiting a 2012 Spanish banking reform and combining administrative credit data with global trade flows, we show that the reform reduced credit supply, leading to a contraction in Spanish imports and cross-border spillovers to exporters serving the Spanish market. Exports decline more for exporters in less financially developed and where trade-credit frictions are stronger, who could not absorb importers' credit cut extending trade-credit, and reallocation across destinations was limited, except for homogeneous goods.

Keywords: bank regulation; international trade; cross-border spillovers; trade frictions; financial development

JEL Codes: F14, F36, F42, G21, G28.

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1 Introduction

The Global Financial Crisis triggered a wave of regulatory reforms that significantly tightened banking regulation worldwide. While these reforms were designed to strengthen financial stability, they may also have generated unintended cross-border spillovers. Existing research has primarily examined *financial channels* of such spillovers, focusing on cross-border capital flows and the lending activities of global banks' foreign subsidiaries.¹ Much less attention has been paid to a distinct channel that can operate even with limited capital market integration: the transmission of domestic financial shocks through international trade linkages.

This paper analyzes whether and how domestic banking regulatory shocks propagate internationally through trade linkages. To that end, we use micro-level credit and trade data to study how a bank regulatory shock affects domestic importers and, through a contraction in import demand, spills over to their foreign trading partners. Exploiting a large and unexpected increase in loan-loss provisions imposed on Spanish banks in 2012, we show that tighter domestic banking regulation reduced import demand and generated economically meaningful cross-border spillovers, which are larger when foreign exporters face stronger frictions to extend trade credit or produce differentiated goods that are difficult to reallocate across destination markets. We develop a stylized framework that delivers these margins as testable predictions, and organize the empirical analysis around them.

While bank credit supply shocks may cause a contraction in firm-level import demand, it is not obvious whether this translates into cross-border spillovers. Domestic importers facing tighter credit conditions may substitute toward alternative sources of financing such as supplier trade credit. If such substitution is limited, less affected importers may capture market share from more affected ones, leaving aggregate import demand largely unchanged. Finally, even if aggregate import demand contracts in the shocked country, foreign exporters

¹For instance, [Aiyar et al. \(2014\)](#); [Bahaj and Malherbe \(2024\)](#); [Buch and Goldberg \(2017\)](#); [Tripathy \(2020\)](#).

may redirect sales toward alternative destinations. The magnitude of cross-border spillovers therefore depends on the extent of these adjustments.

In the first part of the paper, we develop a stylized model of trade to formalize these margins. A continuum of foreign exporters serves two destinations whose importers finance part of each invoice with bank credit and the rest with trade credit. Each exporter builds production capacity financed with long-term debt and extends trade credit funded with short-term debt. The marginal cost of short-term debt increases with total leverage, so that capacity and trade credit interact in the same balance sheet. When the importer's bank credit is cut, the exporter substitutes its own balance sheet for the lost bank credit, but only partially. The share replaced with trade credit is smaller, and the decline in exports is deeper, when the exporter faces stronger frictions in extending trade credit.

The unabsorbed part of the decline lowers marginal revenue in the shocked destination relative to the other, so the exporter redirects quantities to third markets, but whether those reallocated shipments raise the value of exports elsewhere depends on two opposing forces. First, the degree of substitutability across origins, which determines how much quantity must move to keep invoice prices equal across destinations. Second, the exporter's increasing marginal cost of funds, which raises the implicit trade-credit rate it charges across destinations. Reallocation in value therefore requires high enough frictions in the shocked destination, goods substitutable enough, and low enough leverage frictions at the exporter's origin.

The framework delivers three predictions we take to the data. First, exports to Spain fall by more from origins where trade credit frictions are stronger and where the exporter's cost of funds is higher. Second, reallocation across destinations is more likely for more substitutable goods and from origins less able to attenuate. Third, the importer's total trade credit rises independently on the mix of frictions across its suppliers, while its trade-credit share falls only if it shifts purchases toward suppliers that need less external financing. The framework also distinguishes a credit-driven import decline from an expenditure-driven one. When the

importer's expenditure falls, its bank line is intact and there is less invoice to finance, so trade credit falls with imports rather than rising. The shrinking invoice lowers the trade-credit rate, and by more where frictions are higher. On the other hand, the exporter's demand shifts directly, so with capacity committed it redirects sales even without financial frictions, whereas under a credit-driven import shock nothing is redirected unless frictions keep trade credit from absorbing the credit cut.

In the second part of the paper, we test our model's predictions and empirically assess whether domestic banking regulatory shocks generate cross-border spillovers. To do so, we exploit a large and unexpected increase in loan-loss provisioning requirements imposed on Spanish banks in 2012. The regulation targeted exposures to the construction and real estate sectors, non-tradable industries plausibly exogenous to global trade conditions, and generated substantial cross-sectional variation in banks' balance sheet exposure (Jiménez et al., 2017). We follow the identification strategy in Jiménez et al. (2017) to validate the credit-supply effect in our sample of importing firms. Using bank-firm-level data from the Spanish Central Credit Register, we confirm that more exposed banks reduced lending to importers following the reform. Exposed importers exhibit limited scope for substitution toward alternative funding sources and therefore face a contraction in overall financing.

We then combine the Spanish credit registry with transaction-level Balance of Payments data and global bilateral trade flows from BACI. Our analysis proceeds in three steps. First, we document that the regulation reduced firm-level imports among Spanish importers more exposed to affected banks. Specifically, using transaction-level data disaggregated by firm, product, country, and year, and a difference-in-differences framework with rich fixed effects following Paravisini et al. (2014), we find that a one standard deviation increase in exposure reduces imports by 3.6 percent. Consistent with a domestic credit-supply mechanism, the effects are stronger for riskier importers, firms with weaker banking relationships, and those with limited access to credit lines, highlighting the role of bank ties in buffering adverse

financial shocks (Bolton et al., 2016).²

Second, we turn to aggregate import flows using BACI bilateral trade data to assess whether firm-level contractions translate into cross-border spillovers. This level of aggregation is necessary to test for potential business stealing across Spanish importers, which could offset cross-border transmission (Matray et al. 2024; Beaumont et al. 2025). Using our transaction-level data, we define a market as a product-origin pair and construct a measure of exposure to the policy at this level, capturing the extent to which Spanish importers of a given product from a given origin rely on funding from exposed banks. For example, if Spanish importers of textiles from India depend heavily on financing from highly exposed banks, then Spanish imports of textiles from India are highly exposed to the regulation.

We implement a triple difference-in-differences strategy that compares exports to Spain from more- versus less-exposed product-origin markets, relative to exports of the same markets to other destinations, and find little evidence that less-exposed Spanish importers offset the decline in imports by more-exposed importers within a given market. A one standard deviation increase in exposure leads to a 2.1 percent decline in exports to Spain relative to other destinations of the same product and origin, with no evidence of pre-trends. These results remain robust when restricting the counterfactual destinations to comparable high-income countries or excluding GIIP exporters, addressing potential identification concerns related to the European sovereign debt crisis.³ These findings show that the contraction in firm-level import demand translates into an aggregate decline in exports to Spain.

The model’s first prediction is that exports to Spain fall by more where the cost of extending trade credit is higher, because less of the shock is absorbed by the exporter’s own balance sheet. The firm-level data support the mechanism directly: supplier credit to exposed importers rises after the shock, and rises significantly less among firms sourcing

²For completeness, we conduct a parallel analysis for domestic exporters. In contrast to importers, both exporter credit and firm-level export flows are only modestly affected by the regulation, with effects concentrated among smaller firms and negligible at the aggregate level.

³GIIP denotes Greece, Ireland, Italy, and Portugal.

from distant or non-Spanish-speaking origins. Both signs are those of a credit shock; an expenditure-driven fall in importers' demand would have lowered supplier credit and hit distant origins by less. Financial development maps into the exporter's cost of external funds in the framework, so we first examine whether exporters operating in countries with more developed financial systems, proxied by private credit to GDP, are better positioned to absorb the shock.⁴ We find that exporters in financially developed economies are largely unaffected by the contraction, leaving their exports to Spain broadly unchanged. In contrast, exporters in less financially developed countries experience a significant 3.3 percent decline in exports to Spain.

Within the less financially developed group, where the capacity to mitigate the shock is already limited, trade frictions shape the adjustment further. The contraction is significantly stronger for exporters in distant and non-Spanish-speaking countries. A one standard deviation increase in exposure leads to a 3.6 percent decline in exports from distant countries and a 3.9 percent decline when both distance and language frictions are present. These patterns are consistent with the model's implication that distance-related frictions limit exporters' ability to ease financing constraints, thereby amplifying the cross-border transmission of domestic credit conditions.

Third, we turn to the second adjustment margin: reallocation across destinations. Here the outcome is the origin's total exports, the sum of what it ships to Spain and everywhere else, so a decline in total exports means the loss in Spain was not recovered. When exporters are unable to fully absorb the contraction in Spanish demand, they may mitigate its impact by redirecting sales toward alternative markets. To isolate the policy effect, we compare product-origin pairs with similar pre-shock shares of exports to Spain, ensuring that differences in exposure are not mechanically driven by initial trade dependence. On average, we find limited evidence of reallocation. In particular, among product-origin pairs for which

⁴See, for example, [Manova \(2013\)](#) and [Xu \(2022\)](#) on the role of financial development in shaping trade outcomes.

Spain represents a major export market, a one standard deviation increase in exposure leads to a significant 2.8 percent decline in total exports. This contraction is driven entirely by lower exports to Spain, with exports to other destinations remaining broadly unchanged.

The framework predicts that lost sales reappear elsewhere only when goods are substitutable enough and frictions in the shocked destination high enough, so we test the two conditions jointly: we split products by price dispersion within the sample of origins least able to attenuate the shock, namely those that are financially underdeveloped, distant, and without a common language. Consistent with the prediction, homogeneous products show no decline in total exports: the contraction in exports to Spain is offset by increased sales elsewhere. For differentiated products, by contrast, total exports decline by 9 percent, driven by a 16 percent contraction in exports to Spain that is not recovered in other destinations. These patterns indicate that the scope for recovering lost sales by redirecting them to other destinations depends on product substitutability.

Finally, because some Spanish banks have an international presence, one concern is that our results may be confounded by their international lending activities. For instance, if these banks finance firms that export to Spain, the estimated effects could reflect cross-border bank transmission rather than a trade-demand channel. To address this concern, we use BankFocus data to identify countries without Spanish bank presence and re-estimate our baseline specification on this restricted sample. We find no meaningful differences relative to our baseline results. These findings reinforce that the cross-border effects we document operate primarily through trade linkages, rather than through the internal capital markets of Spanish global banks.

Overall, our findings show that tightening domestic financial regulation, even when targeted at local vulnerabilities, generates significant cross-border real effects through the contraction in import demand. The cross-border spillovers we document are shaped by two key adjustment frictions. First, exporters' ability to ease financing constraints faced by importers is limited, particularly in less financially developed economies and in high-friction trade rela-

tionships, so exports to Spain fall. Second, recovering those lost sales by redirecting them to other destinations requires goods substitutable enough that redirected shipments outweigh the higher cost of funds the exporter passes into the rates it charges in third markets; for differentiated goods this condition fails and the loss is not recovered. These results highlight international trade as an important channel of regulatory spillovers and underscore the importance of financial development and cross-border coordination in prudential policy for mitigating unintended disruptions to global trade.

Contribution to the literature. This paper contributes to four strands of literature. First, we extend the literature on the cross-border transmission of financial shocks. Traditional channels emphasize the role of global banks' internal capital markets in propagating funding shocks (Bruno and Shin, 2014; Hale et al., 2019; Bolton and Oehmke, 2018), bank regulations (Bahaj and Malherbe, 2024; Aiyar et al., 2014; Buch and Goldberg, 2017; Baskaya et al., 2017; Tripathy, 2020), and monetary policy (Cetorelli and Goldberg, 2012; Buch et al., 2019; Morais et al., 2019; Brauning and Ivashina, 2020). We identify a distinct trade channel through which domestic banking regulation propagates internationally by contracting import demand. Unlike standard financial channels, this channel does not require banks to operate abroad and instead depends on the characteristics of trade linkages, including financial development in exporting countries and trade costs.

Second, we contribute to the literature on production networks, buyer-supplier relationships, and international trade linkages (Carvalho et al., 2020; Costello, 2020; Alfaro et al., 2021; Huremovic et al., 2025; Burstein et al., 2008; Richmond, 2019; Di Giovanni and Hale, 2022; Brauning and Sheremirov, 2023; Auer et al., 2019; Boehm et al., 2019; Adelino et al., 2022; Ersahin et al., 2024; Romero, 2025). We show that domestic regulatory shocks can be transmitted internationally through trade linkages. The magnitude of this transmission depends not only on financial constraints but also on the structure of trade relationships and the frictions governing them.

Third, we contribute to the literature on export dynamics and cross-market adjustment (Xu, 2022; Almunia et al., 2021; Jiao et al., 2024; Ding, 2024; Almeida et al., 2025). Our analysis is closely related to Xu (2022) and shows how financial and trade frictions jointly shape exporters’ adjustment to financial shocks originating in destination markets. Relative to the venting-out mechanism of Almunia et al. (2021), in which a demand slump at home redirects sales abroad, the shock here is to the importer’s credit. Nothing is redirected unless frictions do not allow trade credit to absorb the bank line cut, and the response in third markets can go either way, reallocation when goods are substitutable enough and frictions in the shocked destination high enough, or contagion through the exporter’s cost of funds. Our findings also complement recent work on trade reallocation during periods of heightened uncertainty (Alfaro and Chor, 2023; Benguria and Saffie, 2024; Gopinath et al., 2025) by emphasizing the role of financial conditions in shaping exporters’ responses.

Fourth, we contribute to the extensive literature on trade finance (Amiti and Weinstein, 2011; Manova, 2013; Del Prete and Federico, 2014; Paravisini et al., 2014; Demir et al., 2017; Niepmann and Schmidt-Eisenlohr, 2017a; Ahn and Sarmiento, 2019; Xu, 2022; Borchert et al., 2023; Moreira and Monteiro, 2023; Dogan and Hjortso, 2024; Matray et al., 2024; Beaumont and Lenoir, 2025). We advance this literature in two directions. First, we document a novel international spillover effect: tighter domestic credit conditions reduce foreign exporters’ sales, particularly when exporters operate in financially underdeveloped economies, face higher trade costs, or produce goods that are difficult to reallocate across destinations. Second, we show that the aggregate and cross-border effects of domestic credit shocks primarily operate through import demand rather than export supply.

While existing models show how financial frictions shape the organization of trade and the choice of financing contracts (e.g., Schmidt-Eisenlohr 2013; Antràs and Foley 2015; Niepmann and Schmidt-Eisenlohr 2017a), we take buyer-supplier relationships as given and focus on their aggregate implications once these trading links are in place. Relatedly, a large literature studies how shocks to exporters’ financing capacity affect trade outcomes (e.g., Demir et al.

2017; Niepmann and Schmidt-Eisenlohr 2017b). In contrast, we study a setting in which the shock originates in the importing country and examine how contractions in import demand propagate to foreign exporters, depending on their capacity to provide financing and on the frictions governing trade relationships.

The rest of the paper is organized as follows: Section 2 develops a trade model with financial frictions and discusses the mechanisms driving the cross-border spillovers. Section 3 provides institutional details of the policy intervention and describes the data. Section 4 presents our firm-level results and Section 5 reports our identification of cross-border spillovers. Section 6 concludes.

2 Conceptual Framework

We develop a three-period model of foreign exporters that serve two destinations whose representative importers are bank-rationed and rely on trade credit from their suppliers. In period 0 an exporter builds capacity, financed with long-term debt. In period 1 goods are produced and shipped, and the exporter extends trade credit funded with short-term debt. In period 2 all loans are repaid and profits are realized. There is no discounting, so all payoffs are expressed in period-2 units. We describe the baseline equilibrium and characterize the effects of an unexpected tightening of the importer’s credit line in one destination.

2.1 Importers

The representative importer in market $i \in \{1, 2\}$ allocates expenditure E_i across a fixed number of sectors in Cobb-Douglas shares; the importer neither produces nor exports, so E_i is exogenous. Within sector s it has CES preferences over a continuum of firm-specific

varieties ω with elasticity of substitution $\sigma_s > 1$, and chooses quantities to solve:

$$\max_{\{q_i(\omega)\}} \left(\int q_i(\omega)^{\frac{\sigma_s-1}{\sigma_s}} d\omega \right)^{\frac{\sigma_s}{\sigma_s-1}} \quad \text{s.t.} \quad \int \tilde{p}_i(\omega) q_i(\omega) d\omega \leq E_{is}$$

where $\tilde{p}_i(\omega)$ is the effective price of variety ω . Firms are located in a finite set of countries j with measure M_j . Within countries firms are identical, and each firm is of measure zero and takes the sectoral index P_{is} as given.

The importer pays the invoice $p_i q_i$ of a given exporter's variety at delivery, financing a fraction $\tau \in (0, 1)$ of it with credit. It has access to trade credit from the exporter at rate r_{xi} and to bank financing attached to the transaction, $\bar{L}_i$. Banks finance imports through letters of credit, documentary collections and import loans written against a specific shipment, so financing terms are set pair by pair and cannot be shifted across suppliers (???). We assume that the domestic bank line is cheaper and the financing need exceeds the line, so the line is exhausted and trade credit is positive:

$$F_i = \tau p_i q_i - \bar{L}_i > 0 \tag{1}$$

The marginal unit is then financed with trade credit, so the effective price of the exporter's variety is $\tilde{p}_i = p_i(1 + \tau r_{xi})$, and its demand is $q_i = A_{is} [p_i(1 + \tau r_{xi})]^{-\sigma_s}$, with $A_{is} \equiv E_{is} P_{is}^{\sigma_s-1}$ and P_{is} the sectoral price index. We hold P_{is} fixed throughout. If every imported variety's line were cut at once, P_{is} would rise and A_{is} with it, a demand shift common to all origins that leaves every relative statement below unchanged.

2.2 Exporter

The exporter is the representative firm of one origin in sector s . It supplies one variety to both destinations and takes A_{is} as given. Trade credit is priced competitively, so the exporter takes interest rates as given.

Technology and financing. In period 0 the exporter commits to capacities D_1 and D_2 , which allows the delivery of $q_i = D_i/(1+t)$ units in period 1, where $1+t$ is the iceberg cost. Total capacity $D = D_1 + D_2$ is financed with long-term zero-coupon debt, at cost $(1+r_0)^2 D$ paid in period 2, where r_0 is the risk-free rate per period. Commitment captures that, at the horizon of a credit shock, production and shipping are fixed by lead times and contracts, so that the exporter can redirect what it has produced but not produce more or less.⁵

In period 1 the exporter produces D units using labor at cost w per unit.⁶ The exporter extends trade credit F_i to destination i , with total $F = F_1 + F_2$. Thus, the exporter faces the following financial need in period 1:

$$B = wD + F - \sum_i p_i q_i \quad (2)$$

This amount is financed with short-term debt, which is priced at a rate that rises with total debt outstanding, such that the cost of short-term debt is convex in total leverage $D + B$, and repaying B costs $(1+r_0)B + \lambda(D + \frac{B}{2})B$ with $\lambda > 0$.⁷ The premium ties the two destinations together: credit extended in one raises the cost of credit extended in the other.

Period-1 problem. Given capacities D_1, D_2 and the trade-credit rates, the exporter chooses how much trade credit to extend to maximize:

$$\Pi_1(D_1, D_2) \equiv \max_{F_1, F_2 \geq 0} \sum_i \left[(1+r_{xi})F_i - \frac{\psi_i}{2} F_i^2 \right] - (1+r_0)^2 D - (1+r_0)B - \lambda(D + \frac{B}{2})B \quad (3)$$

where $\psi_i F_i^2/2$ represents a monitoring cost that captures trade credit frictions, and B is given by (2). The solution to this problem gives the exporter's total profit for given committed

⁵Appendix D shows that this is what links the two destinations on the production side, and that it is the limit of any increasing marginal cost of output.

⁶Committing to capacity commits to the wage bill wD too. Appendix C.2 studies the case of large shocks where this assumption binds.

⁷The marginal cost of one unit of short-term funding is $1 + r_0 + \lambda(D + B)$. The parameter λ can be microfounded by a diversion shock on period-2 revenue, under which the exporter defaults when retained revenue falls short of its obligation and lenders price the default probability.

capacities.

Period-0 problem. Given the trade-credit rates and anticipating its own period-1 choice, the exporter chooses capacities to maximize:

$$\max_{D_1, D_2 \geq 0} \Pi_1(D_1, D_2) \quad (4)$$

2.3 Equilibrium

An equilibrium is a vector of prices $(r_{x1}, r_{x2}, p_1, p_2)$ and allocations $(q_1, q_2, F_1, F_2, D_1, D_2)$ such that, given the prices: (i) the quantities q_1, q_2 solve the importers' problems; (ii) the capacities D_1, D_2 solve the exporter's period-0 problem (4); (iii) given the capacities, the trade-credit quantities F_1, F_2 solve its period-1 problem (3); (iv) trade-credit markets clear, $F_i = \tau p_i q_i - \bar{L}_i$ for each i ; and (v) product markets clear, $(1 + t)q_i = D_i$ for each i . We restrict attention to equilibria in which the exporter borrows in period 1.

2.4 Shock Transmission

In period 1, after capacity and labor are sunk, the importer in destination 1 experiences an unexpected proportional tightening of its bank line, $d \ln \bar{L}_1 = -x$ with $x > 0$. The tightening applies to every facility of the importer in the same proportion; we characterize one exporter's response first and aggregate across exporters in Section 2.5. We write $\varepsilon \equiv x \bar{L}_1$ for the bank credit lost and state responses per euro of it or as elasticities with respect to $\bar{L}_1$. Capacity installed for one destination can serve the other without loss, so after the shock only total capacity matters. Thus, when solving problem (3), the exporter now chooses trade credit and quantities subject to the committed production $Q \equiv q_1 + q_2$.

The shocked equilibrium is a vector of prices $(r_{x1}, r_{x2}, p_1, p_2)$ and allocations (q_1, q_2, F_1, F_2) such that, given the prices and the predetermined capacity D : the quantities solve the im-

porters' problems; trade credit and quantities solve the exporter's period-1 problem subject to $q_1 + q_2 = Q$; and trade-credit markets clear, $F_1 = \tau p_1 q_1 - (1 - x)\bar{L}_1$ and $F_2 = \tau p_2 q_2 - \bar{L}_2$.

Assumption 1 (Symmetric financing). The exporter's buyers finance the same fraction of its invoice with credit in both destinations, $\tau_1 = \tau_2 \equiv \tau$.

The fraction τ may differ across exporters and Proposition 1(iv) compares exporters that differ in it; the assumption is only that the exporter under study faces the same fraction in both destinations. The fraction covered by trade credit still differs across destinations through the bank lines.

Define $\phi_i \equiv |\partial \ln p_i / \partial r_{xi}|$, which measures how strongly the trade-credit rate passes into the invoice price, and $\tilde{\lambda}_i \equiv \partial r_{x,-i} / \partial F_i$, the cross slope of the trade-credit supply curve, which measures how trade credit extended to destination i raises the rate charged in the other destination.

Assumption 2 (Class of equilibria). We consider equilibria where (a) $\lambda(\phi_1 + \phi_2)(V_1 + V_2) < 1$; (b) $\phi_i \psi_i > \tilde{\lambda}_i(\phi_{-i} - \phi_i)$ for $i = \{1, 2\}$; and (c) $\psi_i V_i \leq 1$ for $i = \{1, 2\}$.

Condition (a) ensures that the cross slopes $\tilde{\lambda}_i$ are finite and positive, condition (b) rules out the case in which trade credit to the shocked destination would fall due to leverage contagion, and condition (c) bounds the monitoring premium on each destination's sales by one. Assumptions 1–2 are sufficient conditions under which the following propositions hold.

Let V_i denote the value of exports to destination i , $s_i \equiv q_i/Q$ the quantity share, $a \equiv \partial F_1 / \partial \varepsilon$ the share of a lost euro of bank credit that trade credit replaces, and $\theta \equiv \bar{L}_1 / (\tau V_1) \in (0, 1)$ the share of the financing need covered by the bank line at the baseline. The following propositions treat all derivatives as first-order responses at the baseline.

Proposition 1 (Attenuation). *Under Assumptions 1–2: (i) trade credit to the shocked destination rises but replaces only part of the lost bank credit, $a \in (0, 1)$; (ii) exports to the shocked destination fall with its bank line by the unreplaced share of the cut, $\partial \ln V_1 / \partial \ln \bar{L}_1 =$*

$\theta(1 - a) \in (0, \theta)$; (iii) a is strictly decreasing in ψ_1 ; (iv) when the two destinations carry the same financing wedge, $\phi_1 = \phi_2$, a is strictly decreasing in τ and λ .

Proof. Appendix C.

The proposition describes the exporter's attenuation response. A bank-line tightening forces the exporter to substitute its own balance sheet for the lost bank credit of its importer. It substitutes partially, so the trade credit increase and the exported value decline both absorb the shock, and by (ii) the export decline is the unreplaced share of the cut. Parts (iii) and (iv) are three sources of heterogeneity across exporters. The replaced share falls with the monitoring friction ψ_1 , which raises the rate at which the exporter supplies each additional euro. It falls with the financing need τ , because the rate applies to a larger fraction of every invoice, so the importer's demand for credit is more sensitive to it: the importer that depends most on financing is the one most hurt by its price. And it falls with the exporter's leverage premium λ , which raises the marginal cost of every euro it extends.

Proposition 2 (Reallocation). *Under Assumptions 1–2:*

(i) *Quantities are redirected to the non-shocked destination, and the value and trade credit associated with those quantities move together, $\partial F_2 / \partial \varepsilon = \tau V_2 \partial \ln V_2 / \partial \varepsilon$. Both rise if and only if*

$$(\sigma_s - 1) s_1 [\phi_1 \psi_1 + \tilde{\lambda}_1 (\phi_1 - \phi_2)] > \phi_2 \tilde{\lambda}_1$$

and both fall otherwise. Equivalently, there exist unique thresholds $\bar{\psi}(\sigma_s)$ and $\bar{\sigma}(\psi_1)$ such that both rise if and only if $\psi_1 > \bar{\psi}(\sigma_s)$, or equivalently $\sigma_s > \bar{\sigma}(\psi_1)$.

(ii) *$\partial \ln V_2 / \partial \varepsilon$ is strictly increasing in σ_s , $\bar{\psi}$ is strictly decreasing in σ_s , and $\bar{\sigma}$ is strictly decreasing in ψ_1 .*

(iii) *When $\phi_1 = \phi_2$, $\bar{\sigma} = 1 + \tilde{\lambda}_1 / (s_1 \psi_1)$ is strictly increasing in τ .*

Proof. Appendix C.

The proposition illustrates the three forces that shape reallocation, defined as the increase in exported value in the non-shocked destination. First, the elasticity of substitution σ_s ,

which reduces the reallocation losses driven by price adjustments. Second, the direct pass-through in the shocked destination, which increases the available shipments to reallocate and is driven by ψ_1 . Third, the exporter's higher leverage premium, which raises the rate charged in the non-shocked destination and is driven by λ , amplified by the feedback of invoice prices on the borrowing need. Part (iii) indicates that the exporter whose buyer needs more financing is the one whose balance sheet spills over more, and reallocation is less likely for it.

The monitoring friction in the shocked destination and the leverage premium are the two forces that convert a credit shock into a difference between destination marginal revenues. If we set both to zero, trade credit absorbs the line cut and no difference opens. Finally, the elasticity is what converts the difference into redirected quantity. The value of exports to the non-shocked destination therefore rises only for goods substitutable enough and frictions high enough.

Finally, the two financial channels play distinct roles. The leverage premium transmits the shock across destinations only as contagion, while reallocation comes from committed capacity, and the threshold in Proposition 2 is where the two work together. Appendix D shows that without committed capacity the spillover to third destinations is negative for every good.

2.5 Trade credit at the importer

Propositions 1–2 describe one exporter. The shocked importer sources from many origins, each responding according to its own $(\tau_j, \psi_{1j}, \lambda_j)$, and substitutes across them, buying less from every origin whose effective price rises and least from those that attenuate less. Our firm-level data record trade credit at the importer, aggregated over origins, so we characterize that aggregate in Proposition 3.

Let $F \equiv \sum_j F_{1j}$, $V \equiv \sum_j V_{1j}$, $S \equiv F/V$ the importer's trade-credit share of imports,

$v_j \equiv V_{1j}/V$ the import share of origin j , $\bar{L} \equiv \sum_j \bar{L}_{1j}$, $\varepsilon_j \equiv x\bar{L}_{1j}$ the bank credit lost on origin j , and $\varepsilon \equiv \sum_j \varepsilon_j$.

Proposition 3 (Trade credit at the importer). *Under Assumptions 1–2 for every origin:*

(i) *the importer’s total trade credit rises, $dF = \sum_j a_j \varepsilon_j > 0$, whatever the distribution of $(\tau_j, \psi_{1j}, \lambda_j)$ across origins;* (ii) *its trade-credit share decomposes exactly as*

$$V dS = [\varepsilon + \bar{L} d \ln V] + V \text{Cov}_v(\tau_j, d \ln V_{1j})$$

where the covariance is across origins with import-share weights; the first term is positive whenever $\bar{L} \leq \min_j \tau_j V$; (iii) the covariance is zero when financing needs are common across origins, and negative when origins differ only in τ_j , so the share falls only if origins with higher financing needs contract by more; (iv) the covariance term equals $[\varepsilon + \bar{L} d \ln V] - V dS$, a function of the importer’s bank-credit cut, its pre-shock bank line, its import growth, and the change in its trade-credit share alone.

Proof. Appendix C.

The importer’s trade credit cannot fall at the intensive margin: from every origin, a lost euro of bank credit is met with more trade credit or fewer purchases, never with less credit. Its share can fall, but only through composition, when purchases shift toward origins that need less financing; this is the signature of the importer’s own adjustment, and part (iv) makes it measurable. The bracket is the share change absent any change in the composition of financing needs, built from the importer’s line cut and import decline; its gap with the observed share change is the composition effect. Neither requires data by origin.

2.6 Credit-driven versus expenditure-driven contractions

A cut in the importer’s bank credit and a fall in its expenditure both lower its purchases, but they reach the exporter differently, generating different effects and adjustments. The

first lands on its receivables, the second on its demand. Let expenditure in destination 1 fall, $d \ln E_{1s} = -\delta$, with the bank line unchanged.

Proposition 4 (Expenditure shock). *Under Assumptions 1–2: (i) trade credit to the shocked destination falls with exports, $dF_1 = \tau dV_1 < 0$, and so does the importer’s trade-credit share; (ii) the export decline is strictly smaller for exporters facing a higher monitoring friction ψ_1 ; (iii) exports to the non-shocked destination rise if and only if $\sigma_s > \bar{\sigma}^d \equiv 1/[1 - \lambda(1 - \tau)\phi_2(V_1 + V_2)]$, a threshold that does not depend on ψ_1 and, when $\phi_1 = \phi_2$, lies below $\bar{\sigma}(\psi_1)$ for every ψ_1 .*

Proof. Appendix C.1.9.

With the line intact there is less invoice to finance, so trade credit and the importer’s share fall instead of rising. The shrinking invoice lowers the trade-credit rate, by more where the monitoring friction is higher, so the friction cushions the decline instead of amplifying it. And because the shock shifts the exporter’s demand directly, with capacity committed it redirects sales even without financial frictions, and reallocation fails only through contagion, which is the venting-out mechanism of ? with a fixed quantity; under the credit-driven contraction nothing is redirected unless financial frictions keep trade credit from absorbing the cut, which is why Proposition 2 has a threshold in ψ_1 . Trade credit up, the trade-credit share up, and far origins down by more are therefore the signs of a credit-driven contraction and the opposite of an expenditure-driven one.

3 Institutional Background and Data

3.1 The 2012 Spanish Provisioning Reform

To identify exogenous variation in domestic credit supply that is orthogonal to foreign demand and supply shocks, we exploit the 2012 Spanish provisioning reform as a shock to

bank lending conditions. Because the reform targeted pre-existing exposures to the construction and real estate sectors—non-tradable industries plausibly unrelated to contemporaneous global trade dynamics—it generated cross-sectional variation in banks’ balance sheet exposure that is unlikely to reflect foreign developments.

In early 2012, the Spanish government imposed a one-time, two-phase increase in provisioning requirements for banks’ exposures to the construction and real estate sectors, based on outstanding loans as of late 2011. The policy aimed to reduce uncertainty surrounding the valuation of distressed real estate assets in the aftermath of the Global Financial Crisis.

In the first phase (February), banks were required to provision 7% for all performing loans in the targeted sectors. In the second phase (May), provisioning was differentiated by loan type: an additional 45% for land, 7% for completed properties, and 22% for real estate projects in progress. Provisions for non-performing loans were also raised, and all requirements had to be met by year-end 2012.

This ad hoc policy represented a substantial shock. Banks’ existing provisioning buffers had already been depleted by the global crisis, leaving them unable to meet the new requirements without sharply increasing provisions—by approximately \$85 billion (8.5% of GDP), as documented by Jiménez et al. (2017). As a result, bank profitability fell sharply in 2012.

Following Jiménez et al. (2017), we measure banks’ exposure to the policy using the share of their corporate lending to construction and real estate firms before the reform. While their analysis documents a reduction in credit supply to corporate borrowers by highly exposed banks, we focus on the implications for non-financial firms engaged in international trade, and the resulting effects on trade flows and cross-border spillovers.

Among the 83 banking groups lending to exporters and importers, exposure to construction and real estate sectors varied widely. At the 25th, 50th, and 75th percentiles, banks allocated 26.5%, 34.9%, and 47.0% of corporate credit to these sectors, respectively (see Panel C of Table 1). Figure 1 illustrates the distribution of bank exposure.

3.2 Data

Our analysis relies on three primary datasets: (i) the Spanish Central Credit Register (CIRBE) from the Bank of Spain; (ii) confidential transaction-level Balance of Payments data; and (iii) BACI international trade data ([Gaulier and Zignago, 2010](#)), which provide comprehensive bilateral trade flows at the origin–destination–product level.

3.2.1 Credit Register (CIRBE)

To examine the bank lending channel of the reform, we use quarterly bank–firm credit data from the CIRBE, a comprehensive credit registry maintained by the Bank of Spain. The CIRBE records all loan commitments above €6,000 granted to non-financial firms by any credit institution operating in Spain, and includes detailed information on loan type, drawn and undrawn amounts, collateral status, maturity, currency, past-due status, lender and borrower identities, and firm characteristics such as industry classification and geographic location.

We combine CIRBE data with supervisory bank balance sheets and Balance of Payments data. This allows us to observe bank characteristics alongside firms’ import activities, and to link each lending bank to its exposure to the reform, measured as the share of credit granted to the construction and real estate sectors.

Our final sample includes credit information for 33,361 non-financial importing firms, receiving credit from 83 banking institutions over the period 2011Q1–2013Q2, yielding a total of 1,288,007 observations.⁸ Descriptive statistics for the variables used in this part of the analysis are presented in Panel A of [Table 1](#), while variable definitions can be found in [Table A.1](#).

⁸We account for bank mergers by adopting the industry structure prevailing at the end of the sample period.

3.2.2 Balance of Payments Data

Our analysis of firms’ international trade outcomes relies on confidential transaction-level import records. This administrative dataset is one of the sources used by the Bank of Spain to compile official Balance of Payments statistics. It captures most trade flows into and out of Spain and provides detailed information for each importer, including the origin country, the year, and the product coded at the 2-digit Harmonized System (HS) level.

Our analysis covers the period 2009–2013, during which all transactions exceeding €50,000 were subject to mandatory reporting. Despite this reporting threshold, the dataset captures 91.3% of total trade flows relative to official Customs statistics (Almunia et al., 2021; Prades and Villegas-Sánchez, 2026; Gutiérrez and Moral-Benito, 2024).

The dataset covers 558,540 import observations for 23,029 firms spanning 99 products and 167 import origins (Panel D of Table 1).

3.2.3 BACI Data

To assess international spillover effects, we use BACI trade data (Gaulier and Zignago, 2010), an annual panel of bilateral trade flows at the origin–destination–product level. We merge these data with a measure of exposure to the reform at the product–origin level, constructed using Balance of Payments and CIRBE data.⁹

To enrich the analysis, we incorporate information from Moody’s Analytics BankFocus, which provides annual balance sheet data for more than 38,000 banks worldwide and allows us to identify countries in which Spanish banks have a local presence. Additionally, we use the CEPII Gravity database, which provides country-pair characteristics relevant for trade costs, such as geographical distance, common language, and trade agreements.

The BACI sample includes 3,141,210 trade flow observations for 5,216 HS 6-digit products grouped into 96 HS 2-digit goods, covering bilateral trade among 190 countries over the

⁹For the years 2005–2013, Spanish import flows obtained from Balance of Payments data at the 2-digit product–country level account for approximately 90% of the corresponding flows in BACI.

period 2009–2013 (see Panel E of [Table 1](#)).

4 Credit Shock and Import Contraction

We begin by documenting the domestic effects of the bank regulatory reform on bank lending to importing firms and on firm-level imports. We then turn to aggregate trade data to assess whether these firm-level effects translated into a broader contraction in Spanish import demand by testing whether less-exposed Spanish importers offset the decline in imports by more-exposed firms.

4.1 Domestic Credit Contraction

First, we examine whether the 2012 provisioning reform generated a contraction in credit supply for importing firms. While the credit effects of the reform have been documented in [Jiménez et al. \(2017\)](#), we confirm that similar patterns arise for firms engaged in international trade. In particular, banks more exposed to the reform reduced lending to a larger extent.¹⁰

Using bank–firm data from the CIRBE over the period 2011Q1–2013Q2, we estimate:

$$\log y_{f,b,t} = \theta \times Exposure_b \times Policy_t + controls_{b,f,t} + \alpha_{f,t} + \alpha_{f,b} + \varepsilon_{f,b,t}, \quad (5)$$

where $y_{f,b,t}$ denotes committed credit from bank b to firm f in quarter t . $Exposure_b$ measures bank b ’s pre-reform share of lending to construction and real estate (as of the end of 2011), and $Policy_t$ equals one from 2012Q1 onward. $Controls_{b,f,t}$ account for various bank and bank-firm characteristics measured as of the end of 2011 interacted with time fixed effects. Specifically, bank–firm controls comprise the collateralization rate, the share of long-term lending, the fraction of firm f ’s non-performing loans at bank b , and bank b ’s share in the firm’s total borrowing. Bank-level controls consist of indicator variables for quartiles of bank

¹⁰Our analysis excludes firms operating in the construction and real estate sectors.

size (by total assets) and quartiles of trade-sector exposure, measured as the share of lending to exporters and importers in total corporate lending. Importantly, our balance-of-covariates analysis indicates that, after controlling for these bank bins, the bank exposure measure is statistically orthogonal to other key bank characteristics, such as ROA, NPLs, liquidity, capital ratios, and credit to the public sector divided by total assets (see [Figure A.1](#)).¹¹

Firm–time fixed effects absorb credit demand shocks, so identification stems from within-firm variation across banks with differential exposure ([Khwaja and Mian, 2008](#)).¹² Exposure is standardized, so θ captures the effect of a one-standard-deviation increase.

[Table 2](#) presents the bank–firm level estimates for domestic importers. More exposed banks significantly reduced credit supply following the reform. A one standard deviation increase in bank exposure is associated with a 3.1 percent decline in committed credit relative to other lenders to the same firm. The contraction operates along both the intensive and extensive margins: exposed banks reduced credit volumes and were more likely to terminate lending relationships. Event-study estimates in [Figure 2](#) show no evidence of differential pre-trends and a sharp contraction beginning in 2012, consistent with the timing of the regulatory shock.

We next assess whether importers were able to offset this shock by reallocating borrowing across banks. We estimate:

$$\log y_{f,t} = \theta \times Exposure_f \times Policy_t + controls_{f,t} + \hat{\alpha}_{f,t} + \alpha_f + \varepsilon_{f,t}, \quad (6)$$

where $y_{f,t}$ denotes total bank credit to firm f , and $Exposure_f$ is the credit-weighted average

¹¹This final check addresses the concern that our exposure measure may proxy for banks’ sovereign debt holdings and therefore capture differential exposure to the sovereign debt crisis rather than to the provisioning shock. Public disclosures released as part of the European Banking Authority’s 2011 stress test indicate that Spanish banks’ sovereign exposures were heavily home-biased and held predominantly in the banking book. A proxy for banking-book sovereign exposure, defined as the ratio of public sector credit to total assets and computed using CIRBE data, is uncorrelated with our measure of bank regulatory exposure.

¹²Firms engaged in international trade in our sample typically maintain multiple banking relationships. In 2011, only about 30% of these firms borrowing from banks relied on a single lender.

exposure of its lenders prior to the reform. As controls, we interact quarter dummies with firm-level characteristics constructed as credit-weighted averages of the bank–firm variables used in specification (5), weighted by each bank’s outstanding credit to the firm at end-2011. We also include province and industry indicators. To account for firm-level demand shocks, we incorporate the estimated firm–quarter fixed effects from equation (5), $\hat{\alpha}_{f,t}$, following [Abowd et al. \(1999\)](#) and [Bonaccorsi di Patti and Sette \(2016\)](#).

[Table 3](#) shows that importers more exposed to affected banks experienced a 2.8 percent decline in total bank debt, indicating limited substitution toward less exposed lenders. Again, we find no evidence of differential pre-trends (see [Figure 3](#)).

Moreover, [Table 4](#) shows no evidence of substitution toward non-bank financing (columns (2)-(3)): importers more exposed to affected banks experience declines not only in total bank debt but also in total debt and total debt plus equity, indicating limited access to alternative funding sources. The contraction in credit is also accompanied by negative real effects—reductions in employment and tangible assets—reinforcing the economic significance of the shock (columns (4)-(5)).

Tables [A.2](#) and [A.3](#) in the Appendix present additional robustness tests showing that our results are not driven by bank specialization in foreign markets or by foreign demand shocks.

Having established that the reform tightened credit conditions for importers, we next examine whether this led to a decline in imports and how these effects propagated across borders.

4.2 Effect on Firm-Level Trade Flows

Using confidential Balance of Payments data, we observe firm-level trade flows by origin country, 2-digit product, and year. This granularity allows us to estimate the impact of the policy while flexibly controlling for foreign supply shocks and local demand conditions.

We estimate the following specification:

$$y_{f,c,g,t} = \alpha_{f,c,g} + \alpha_{c,g,t} + \alpha_{p(f),i(f),t} + \theta \times Exposure_f \times Policy_t + \varepsilon_{f,c,g,t}, \quad (7)$$

where $y_{f,c,g,t}$ denotes the mid-point growth rate of firm f 's imports of product g from country c in year t , relative to its pre-policy average.¹³ This specification captures both intensive and extensive margins, as zero trade flows reflect market entry and exit (Fonseca and Matray, 2024; Matray et al., 2024; Beaumont et al., 2025). $Exposure_f$ is defined as in equation (6), and $Policy_t$ equals one in 2012–2013. Firm–country–product fixed effects absorb time-invariant relationship characteristics, while country–product–year fixed effects control for foreign supply shocks. Province–industry–year fixed effects account for local economic conditions.¹⁴

Table 6 reports the results. A one standard deviation increase in exposure to affected banks is associated with a 3.6 percent decline in imports following the reform (column 1), roughly 14 percent of the year-on-year import growth rate in the year prior to the policy. The contraction operates along both margins: more exposed firms are significantly more likely to exit import markets and less likely to enter new ones. Event-study estimates confirm no pre-trends and a sharp post-reform decline in import growth (see Figure 4).

We further examine whether the import contraction varies with firms' financial conditions, as implied by a credit-driven mechanism. Consistent with this interpretation, Table A.4 shows that importers more reliant on a single relationship bank experience smaller trade contractions, as relationship lenders tend to shield clients during downturns (Bolton et al., 2016). Similarly, firms with greater pre-existing credit line availability exhibit weaker

¹³Specifically,

$$y_{f,c,g,t} = \frac{Z_{f,c,g,t} - \bar{Z}_{f,c,g}}{1/2(Z_{f,c,g,t} + \bar{Z}_{f,c,g})},$$

where $\bar{Z}_{f,c,g} = 1/5 \sum_{t=2009}^{2013} Z_{f,c,g,t}$ where $Z = \text{Imports}$.

¹⁴Given the granularity of our data, we exclude origin–product flows whose cumulative value over 2009–2013 represents less than 1% of total imports, corresponding to an average of less than 35,000 euros per year, to avoid results driven by very small or highly volatile trade relationships.

import declines, while those with higher non-performing loan ratios experience sharper contractions.

Appendix A reports additional robustness tests confirming that the results are not driven by specific trading partners or product categories (**Figure A.2**), firms' exposure to countries affected by the European sovereign debt crisis (**Table A.5**), or sectors indirectly linked to construction and real estate through input-output linkages (**Table A.6**). The results are also robust to alternative functional forms: the baseline log specification points to economically meaningful adjustment along the intensive margin, while PPML estimates, which account for zero trade flows and thus incorporate both intensive and extensive margins, yield similar conclusions to our baseline specification (see **Table A.7**).

Exporters. For completeness, **Appendix B** presents the analogous analysis for exporters. We find that more exposed banks also reduced lending to exporters (**Table B.1**), following a pattern similar to that for importers (**Figure B.1**). However, the overall impact on firms' total debt is somewhat dampened, consistent with exporters being able to smooth the shock by borrowing from other banks (**Table B.2, Figure B.2**). Using firm-level balance sheet data, we further show that exporters do not experience a decline in total debt nor in real outcomes (**Table B.3**), suggesting that they are able to offset the tightening in credit conditions.¹⁵

Turning to trade activity, we find a negative effect on exports (**Table B.7**), consistent with the trade finance literature (e.g., **Amiti and Weinstein 2011; Paravisini et al. 2014**). However, in line with our credit results, the impact on export activity is weaker and primarily driven by small transactions, with little evidence of an effect on larger flows. **Table B.8** presents trade-weighted regressions, which render the export effects statistically insignificant, whereas the impact on imports remains robust. In addition, **Table B.9** shows that the export response is concentrated among small exporters (bottom 25% by export value), with no significant

¹⁵Our baseline importer-exporter classification allows for overlap. **Table B.6** in the Appendix separates firms that are only importers, only exporters, or both, and confirms that exclusively importing firms experience the largest credit declines, whereas exclusively exporting firms better cushion the shock through multiple banking relationships. Moreover, **Tables B.4 and B.5** show that exporters are on average larger and financially stronger than importers, with greater debt capacity and more stable funding structures.

effect for larger firms that account for the bulk of aggregate exports.¹⁶ By contrast, the contraction in imports is broad-based across firm size groups.

For these reasons, our analysis of cross-border spillovers focuses on the contraction in Spanish import demand, which emerges as the more robust margin through which the regulatory shock propagates internationally.

4.3 Aggregate Cross-Border Effects

We now examine whether the firm-level contraction in imports translates into an aggregate decline in Spanish imports. As discussed above, lower imports among firms more exposed to the policy through their relationships with affected banks need not imply a fall in total imports. In principle, reduced demand for machinery by highly exposed firms in the wine industry could be offset by increased demand from less exposed firms whose banks were less affected by the reform.¹⁷ A modest—or even zero—response in aggregate imports would therefore still be fully consistent with our firm-level evidence.

To assess whether these offsetting forces matter in the aggregate, we use annual BACI data for 2009–2013 and construct a measure of exposure to the policy at the product–country–of-origin level:

$$\text{Exposure}_{go} = \sum_b w_b^{go} \times \text{Exposure}_b, \quad \text{where} \quad w_b^{go} = \frac{\sum_f L_{fb} \times M_f^{go}}{\sum_b \sum_f L_{fb} \times M_f^{go}} \quad (8)$$

Here, L_{fb} denotes the outstanding debt of firm f with bank b , and M_f^{go} denotes firm f 's imports of product g from country o . The resulting measure captures the weighted average

¹⁶A potential concern is that these findings are confounded by the export boom documented by [Almunia et al. \(2021\)](#), who show that post-crisis competitiveness gains led to a 31% increase in Spanish exports between 2009 and 2013. We view this as unlikely for two reasons. First, our specification includes province- and industry-year fixed effects that absorb the local shocks highlighted in their analysis. Second, we find no evidence of geographic sorting in our bank exposure measure: provinces experiencing larger domestic sales declines were not disproportionately served by banks more exposed to the regulation ([Figure A.3](#)).

¹⁷Business stealing may be particularly relevant in international trade. Internationally active firms are often larger and may be better positioned to capture market share from competitors facing negative shocks ([Matray et al. 2024](#); [Beaumont et al. 2025](#)).

exposure of the banks financing imports in product-origin pair go , where the weights w_b^{go} reflect each bank’s relative importance in funding those imports. We construct this measure using end-2006 information, a predetermined baseline that predates our sample period by three years.¹⁸ A product–country-of-origin pair is highly exposed to the policy if Spanish imports from that pair are disproportionately purchased by firms that rely more heavily on banks most exposed to the regulation. [Figure 5](#) shows this measure’s substantial cross-sectional variation across pairs.

We begin by estimating the regulation’s impact on Spain’s total imports, observed in BACI as other countries’ exports to Spain. Specifically, we compare how exports to Spain changed for more- versus less-exposed product-origin pairs, relative to how exports from the same pairs evolved in non-Spain destinations before and after the reform. Formally, we implement a triple difference-in-differences design:

$$\frac{X_{godt} - \bar{X}_{god}}{(X_{godt} + \bar{X}_{god})/2} = \gamma \times \text{Exposure}_{go} \times \mathbb{1}[d = \text{Spain}] \times \text{Post}_t + \delta_{god} + \delta_{gt} + \delta_{ot} + \delta_{dt} + u_{godt}, \quad (9)$$

where X_{godt} denotes exports of product g from country o to country d in year t . Thus, our dependent variable is the mid-point growth rate of exports relative to the corresponding average value of X_{godt} over the sample period for each product-origin-destination triplet. Exposure_{go} is defined as in equation (8) and is interacted with an indicator equal to one when Spain is the destination country and with Post_t , which equals one in the post-reform years 2012 and 2013. We include time-invariant product×origin×destination fixed effects denoted by δ_{god} , and time-varying fixed effects at the product, origin, and destination level, denoted by δ_{gt} , δ_{ot} , and δ_{dt} , respectively. Given the granularity of our data, we exclude product-origin pairs for which Spain accounts for less than 1.1 percent of total exports

¹⁸From 2008 onward, the representativeness of the Balance of Payments data by country and product declines due to a change in reporting procedures; see [Prades and Villegas-Sánchez \(2026\)](#) for details. We therefore use 2006 as the baseline year to compute the exposure measure at the product-origin level and match it with BACI trade flows. For all other exposure measures—bank-level and firm-level—we employ end-of-year 2011 data.

across destinations, the median share in the raw data, thereby reducing noise from markets in which Spain is only a negligible destination. Even after this restriction, the sample still covers 91 percent of Spain’s total imports.

Our parameter of interest, γ , captures whether more exposed product-origin pairs experience a larger post-reform decline in exports to Spain relative to exports to other destinations. If business stealing across Spanish firms fully offsets the decline in imports by more exposed firms, aggregate exports to Spain should not fall differentially across product-origin pairs with different levels of exposure, and γ should be zero. The triple-difference design includes product-year, origin-year, and destination-year fixed effects, which absorb product-specific shocks, origin-specific shocks, and destination-specific shocks that coincide with the increase in loan-loss provisions. Our identifying assumption is therefore that, absent the reform, the differential evolution of exports to Spain between high- and low-exposure product-origin pairs would have been similar to the corresponding difference in other importing countries.

As a further robustness check, we re-estimate the triple-difference specification restricting the set of comparison destinations to markets comparable to Spain. Specifically, we consider EU member states and the advanced economies classified by Autor et al. (2013).¹⁹ In addition, we exclude Greece, Ireland, Italy, and Portugal (GIIP) from the set of exporters to address potential identification concerns arising from the decline in their exports to Spain during the European sovereign debt crisis.

Table 7 reports the results. Column (1) shows that a one-standard-deviation increase in exposure is associated with a 3.2 percent decline in exports to Spain relative to exports to other destinations in a parsimonious specification with origin-destination and destination-year fixed effects. Column (2) yields a quantitatively similar estimate in a more saturated specification that additionally includes product-year fixed effects. Our benchmark specification, reported in Column (3), further adds origin-year fixed effects to absorb time-varying

¹⁹That is, high-income countries with comparable trade data: Australia, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Japan, Luxembourg, Netherlands, New Zealand, Portugal, Spain, Switzerland, and the United States.

supply shocks. The estimate implies that a one-standard-deviation increase in product-origin exposure reduces exports to Spain by 2.1 percent in the post-reform period, relative to the same product-origin pair’s exports to other destinations. Column (4) shows similar results when restricting the sample of destination countries to the set of high-income countries with comparable trade data used by Autor et al. (2013). Finally, Column (5) yields similar estimates when Greece, Ireland, Italy, and Portugal (GIIP) are excluded from the set of exporters.

Figure 6 presents the event-study estimates for our benchmark specification. Prior to the reform, exposure does not predict differential movements in exports to Spain, consistent with our identifying assumption. After the reform, exports to Spain from highly exposed product-origin pairs decline persistently relative to exports from the same pairs to other destinations. Taken together, these results indicate that business stealing across Spanish importing firms was limited: the firm-level contraction documented in Section 4.2 was not offset in the aggregate and translated into a decline in total imports.

Despite the advantages of our triple-differences specification in controlling for multiple demand and supply shocks beyond the increase in Spanish loan loss provisions, the narrow level of disaggregation in our data raises the concern that results may be disproportionately driven by small destinations (i.e. partners with negligible trade volumes), which are likely to display noisier dynamics than larger markets. In addition, the analysis of the attenuation and reallocation margins in the following section requires us to aggregate the data to the exporter-product level.

Thus, we estimate the following standard difference-in-differences specification separately for exports to Spain and for exports to all other destinations, as a validation exercise to the

triple-difference results:²⁰

$$\frac{X_{got} - \bar{X}_{go}}{(X_{got} + \bar{X}_{go})/2} = \beta \times \text{Exposure}_{go} \times \text{Post}_t + \delta_{go} + \delta_{x(g)t} + \delta_{x(o)t} + \delta_{x(o,g)t} + u_{got}, \quad (10)$$

where X_{got} denotes exports of product g from country o either to Spain or all other destinations in year t . Because this aggregate analysis is conducted at the exporter-product level, we cannot include the full set of high-dimensional fixed effects used in the triple-difference specification. Instead, we include time-varying fixed effects for groups of exporters and products. Specifically, $\delta_{x(o)t}$ denotes exporter-group-by-year fixed effects based on characteristics such as distance to Spain and official language, while $\delta_{x(g)t}$ denotes product-group-by-year fixed effects based on product differentiation.²¹ Finally, we include time-varying fixed effects for deciles of Spanish participation at the product-origin level, $\delta_{x(o,g)t}$, to control for Spain's relative importance in each market.

Our results are consistent with the triple-difference estimates. The benchmark specification reported in Column (3) of Table 8 implies a 2.7 percent decline in exports to Spain. Column (7) serves as a placebo: exports to other destinations do not respond significantly, consistent with the shock being specific to Spain. The estimates remain very similar when excluding GIIP exporters (Columns (4) and (8)). The event-study in Figure 7 shows a clear post-reform decline and no evidence of differential pre-trends. Overall, these findings reinforce the evidence from Table 7.

5 Cross-Border Spillovers

Having established that the policy reduced Spanish import demand, we now examine how foreign exporters adjusted to this contraction. The analysis follows the two margins of

²⁰The specification for exports to all other destinations serves as a placebo test, since the credit shock originated in Spain and should not systematically affect exports elsewhere.

²¹As explained below, we construct quartiles of within-product price dispersion to proxy for product differentiation.

Section 2. Under attenuation, the exporter substitutes its own trade credit for the importer’s lost bank credit. Under reallocation, the shipments that are not absorbed in this way are redirected toward other destinations. Proposition 1 predicts that exports to Spain fall by more where the cost of extending trade credit is higher, while Proposition 2 predicts that lost sales reappear elsewhere only for goods substitutable enough and origins frictional enough.

5.1 Attenuation

5.1.1 Firm-level evidence on supplier credit use

In the model, exporters attenuate the decline in trade flows by extending trade credit to financially constrained importers, so the international propagation of the shock depends on the relationships linking importers and foreign suppliers.

Empirically, however, directly observing bilateral financing arrangements between importers and exporters is challenging. Our data contain firm-level measures of short-term non-interest-bearing external financing received by Spanish firms, which primarily reflects commercial liabilities to suppliers and is therefore most closely related to trade credit (or supplier credit), rather than bank-intermediated trade-finance instruments such as letters of credit. Because the data do not identify the counterparties associated with these liabilities, we cannot directly observe financing provided by individual foreign suppliers.²² Accordingly, we interpret supplier credit as a reduced-form proxy for financing and relationship-specific arrangements embedded in importer–supplier relationships. More broadly, it may also capture the strength of commercial relationships between importers and suppliers, which previous work has shown to play an important role in sustaining trade during adverse shocks (Monarch and Schmidt-Eisenlohr, 2023; Macchiavello and Morjaria, 2015).

Table A.9 shows that importing firms with greater pre-existing reliance on supplier credit

²²Although our measure of supplier credit does not distinguish between domestic and foreign counterparties, Figure A.6 shows that firms more reliant on imports also rely more heavily on supplier credit, suggesting that foreign suppliers contribute importantly to this financing margin.

experience significantly smaller declines in imports following the banking shock. While this pattern is consistent with supplier credit attenuating financing disruptions, it may also reflect the fact that firms engaged in stronger supplier relationships are less likely to sever sourcing links following temporary financial shocks.

Figure A.7 further shows that, conditional on import intensity and restricting attention to importing firms, supplier credit is lower among firms sourcing from more distant partners and from financially developed economies. These patterns are consistent with the idea that financing arrangements embedded in supplier relationships are easier to sustain when cross-border frictions are lower. By contrast, transactions involving suppliers located in financially developed economies appear to rely relatively less on inter-firm financing, consistent with the greater availability of formal external finance in those countries.

We next examine whether firms differentially adjust supplier-credit usage following the domestic banking shock depending on the characteristics of their sourcing relationships. Specifically, we estimate:

$$\begin{aligned} \Delta TC_f = & \beta_1 Exposure_f + \beta_2 X_f + \beta_3 (Exposure_f \times X_f) + \Gamma' Z_f \\ & + \alpha_{s(f)} + \alpha_{p(f)} + \alpha_{c(f)} + \alpha_{u(f)} + \alpha_{q(f)} + \varepsilon_f \end{aligned} \quad (11)$$

where ΔTC_f denotes the midpoint growth rate of supplier credit for firm f , $Exposure_f$ measures the firm's exposure to banks affected by the 2012 provisioning reform, and X_f captures characteristics of the firm's sourcing structure. Specifically, X_f corresponds, depending on the specification, to the share of imports sourced from geographically distant origins, non-Spanish-speaking countries, or distant non-Spanish-speaking origins. Z_f includes firm-level controls, while the fixed effects absorb industry, province, main origin, share of imports to total purchases and number of sourcing countries quintile effects.

The results reported in Table A.10 show two main findings. First, among firms sourcing primarily from financially developed countries, supplier credit exhibits little response to the

domestic banking shock. In contrast, among firms sourcing from less financially developed origins, supplier credit increases significantly for firms more exposed to affected banks. Second, this adjustment weakens with trade frictions. Within the low-financial-development sample, firms sourcing more intensively from distant origins or from distant non-Spanish-speaking countries exhibit significantly smaller increases in supplier credit following the shock.

5.1.2 Aggregate evidence

We next examine whether the patterns documented at the firm level are also reflected in aggregate trade flows. [Table 9](#) reports estimates of equation (10) using exports to Spain as the dependent variable. Column (1) shows that exporters located in financially developed economies are largely unaffected by the contraction in Spanish demand. While the firm-level evidence suggests that this resilience is not driven by greater supplier-credit adjustment, it points to the presence of other mechanisms that help sustain trading relationships following a demand disruption. By contrast, column (2) indicates that a one-standard-deviation increase in product-origin exposure reduces exports to Spain by 3.3 percent when exporters belong to countries with lower levels of financial development.²³

We next examine whether trade frictions further shape these effects. Consistent with the patterns documented at the firm level, distance raises the monitoring, enforcement, and working-capital costs of extending trade credit, which the model captures through the friction ψ , while language barriers may impede communication and relationship-specific contracting. Accordingly, the decline in exports from less financially developed countries is concentrated among exporters facing higher trade frictions.

Columns (3) and (4) show that a one-standard-deviation increase in exposure leads to a statistically insignificant 2.2 percent decline in exports for countries located close to Spain,

²³[Figure A.4](#) plots the density of product-origin exposure by financial-development group and shows that the distributions are similar, alleviating concerns that the results are driven by differences in exposure across country groups.

but a significant 3.6 percent decline for more distant exporters. Similarly, columns (5) and (6) show no significant effect for Spanish-speaking countries, whereas exports from non-Spanish-speaking countries decline significantly following the shock.²⁴ Finally, column (8) shows a significant 3.9 percent contraction in exports for countries that are both distant and non-Spanish-speaking, whereas column (7) reveals a smaller and statistically insignificant effect for countries that are either close to Spain or share a common language.

Table 10 reports results from the same analysis but using exports to destinations other than Spain as the dependent variable. This serves as a placebo test: since the credit shock originated in Spain, we should not expect financial development or trade costs to play a significant role for other destinations, and indeed, we find no such effects.

One concern is that these estimates might partly reflect a reduction in lending by Spanish banks abroad, similar to the mechanism documented by [Tripathy \(2020\)](#). However, our results remain robust to the inclusion of exporter-year fixed effects, which already account for the presence of Spanish banks in foreign countries. That said, the trade channel could still be amplified by the Spanish banking footprint in foreign markets, for instance, if Spanish banks are specialized in financing firms that export to Spain ([Paravisini et al., 2023](#)).

To assess this mechanism, we use BankFocus data to map the international footprint of Spanish banks and compare export responses across countries with and without a Spanish bank presence. The results in **Table A.8** show no differential response. Exports from financially developed countries remain unaffected regardless of Spanish bank presence, while the decline observed in less financially developed countries does not vary with local Spanish bank operations. Moreover, there is no evidence that Spanish bank presence interacts with trade costs. These findings rule out multinational banking networks as the primary driver of our results.

²⁴**Figure A.5** plots the density of product-origin exposure across country groups and shows that the distributions are comparable.

5.2 Export Reallocation

If foreign exporters cannot fully attenuate the contraction in Spanish demand through trade credit, shipments are redirected toward alternative destinations. Whether this raises the value of exports elsewhere is not automatic. Redirected quantity arrives at a lower price, and the exporter’s higher cost of funds raises the trade-credit rate it charges in every destination it serves. Reallocation in value therefore depends on how substitutable the product is and on how important Spain is as a destination.

To capture this, we note that the effect of our treatment measure, defined in equation (8), on total exports may be amplified or dampened by a product-origin pair’s reliance on Spanish demand. For instance, a pair may appear highly exposed if its Spanish importers rely heavily on treated banks, but if Spain represents only a small share of its overall exports, the aggregate exposure is effectively low. In the model, the elasticity threshold above which lost sales reappear elsewhere falls with the shocked destination’s share of the exporter’s sales, so that when Spain is a minor buyer the rise in the exporter’s cost of funds dominates the redirected quantity and reallocation does not raise the value of exports elsewhere. To account for this, we construct the following measure of exporter exposure to the regulation in Spain, which weights our product-origin exposure by the share of each pair’s exports directed to Spain:

$$\text{Aggregate Exposure}_{go} = \frac{X_{go}^{\text{Spain}}}{X_{go}} \times \text{Exposure}_{go}. \quad (12)$$

where $\frac{X_{go}^{\text{Spain}}}{X_{go}}$ denotes Spain’s share of country o ’s exports of product g prior to the policy.²⁵ Under this definition, a foreign exporter o of product g is highly exposed when Spanish importers of g rely heavily on treated banks and Spain represents a substantial share of o ’s exports of that product. We then re-estimate the difference-in-differences equation (10), using this new exposure measure as the independent variable and total exports to all desti-

²⁵Shares are calculated using 2006 data, consistent with the exposure measure in equation (8).

nations for each product-origin pair as the dependent variable:

$$\frac{X_{got} - \bar{X}_{go}}{(X_{got} + \bar{X}_{go})/2} = \rho \times \text{Aggregate Exposure}_{go} \times Post_t + \delta_{go} + \delta_{x(g)t} + \delta_{x(o)t} + u_{got}, \quad (13)$$

Thus, our parameter of interest, ρ , measures the cross-border effects of Spanish loan loss provisions and is identified by comparing total sales from highly exposed exporters to those from less exposed ones, before and after the regulation. If Spain's trade partners faced no frictions in reallocating sales away from Spain, our exposure measure would have no effect on foreign countries' total exports. Conversely, if redirected shipments do not compensate the loss, we would observe a significant decline in trade partners' total exports.

We report our results in [Table 11](#). Column (1) shows that a one-standard deviation higher aggregate exposure is associated with a 1.6 percent decline in total exports after the policy. Although not statistically significant at conventional levels, this aggregate effect reflects a significant 3.4 percent contraction in exports to Spain and an insignificant response of exports to other destinations, as shown in columns (2) and (3). To explore heterogeneity, we split product-country-of-origin cells into two groups based on Spain's share in their total sales, such that each group accounts for 50 percent of Spain's total imports.²⁶ Columns (4)-(6) show that aggregate exposure has no statistically significant effect among exporters for whom Spain represents only a minor share of sales. By contrast, columns (7)-(9) indicate that when Spain is a major buyer, aggregate exposure leads to a 2.8 percent decline in total exports, driven almost entirely by a 4.5 percent contraction in sales to Spain, with no evidence of reallocation to other destinations.²⁷

[Figure 8](#) presents the event-study graph corresponding to the average effect reported in

²⁶As a result of this partition, pairs with a high Spanish share are those where exports to Spain account for more than 6 percent of their total sales.

²⁷[Table A.11](#) in the Appendix conducts a similar analysis, estimating the impact of our aggregate exposure in countries that are geographically distant from Spain and do not share a common language. Columns (2), (5), and (8) show that exports to Spain decline sharply in these higher-trade-cost countries, consistent with our main results, which in turn drives the aggregate export response in columns (1), (4), and (7). By contrast, the response of exports to other destinations is unrelated to the trade costs of exporting to Spain.

column (7) of [Table 11](#). The decline in total exports from highly exposed product-origin pairs appears only after the policy shock, while exports from differently exposed pairs followed parallel trends before 2011, supporting our identifying assumption. These results indicate that the increase in loan loss provisions in Spain generated aggregate effects abroad through the import demand channel, as reallocation to alternative destinations was limited for the average exporter-product pair.

To assess the role of product substitutability discussed in our model, we conduct a heterogeneity analysis at the product level. We test the hypothesis that lost sales of homogeneous products, such as commodities, are recovered in other destinations relatively easily, whereas heterogeneous products, for which buyers demand differentiated varieties, are not. In the model, substitutability governs how much quantity moves to keep invoice prices equal across destinations, so homogeneous products clear the reallocation threshold at lower frictions than differentiated ones. Using BACI data, we compute an implicit price for each HS 6-digit product–country pair by dividing total value by total weight, and then calculate the variance of these prices at the HS 2-digit level. Products with price dispersion in the bottom quartile are classified as homogeneous, while those above this cutoff are defined as heterogeneous.

We report our results in [Table 12](#), focusing on countries that were less able to attenuate the import demand shock from Spain—those with low financial development, greater geographic distance, and no common language—where the need for reallocation is expected to be strongest. Columns (1)-(3) present results for products with greater price dispersion (i.e., more heterogeneous goods). In this case, total exports fall by 9%, driven entirely by a contraction in sales to Spain, which is not offset by sales to other destinations. By contrast, column (4) shows that when products are relatively homogeneous, total exports remain unaffected. Columns (5)-(6) confirm that although exports to Spain still decline, homogeneous products are reallocated to other destinations, fully compensating for the decline in Spanish imports.²⁸

²⁸[Figure A.8](#) plots the density of aggregate exposure across product groups defined by price dispersion

Overall, these findings confirm that the extent of cross-border transmission depends on two conditions. The first is whether the exporter can absorb the shock through trade credit, which fails in financially underdeveloped and high-trade-cost economies. The second is whether redirected shipments compensate the loss, which fails for differentiated products.

6 Conclusions

This paper provides evidence that domestic banking regulations generate cross-border spillovers through international trade, with unintended consequences for the trade flows of foreign partners.

Our analysis focuses on the sharp and unexpected 2012 increase in loan-loss provisioning requirements for Spanish banks' exposures to the construction and real estate sectors, an intervention plausibly unrelated to global trade conditions. Using the CIR, confidential firm-level trade transactions, and bilateral trade data from BACI, we examine the effects of this policy along three dimensions: credit supply to firms engaged in international trade, firm-level import outcomes, and the aggregate exports of Spain's trade partners.

We document that banks more exposed to the regulation reduced credit supply to their borrowers; importers were unable to fully substitute toward alternative lenders or funding sources, translating into a broad-based import decline. Moreover, less exposed importers did not compensate for this decline, indicating limited reallocation within Spain.

Turning to the international implications, we show that the resulting decline in Spanish import demand generated significant spillovers to Spain's trade partners. Consistent with the mechanisms emphasized in our conceptual framework, exporters adjusted along two margins. First, exporters could attenuate the shock by extending trade credit to constrained importers. We find that this attenuation margin depends on exporter-side financial conditions: exporters located in financially developed economies were able to absorb the

and shows that the distributions are similar.

shock, whereas exporters in less financially developed countries experienced significantly larger export declines. Moreover, these effects are amplified when trade costs are higher, in geographically distant or non-Spanish-speaking countries.

Second, exporters could respond by reallocating sales toward alternative destinations. However, we find limited evidence of systematic reallocation. Exporters were able to redirect sales only for relatively homogeneous products shipped from the origins least able to attenuate the shock, while differentiated goods experienced larger declines in total exports. These patterns highlight the role of product substitutability in shaping the international transmission of domestic financial shocks.

Taken together, our findings show that the cross-border consequences of banking regulation extend beyond international capital flows or multinational bank networks. Domestic financial regulations can also propagate internationally through trade linkages, with the magnitude of spillovers shaped by exporters' financial development, trade costs, and product differentiation.

The evidence presented in this paper carries important policy implications. Domestic policymakers should account for externalities beyond their borders when designing prudential policy, and international coordination may therefore be critical—not only to safeguard financial stability, but also to minimize unintended disruptions to global trade.

References

- Abowd, J. M., Kramarz, F., and Margolis, D. N. (1999). High Wage Workers and High Wage Firms. *Econometrica*, 67(2):251–333.
- Adelino, M., Ferreira, M. A., Giannetti, M., and Pires, P. (2022). Trade Credit and the Transmission of Unconventional Monetary Policy. *The Review of Financial Studies*, 36(2):775–813.
- Ahn, J. and Sarmiento, M. (2019). Estimating the Direct Impact of Bank Liquidity Shocks on the Real Economy: Evidence from Letter-of-Credit Import Transactions in Colombia. *Review of International Economics*, 27(5):1510–1536.
- Aiyar, S., Calomiris, C. W., Hooley, J., Korniyenko, Y., and Wieladek, T. (2014). The International Transmission of Bank Capital Requirements: Evidence from the UK. *Journal of Financial Economics*, 113(3):368–382.
- Alfaro, L. and Chor, D. (2023). Global Supply Chains: The Looming ‘Great Reallocation’. Working Paper 31661, National Bureau of Economic Research.
- Alfaro, L., García-Santana, M., and Moral-Benito, E. (2021). On the direct and indirect real effects of credit supply shocks. *Journal of Financial Economics*, 139(3):895–921.
- Almeida, H., Carvalho, D. R., and Kim, Y. (2025). How Do Financing Frictions Shape Export Activity? The Firm Balance-Sheet Channel. Working paper.
- Almunia, M., Antràs, P., Lopez-Rodriguez, D., and Morales, E. (2021). Venting Out: Exports during a Domestic Slump. *American Economic Review*, 111(11):3611–3662.
- Amiti, M. and Weinstein, D. E. (2011). Exports and Financial Shocks. *Quarterly Journal of Economics*, 126(4):1841–1877.
- Antràs, P. and Foley, C. F. (2015). Poultry in motion: A study of international trade finance practices. *Journal of Political Economy*, 123(4):853–901.
- Auer, R. A., Levchenko, A. A., and Saure, P. (2019). International Inflation Spillovers through Input Linkages. *The Review of Economics and Statistics*, 101(3):507–521.
- Autor, D. H., Dorn, D., and Hanson, G. H. (2013). The China Syndrome: Local Labor Market Effects of Import Competition in the United States. *American Economic Review*, 103(6):2121–2168.
- Bahaj, S. and Malherbe, F. (2024). The Cross-Border Effects of Bank Capital Regulation. *Journal of Financial Economics*, 160.
- Baskaya, Y. S., Binici, M., and Kenç, T. (2017). International Banking and Cross-Border Effects of Regulation: Lessons from Turkey. *International Journal of Central Banking*, 13(1):317–373.
- Beaumont, P. and Lenoir, C. (2025). Customer Capital in Times of Crisis. Working paper.

- Beaumont, P., Matray, A., and Xu, C. (2025). Entry, Exit, and Aggregation in Trade Data: A New Estimator. Working paper.
- Benguria, F. and Saffie, F. (2024). Escaping the trade war: Finance and relational supply chains in the adjustment to trade policy shocks. *Journal of International Economics*, 152:103987.
- Boehm, C. E., Flaaen, A., and Pandalai-Nayar, N. (2019). Input Linkages and the Transmission of Shocks: Firm-Level Evidence from the 2011 Tohoku Earthquake. *The Review of Economics and Statistics*, 101(1):60–75.
- Bolton, P., Freixas, X., Gambacorta, L., and Mistrulli, P. E. (2016). Relationship and Transaction Lending in a Crisis. *The Review of Financial Studies*, 29(10):2643–2676.
- Bolton, P. and Oehmke, M. (2018). Bank Resolution and the Structure of Global Banks. *The Review of Financial Studies*, 32(6):2384–2421.
- Bonaccorsi di Patti, E. and Sette, E. (2016). Did the securitization market freeze affect bank lending during the financial crisis? Evidence from a credit register. *Journal of Financial Intermediation*, 25:54–76.
- Borchert, L., de Haas, R., Kirschenmann, K., and Schultz, A. (2023). Broken Relationships: De-Risking by Correspondent Banks and International Trade. Working Paper 285, European Bank for Reconstruction & Development.
- Brauning, F. and Ivashina, V. (2020). Monetary Policy and Global Banking. *The Journal of Finance*, 75(6):3055–3095.
- Brauning, F. and Sheremirov, V. (2023). The Transmission Mechanisms of International Business Cycles: International Trade and the Foreign Effects of US Monetary Policy. *IMF Economic Review*, 71:300–325.
- Bruno, V. and Shin, H. S. (2014). Cross-border banking and global liquidity. *The Review of Economic Studies*, 82(2):535–564.
- Buch, C. M., Bussiere, M., Goldberg, L., and Hills, R. (2019). The international transmission of monetary policy. *Journal of International Money and Finance*, 91:29–48.
- Buch, C. M. and Goldberg, L. (2017). Cross-Border Prudential Policy Spillovers: How Much? How Important? Evidence from the International Banking Research Network. *International Journal of Central Banking*, 13(S1):505–558.
- Burstein, A., Kurz, C., and Tesar, L. (2008). Trade, production sharing, and the international transmission of business cycles. *Journal of Monetary Economics*, 55(4):775–795.
- Carvalho, V. M., Nirei, M., Saito, Y. U., and Tahbaz-Salehi, A. (2020). Supply Chain Disruptions: Evidence from the Great East Japan Earthquake. *The Quarterly Journal of Economics*, 136(2):1255–1321.

- Cetorelli, N. and Goldberg, L. S. (2012). Banking globalization and monetary transmission. *The Journal of Finance*, 67(5):1811–1843.
- Costello, A. M. (2020). Credit market disruptions and liquidity spillover effects in the supply chain. *Journal of Political Economy*, 128(9):3434–3468.
- Del Prete, S. and Federico, S. (2014). Trade and finance: Is there more than just ‘trade finance’? Evidence from matched bank-firm data. *Temi di Discussione 248, Bank of Italy*.
- Demir, B., Michalski, T. K., and Ors, E. (2017). Risk-based capital requirements for banks and international trade. *The Review of Financial Studies*, 30(11):3970–4002.
- Di Giovanni, J. and Hale, G. (2022). Stock Market Spillovers via the Global Production Network: Transmission of U.S. Monetary Policy. *The Journal of Finance*, 77(6):3373–3421.
- Ding, X. (2024). Industry Linkages from Joint Production. *CEPR Working Paper 056*.
- Dogan, A. and Hjortso, I. (2024). The Role of Finance for Export Dynamics: Evidence from the UK. *CEPR Working Paper 19027*.
- Ersahin, N., Giannetti, M., and Huang, R. (2024). Trade credit and the stability of supply chains. *Journal of Financial Economics*, 155:103830.
- Fonseca, J. and Matray, A. (2024). Financial inclusion, economic development, and inequality: Evidence from Brazil. *Journal of Financial Economics*, 156:103854.
- Gaulier, G. and Zignago, S. (2010). BACI: International Trade Database at the Product-Level. The 1994-2007 Version. Working Papers 2010-23, CEPII research center.
- Gopinath, G., Gourinchas, P.-O., Presbitero, A. F., and Topalova, P. (2025). Changing global linkages: A new Cold War? *Journal of International Economics*, 153:104042.
- Gutiérrez, E. and Moral-Benito, E. (2024). Trade and credit: Revisiting the evidence. *Review of International Economics*, 32(3):1149–1173.
- Hale, G., Kapan, T., and Minoiu, C. (2019). Shock Transmission Through Cross-Border Bank Lending: Credit and Real Effects. *The Review of Financial Studies*, 33(10):4839–4882.
- Huremovic, K., Jiménez, G., Moral-Benito, E., Peydró, J.-L., and Vega-Redondo, F. (2025). Production and Financial Networks in Interplay. Working paper.
- Jiao, Y., Liu, Z., Tian, Z., and Wang, X. (2024). The Impacts of the U.S. Trade War on Chinese Exporters. *The Review of Economics and Statistics*, 106(6):1576–1587.
- Jiménez, G., Ongena, S., Peydró, J.-L., and Saurina, J. (2017). Macroprudential Policy, Countercyclical Bank Capital Buffers, and Credit Supply: Evidence from the Spanish Dynamic Provisioning Experiments. *Journal of Political Economy*, 125(6):2126–2177.

- Khwaja, A. I. and Mian, A. (2008). Tracing the Impact of Bank Liquidity Shocks: Evidence from an Emerging Market. *American Economic Review*, 98(4):1413–1442.
- Macchiavello, R. and Morjaria, A. (2015). The value of relationships: Evidence from a supply shock to kenyan rose exports. *American Economic Review*, 105(9):2911–2945.
- Manova, K. (2013). Credit constraints, heterogeneous firms, and international trade. *Review of Economic Studies*, 80(2):711–744.
- Matray, A., Müller, K., Xu, C., and Kabir, P. (2024). EXIM’s Exit: The Real Effects of Trade Financing by Export Credit Agencies. *Working Paper 32019, NBER*.
- Monarch, R. and Schmidt-Eisenlohr, T. (2023). Longevity and the value of trade relationships. *Journal of International Economics*, 145:103842.
- Morais, B., Peydró, J.-L., Roldan-Pena, J., and Ruiz-Ortega, C. (2019). The International Bank Lending Channel of Monetary Policy Rates and QE: Credit Supply, Reach-for-Yield, and Real Effects. *The Journal of Finance*, 74(1):55–90.
- Moreira, P. D. and Monteiro, J. (2023). The Impact of a Higher Cost of Credit on Exporters: Evidence from a Change in Banking Regulation. Working Papers 202320, Banco de Portugal, Economics and Research Department.
- Niepmann, F. and Schmidt-Eisenlohr, T. (2017a). International trade, risk and the role of banks. *Journal of International Economics*, 107(C):111–126.
- Niepmann, F. and Schmidt-Eisenlohr, T. (2017b). No guarantees, no trade: How banks affect export patterns. *Journal of International Economics*, 108:338–350.
- Paravisini, D., Rappoport, V., and Schnabl, P. (2023). Specialization in Bank Lending: Evidence from Exporting Firms. *The Journal of Finance*, 78(4):2049–2085.
- Paravisini, D., Rappoport, V., Schnabl, P., and Wolfenzon, D. (2014). Dissecting the Effect of Credit Supply on Trade: Evidence from Matched Credit-Export Data. *The Review of Economic Studies*, 82(1):333–359.
- Prades, E. and Villegas-Sánchez, C. (2026). Made in Spain: Import Switching and Productivity. Working paper.
- Richmond, R. J. (2019). Trade Network Centrality and Currency Risk Premia. *The Journal of Finance*, 74(3):1315–1361.
- Romero, D. (2025). Domestic linkages and the transmission of commodity price shocks. *Journal of International Economics*, 153:104041.
- Schmidt-Eisenlohr, T. (2013). Towards a theory of trade finance. *Journal of International Economics*, 91(1):96–112.
- Tripathy, J. (2020). Cross-border effects of regulatory spillovers: Evidence from Mexico. *Journal of International Economics*, 126:103350.

Xu, C. (2022). Reshaping Global Trade: The Immediate and Long-Run Effects of Bank Failures. *The Quarterly Journal of Economics*, 137(4):2107–2161.

Figures and tables

Table 1: Summary statistics

	Obs.	Mean	sd	p25	p50	p75
Panel A: Bank-Firm Credit Register Data						
Log of Credit	1,288,007	5.65	1.65	4.61	5.64	6.65
Long-term credit ratio	1,288,007	54.33	41.08	6.07	59.32	100.00
Collateralization rate	1,288,007	13.40	30.25	0.00	0.00	0.00
NPL ratio	1,288,007	8.07	26.87	0.00	0.00	0.00
Credit share	1,288,007	23.30	22.83	6.34	15.29	32.85
Termination dummy	137,800	0.18	0.38	0.00	0.00	0.00
Panel B: Firm-Level Credit Data						
Log of Total Bank Credit	316,925	7.17	1.60	6.12	7.09	8.13
Panel C: Bank-Level data						
Exposure	83	36.76	16.71	25.70	34.95	47.36
Size	83	14.69	2.38	12.84	14.13	16.63
Capital ratio	83	9.56	6.46	6.66	8.04	10.31
Liquidity ratio	83	5.70	5.73	1.83	3.49	8.12
ROA	83	0.08	1.07	0.11	0.27	0.45
NPL ratio	83	5.39	4.26	2.75	4.84	6.61
Rural bank dummy	83	0.47	0.50	0.00	0.00	1.00
% Lending to X or M	83	34.66	21.11	18.57	34.18	48.59
Local Govt. Credit/Assets	83	2.67	6.20	0.32	1.50	3.08
Panel D: Balance of Payments Import Data						
Imports, mid-point growth	558,540	-0.84	1.28	-2.00	-1.93	0.39
Imports, Entry dummy	214,098	0.35	0.48	0.00	0.00	1.00
Imports, Exit dummy	221,390	0.33	0.47	0.00	0.00	1.00
Panel E: BACI Bilateral Trade Data						
Bilateral trade flow, mid-point growth	3,141,210	-0.54	1.07	-1.90	-0.28	0.23
Exports, mid-point growth	24,945	-0.07	0.40	-0.17	-0.00	0.11

In Panel A, bank–firm controls are constructed using credit information as of end-2011. Panel B aggregates CIRBE data to the firm level, summing the credit extended to each firm by all Spanish banks. In Panel C, bank-level variables are constructed using supervisory financial statements as of end-2011. Panel D is based on transaction-level Balance of Payments data at the importer–HS2–origin level. Panel E uses BACI bilateral trade data aggregated to the HS2–origin–destination level. All variables are defined in [Table A.1](#).

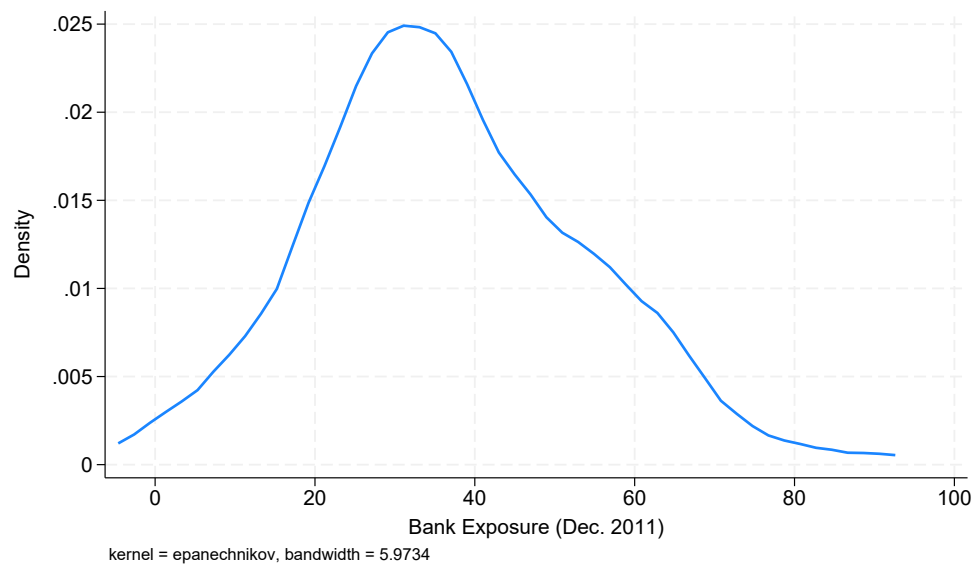


Figure 1: Bank Exposure: Kernel Density Estimate

This figure plots the kernel density of bank exposure defined by the ratio of construction and real estate lending over total corporate loans as of December 2011.

Table 2: Domestic credit contraction: intensive and extensive margins

	Intensive Margin		Extensive Margin	
	(1)	(2)	(3)	(4)
<i>Exposure</i> $\times$ <i>Policy</i>	-0.025** (0.011)	-0.031** (0.013)	0.016* (0.008)	0.019* (0.010)
Controls $\times$ time	No	Yes	–	–
Controls	–	–	No	Yes
Firm-time FE	Yes	Yes	–	–
Firm FE	–	–	Yes	Yes
Bank-firm FE	Yes	Yes	–	–
Obs.	962,963	962,963	102,788	102,788
R^2	0.97	0.97	0.52	0.56

This table presents regression results on lending from bank b to firm f at the intensive margin (columns 1–2) and the extensive margin (columns 3–4). In columns 1–2, the dependent variable is the logarithm of credit granted by bank b to firm f in quarter t . In columns 3–4, conditional on the existence of the lending relationship at the end of 2011, the dependent variable is a dummy variable that takes the value of one if the lending relationship was terminated in the aftermath of the policy and zero otherwise. *Exposure* is computed as the ratio of construction and real estate lending to total corporate sector lending by bank b as of the end of 2011. *Controls* are measured as of the end of 2011 and include bank-firm characteristics (the collateralization rate, the ratio of long-term lending, the share of firm f 's NPLs with the bank, and the share of lending coming from bank b out of firm f 's total bank lending) and bank characteristics (dummies for quartiles of size, measured by total assets, and quartiles of exposure to exporter/importer companies, measured as the lending share to exporters and importers relative to total corporate sector lending). All variables are defined in [Table A.1](#). We exclude firms in the construction or real estate sectors. Standard errors are double clustered at the bank and firm levels and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

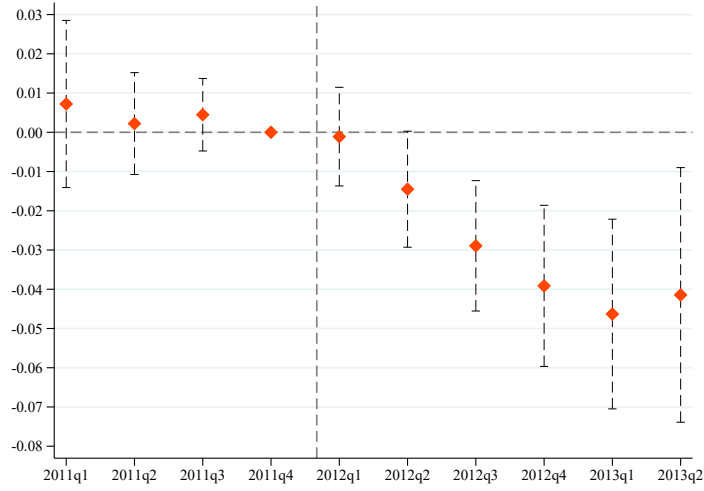


Figure 2: Impact on Lending to Importing Firms at the Intensive Margin

This figure plots the event study associated with specification (5), where we replace *Policy* with quarter dummies. The dependent variable is the logarithm of credit. We drop firms in construction or real estate sectors. Each dot represents the estimated coefficient of the interaction of bank exposure and a quarter dummy. We normalize the coefficient of 2011Q4 to zero. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors double clustered at the bank and firm levels.

Table 3: Impact of Provisioning Shift on Total Bank Debt to Importing Firms,
Firm Level Analysis

	(1)	(2)
<i>Exposure</i> $\times$ <i>Policy</i>	-0.019*** (0.006)	-0.028*** (0.007)
Controls $\times$ time	No	Yes
Firm FE	Yes	Yes
Time FE	Yes	No
$\hat{\alpha}_{f,t}$	No	Yes
Obs.	228,786	228,786
R^2	0.98	0.98

This table presents firm-level regression results for firms' total bank debt, estimated using equation (6). The dependent variable is the logarithm of firm f 's total bank debt. *Exposure* is calculated as the weighted average of the exposure of banks lending to firm f , using outstanding credit at the end of 2011 as weights. To account for credit demand, regressions in columns (2) and (4) include estimated firm-quarter fixed effects obtained from specification (5), while regressions in columns (1) and (3) include quarter fixed effects instead. *Controls* are the weighted average of bank-firm characteristics (collateralization rate, ratio of long-term lending, share of firm f 's NPLs with the bank) and bank characteristics (quartiles of size and exposure to exporter/importer companies), measured as of the end of 2011 and weighted by banks' lending to firm f as of 2011. Additionally, we include industry and province indicators as control variables. All variables are defined in Table A.1. Firms in the construction or real estate sectors are excluded. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

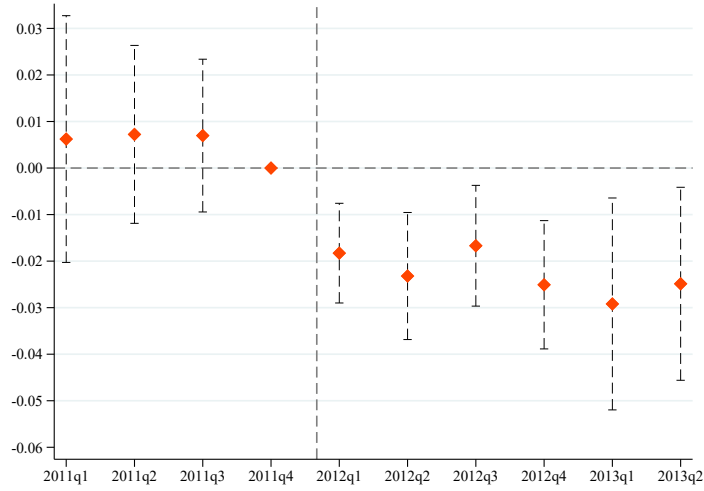


Figure 3: Impact on Total Bank Debt to Importing Firms

This figure plots the event study associated with specification (6), where *Policy* is replaced with quarter dummies. The dependent variable is the logarithm of firms' total bank credit. To account for credit demand, all regressions include the estimated firm-quarter fixed effects obtained from our bank-firm specification in (5). Firms in the construction or real estate sectors are excluded. Each dot represents the estimated coefficient of the interaction between a quarter dummy and the firm-level exposure, which is computed as the weighted average of the exposure of banks lending to firm f , using the outstanding credit at the end of 2011 as weights. The coefficient for 2011Q4 is normalized to zero. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors double-clustered at the main bank level and firm level.

Table 4: Effect of Shift in Provisions on annual Firm-level Outcomes

	Bank Debt	Total Debt	Total Debt +Equity	Employment	Tangible Assets
	(1)	(2)	(3)	(4)	(5)
<i>Exposure</i>	-0.036*** (0.013)	-0.023* (0.013)	-0.015* (0.009)	-0.008* (0.004)	-0.014** (0.005)
Firm controls	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Main Buyer FE	Yes	Yes	Yes	Yes	Yes
Obs.	17,269	17,269	17,269	17,269	17,269
R^2	0.02	0.02	0.02	0.02	0.02

This table reports regression results on firm-level outcomes using annual financial statements. The dependent variables are the mid-point growth rates between 2011 and 2012 of total bank debt, total debt, total debt plus equity, employment, and tangible assets. The main explanatory variable, *Exposure*, is computed as the weighted average exposure of banks lending to firm f , using outstanding credit at the end of 2011 as weights. Controls (measured in 2011) include weighted averages of bank characteristics—total assets and the share of lending to importing and exporting firms—as well as firm characteristics (log total assets, cash-to-assets, sales-to-assets, and own funds-to-assets ratios). Regressions include the fixed effects listed in the table, accounting for industry, province, and the firm’s main export destination. All variables are defined in [Table A.1](#). Standard errors are double-clustered at the main bank and firm levels. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5: Impact of Provisioning Shift on Firms' Import Flows

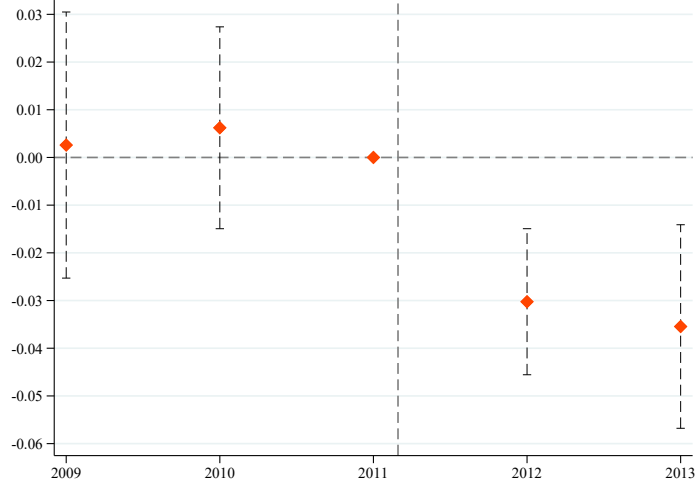
	Mid-point growth	Entry Dummy	Exit Dummy
	(1)	(2)	(3)
$Exposure \times Policy$	-0.036*** (0.011)	-0.007** (0.003)	0.014*** (0.004)
Product-Country-Year FE	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes
Product-Country-Firm FE	Yes	No	No
Firm FE	No	Yes	Yes
Obs.	558,540	214,098	221,390
R^2	0.17	0.18	0.23

This table presents regression estimates for firms' import flows at the product-country-year level. In column 1, we compute the mid-point growth rate between firm f 's imports of product g from country c and the 5-year average of imports of the same product from that country. In column 2, we use a binary indicator that, conditional on not participating in the market in the previous year, takes the value of one if the firm enters the market. In column 3, we use a binary indicator that takes the value of one if the firm exits the market, conditional on participation in the same market in the previous year. *Exposure* is computed as the average exposure of banks lending to firm f , weighted by outstanding credit as of the end of 2011. *Policy* is a dummy that takes the value of one after the provisioning increase. Fixed effects are reported in the lower part of the table. All variables are defined in [Table A.1](#). Firms in construction or real estate sectors are excluded. Standard errors are double clustered at the main bank and firm levels and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

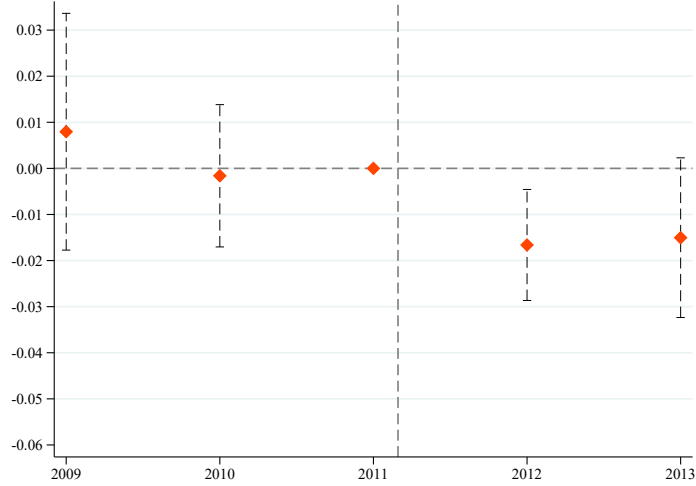
Table 6: Impact of Provisioning Shift on Firms' Import Flows

	Mid-point growth	Entry Dummy	Exit Dummy
	(1)	(2)	(3)
$Exposure \times Policy$	-0.036*** (0.011)	-0.007** (0.003)	0.014*** (0.004)
Product-Country-Year FE	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes
Product-Country-Firm FE	Yes	No	No
Firm FE	No	Yes	Yes
Obs.	558,540	214,098	221,390
R^2	0.17	0.18	0.23

This table presents regression estimates for firms' import flows at the product-country-year level. In column 1, we compute the mid-point growth rate between firm f 's imports of product g from country c and the 5-year average of imports of the same product from that country. In column 2, we use a binary indicator that, conditional on not participating in the market in the previous year, takes the value of one if the firm enters the market. In column 3, we use a binary indicator that takes the value of one if the firm exits the market, conditional on participation in the same market in the previous year. *Exposure* is computed as the average exposure of banks lending to firm f , weighted by outstanding credit as of the end of 2011. *Policy* is a dummy that takes the value of one after the provisioning increase. Fixed effects are reported in the lower part of the table. All variables are defined in [Table A.1](#). Firms in construction or real estate sectors are excluded. Standard errors are double clustered at the main bank and firm levels and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.



(a) Imports



(b) Exports

Figure 4: Impact of Provisioning Shift on Imports and Exports at the Intensive Margin

This figure plots coefficient estimates from a modified version of the specification in (7). For each year, the coefficient corresponds to the interaction of *Exposure* with a year dummy. Each coefficient measures the impact of a one standard deviation increase in firm f 's exposure to the policy on its imports (or exports) growth of good g from (or to) country c , relative to the year before the policy implementation (2011). Panel (a) presents the estimates for importers, whereas Panel (b) presents the estimates for exporters. Importers and exporters in the construction or real estate sectors are excluded. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors double clustered at the main bank and firm levels.

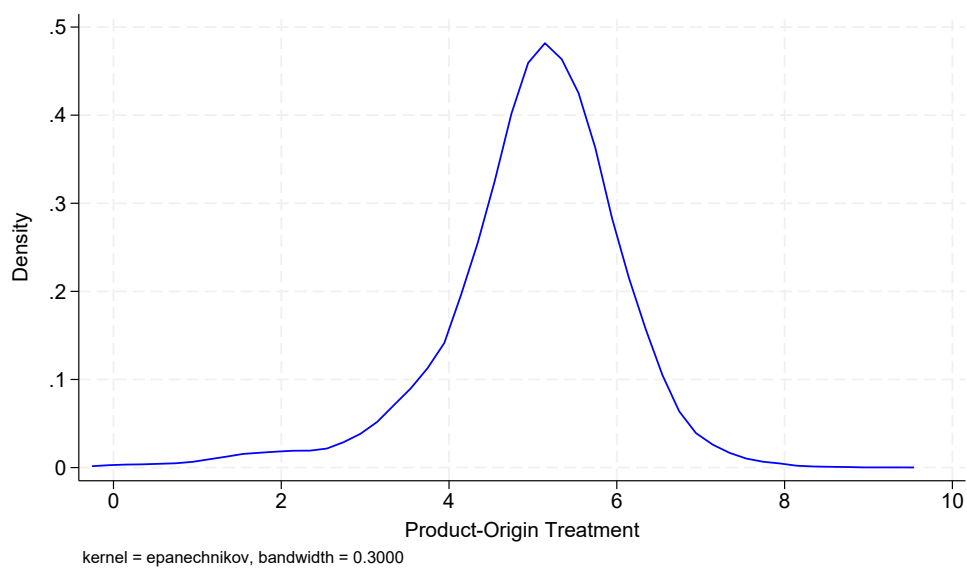


Figure 5: Product-origin Exposure: Kernel Density Estimate

This figure plots the kernel density of the standardized product-origin treatment, as defined in [Equation 8](#).

Table 7: Impact of Provisioning Shift on Exports to Spain Relative to Other Destinations, Triple Difference-in-Differences Specification

	Mid-point growth				
	All destinations			Comparable destinations	All dest. Excl. GIIP exporters
	(1)	(2)	(3)	(4)	(5)
$Exposure_{go} \times \mathbb{1}[d = \text{Spain}] \times Post_t$	-0.032** (0.013)	-0.027** (0.013)	-0.021* (0.012)	-0.022* (0.013)	-0.023* (0.013)
Observations	1,600,910	1,600,910	1,600,910	144,500	1,394,170
Fixed Effects					
Exporter $\times$ Importer $\times$ Product	Yes	Yes	Yes	Yes	Yes
Importer $\times$ Year	Yes	Yes	Yes	Yes	Yes
Product $\times$ Year	No	Yes	Yes	Yes	Yes
Exporter $\times$ Year	No	No	Yes	No	Yes

This table presents the regression results for the mid-point growth rate of exports of product g from country o to country d in year t relative to its five-year average. The analysis is based on a product-origin level measure of exposure, interacted with a dummy variable that equals one when Spain is the importing country, and incorporates different sets of fixed effects, as specified in equation (9). Product-origin exposure is calculated as the weighted average of the exposures of Spanish banks lending to firms importing product g from country o , following equation (8). The fixed effects included in each regression are listed in the lower part of the table. Columns (1) to (3) include all importers, while column (4) focuses on a set of comparable destinations, as those in Autor et al. (2013). Column (5) includes all importers and excludes GIIP exporters. All variables and comparable countries are defined in Table A.1. Standard errors shown in parentheses are clustered by product and country of origin.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

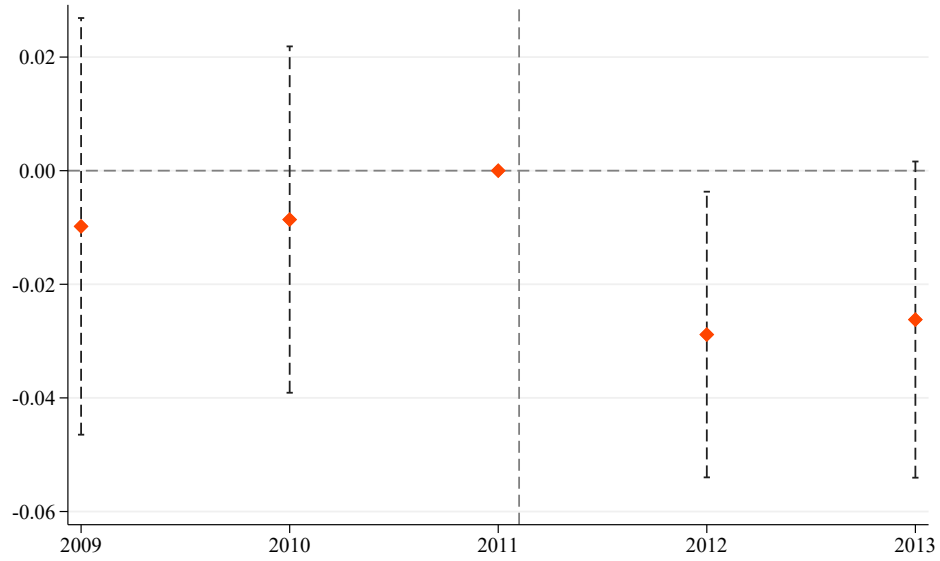


Figure 6: Event Study for the Average Effect of Product Exposure to Provisioning Shift on Exports to Spain Relative to Other Destinations

This figure plots coefficient estimates from a modified version of the specification in equation (9), which coincides with our benchmark specification in column (3) of Table 7. For each year, the coefficient corresponds to the interaction of the product-level exposure, the year dummy, and an indicator equal to one when Spain is the importing country. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors clustered by product and country of origin.

Table 8: Impact of Provisioning Shift on Exports to Spain and Other Destinations

	Mid-point growth							
	Exports to Spain				Other Destinations			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Exposure_{go} \times Post_t$	-0.034** (0.013)	-0.035*** (0.013)	-0.027** (0.014)	-0.028* (0.015)	-0.003 (0.007)	-0.004 (0.007)	-0.010 (0.007)	-0.010 (0.008)
Observations	11,363	11,363	11,363	9,948	11,363	11,363	11,363	9,948
Fixed Effects								
Exporter $\times$ Product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Spanish share decile $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product charact. $\times$ Year	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Country charact. $\times$ Year	No	No	Yes	Yes	No	No	Yes	Yes

This table presents regression results for the mid-point growth rate of exports of product g from country o relative to its five-year average: to Spain (columns 1-3) and to destinations other than Spain (columns 5-7). Columns (4) and (8) report results excluding GIIP exporters. The analysis is based on a product-origin level measure of exposure, and incorporates different sets of fixed effects, as specified in equation (10). Product-origin exposure is calculated as the weighted average of the exposures of Spanish banks lending to firms importing product g from country o , following equation (8). The fixed effects included in each regression are listed in the lower part of the table. Product characteristics include a dummy for high or low heterogeneity: i.e, low, if price dispersion of the product is at the bottom quartile of the price dispersion distribution. Exporter country characteristics include: distance to Spain indicator (close to Spain if it belongs to the bottom quartile of the distance-to-Spain distribution), official language (Spanish or not). Spanish share corresponds to the participation of Spain in total exports of a specific origin $\times$ product pair. All variables are defined in Table A.1. Standard errors shown in parentheses are clustered by product and country of origin.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

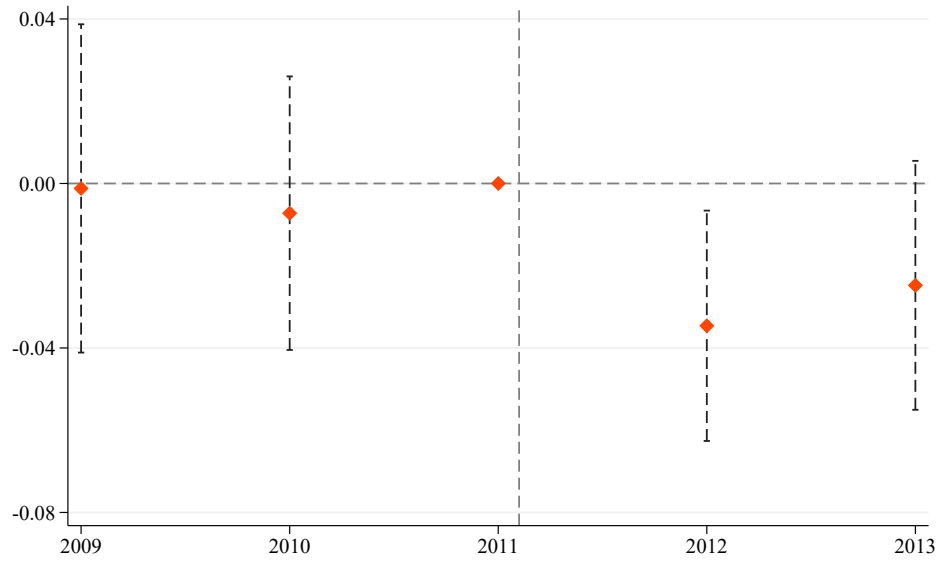


Figure 7: Event Study for the Average Effect of Product Exposure to Provisioning Shift on Exports to Spain

This figure plots coefficient estimates from a modified version of the specification in equation (10), which coincides with our benchmark specification in column (3) of Table 8. For each year, the coefficient corresponds to the interaction of the product-level exposure and the year dummy. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors clustered by product and country of origin.

Table 9: Impact of Provisioning Shift on Exports to Spain, by Origin-Country Characteristics

	Mid-point growth Exports to Spain							
	High FD	Low Financial Development						
		All	Distance		Language		Far & Not Spanish	
			Close	Far	Spanish	Other	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Exposure_{go} \times Post_t$	0.005 (0.024)	-0.033** (0.016)	-0.022 (0.034)	-0.036** (0.018)	-0.026 (0.055)	-0.035** (0.017)	-0.023 (0.031)	-0.039** (0.019)
Observations	2,703	8,271	1,620	6,651	1,040	7,231	2,660	5,611
Fixed Effects								
Exporter $\times$ Product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product charact. $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country charact. $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	—	—
Spanish share decile $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table presents the regression results for the mid-point growth rate of exports of product g from country o to Spain, relative to its five-year average. Column (1) reports estimates for exports to Spain from countries with high financial development, defined as those in the top quartile of the private credit-to-GDP ratio. Columns (2)-(8) report estimates for countries with low financial development, those at the bottom three quartiles. Column (2) includes all origins. Columns (3) and (4) correspond to exporters that are geographically close to Spain (bottom quartile of the distance-to-Spain distribution), or far from Spain, respectively. Columns (5) and (6) report estimates for exporters from Spanish-speaking and non-Spanish-speaking countries, respectively. Column (7) includes exporters that are either geographically close to Spain or Spanish-speaking, while Column (8) covers exporters that are both geographically distant and non-Spanish-speaking. The analysis is based on a product-origin level measure of exposure, and incorporates different sets of fixed effects, as specified in equation (10). Product-origin exposure is calculated as the weighted average of the exposures of Spanish banks lending to firms importing product g from country o , following equation (8). The fixed effects included in each regression are listed in the lower part of the table. Product characteristics include a dummy for high or low heterogeneity: i.e, low, if price dispersion of the product is at the bottom quartile of the price dispersion distribution. Exporter country characteristics include: distance to Spain indicator (close to Spain or not), official language (Spanish or non Spanish). Spanish share corresponds to the participation of Spain in total exports of a specific origin $\times$ product pair. All variables are defined in Table A.1. Standard errors shown in parentheses are clustered by product and country of origin.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10: Impact of Provisioning Shift on Exports to Destinations other than Spain, by Origin-Country Characteristics

	Mid-point growth Exports to Destinations other than Spain							
	High FD	Low Financial Development						
		All	Distance		Language		Far & Not Spanish	
			Close	Far	Spanish	Other	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Exposure_{go} \times Post_t$	0.001 (0.013)	-0.008 (0.008)	-0.015 (0.020)	-0.008 (0.009)	-0.034 (0.026)	-0.005 (0.008)	-0.022 (0.015)	-0.004 (0.009)
Observations	2,703	8,271	1,620	6,651	1,040	7,231	2,660	5,611
Fixed Effects								
Exporter $\times$ Product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product charact. $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country charact. $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	—	—
Spanish share decile $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table presents the regression results for the mid-point growth rate of exports of product g from country o to destinations other than Spain, relative to its five-year average. Column (1) reports estimates for exports to Spain from countries with high financial development, defined as those in the top quartile of the private credit-to-GDP ratio. Columns (2)-(8) report estimates for countries with low financial development, those at the bottom three quartiles. Column (2) includes all origins. Columns (3) and (4) correspond to exporters that are geographically close to Spain (bottom quartile of the distance-to-Spain distribution), or far from Spain, respectively. Columns (5) and (6) report estimates for exporters from Spanish-speaking and non-Spanish-speaking countries, respectively. Column (7) includes exporters that are either geographically close to Spain or Spanish-speaking, while Column (8) covers exporters that are both geographically distant and non-Spanish-speaking. The analysis is based on a product-origin level measure of exposure, and incorporates different sets of fixed effects, as specified in equation (10). Product-origin exposure is calculated as the weighted average of the exposures of Spanish banks lending to firms importing product g from country o , following equation (8). The fixed effects included in each regression are listed in the lower part of the table. Product characteristics include a dummy for high or low heterogeneity: i.e, low, if price dispersion of the product is at the bottom quartile of the price dispersion distribution. Exporter country characteristics include: distance to Spain indicator (close to Spain or not), official language (Spanish or non Spanish). Spanish share corresponds to the participation of Spain in total exports of a specific origin $\times$ product pair. All variables are defined in Table A.1. Standard errors shown in parentheses are clustered by product and country of origin.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11: Impact of Provisioning Shift on the Total Exports of Spain's Trade Partners, Conditional on Spain's share in their total sales

	Mid-point growth rate of Exports								
	All Product $\times$ Origin Pairs			Low Spanish Share			High Spanish Share		
	All	Spain	Rest	All	Spain	Rest	All	Spain	Rest
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Agg. Exposure_{go} $\times$ Post_t</i>	-0.016 (0.011)	-0.034** (0.015)	0.007 (0.012)	-0.041 (0.075)	-0.082 (0.119)	-0.034 (0.076)	-0.028* (0.015)	-0.045** (0.018)	-0.002 (0.015)
Observations	11,363	11,363	11,363	8,525	8,525	8,525	2,838	2,838	2,838
Fixed Effects									
Exporter $\times$ Product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product charact. $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country charact. $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table reports regression results for the mid-point growth rate of exports of product g from country o , relative to its five-year average. Columns (1), (4), and (7) present results for total exports to all destinations; columns (2), (5), and (8) for exports to Spain; and columns (3), (6), and (9) for exports to destinations other than Spain. Columns (1)-(3) include all product-origin pairs. Columns (4)-(6) restrict the sample to product-origin pairs with a low Spanish share, (i.e. Spanish purchases account for less than 6 percent of pair's total exports). Columns (7)-(9) restrict the sample to product-origin pairs with a high Spanish share. The analysis is based on an aggregate measure of exposure, and incorporates different sets of fixed effects, as specified in equation (13). Aggregate exposure is calculated as the product-origin level exposure defined in equation (8) multiplied by Spain's share of country o 's exports of product g , following equation (12). The fixed effects included in each regression are listed in the lower part of the table. Product characteristics include a dummy for high or low heterogeneity: i.e, low, if price dispersion of the product is at the bottom quartile of the price dispersion distribution. Exporter country characteristics include: distance to Spain indicator (close to Spain if it belongs to the bottom quartile of the distance-to-Spain distribution), official language (Spanish or non Spanish). All variables are defined in Table A.1. Standard errors shown in parentheses are clustered by product and country of origin.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

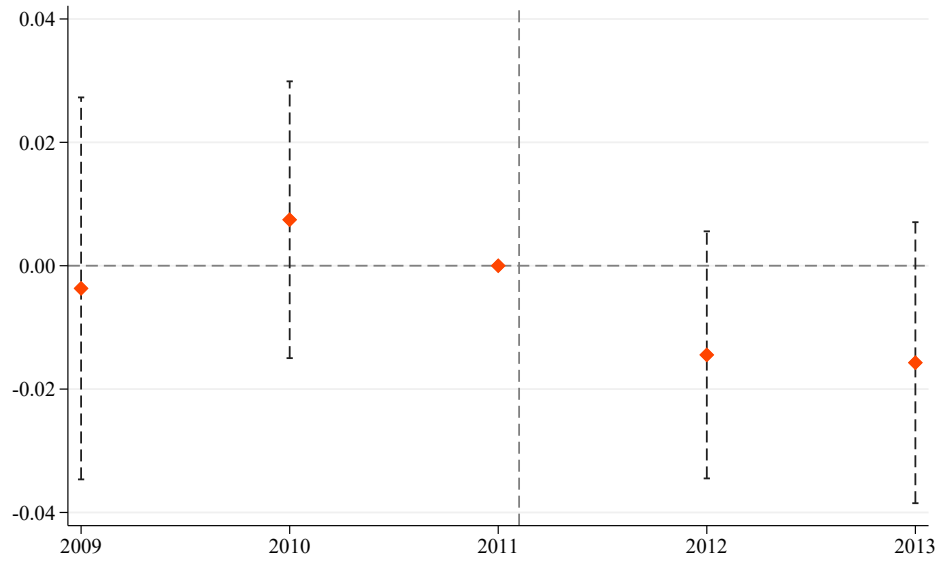


Figure 8: Event Study for the Average Effect of Aggregate Exposure to Provisioning Shift on Total Exports of Spain's Trade Partners, conditional on high Spanish share

This figure plots coefficient estimates from a modified version of the specification in equation (13), which coincides with our benchmark specification in column (7) of Table 11. For each year, the coefficient corresponds to the interaction of the aggregate exposure and the year dummy. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors clustered by product and country of origin.

Table 12: Impact of Provisioning Shift on the Total Exports of Spain's Trade Partners and Product Differentiation: Low Financial Development and High Trade Costs Origins

	Mid-point growth rate of Exports					
	High Price Dispersion			Low Price Dispersion		
	All	Spain	Rest	All	Spain	Rest
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Agg. Exposure_{go} × Post_t</i>	-0.088** (0.037)	-0.160*** (0.049)	-0.026 (0.025)	0.057 (0.052)	-0.291*** (0.098)	0.121** (0.059)
Observations	4,172	4,172	4,172	1,439	1,439	1,439
Fixed Effects						
Exporter × Product	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes

This table reports regression results for the mid-point growth rate of exports of product g from country o , relative to its five-year average. Columns (1) and (4) present results for total exports to all destinations; columns (2) and (5) for exports to Spain; and columns (3) and (6) for exports to destinations other than Spain. Columns (1)-(3) restrict the sample to products with high price dispersion, defined as those in the top three quartiles of the distribution. Columns (4)-(6) restrict the sample to products with low price dispersion (bottom quartile). The analysis is based on an aggregate measure of exposure, and incorporates different sets of fixed effects, as specified in equation (13). Aggregate exposure is calculated as the product-origin level exposure defined in equation (8) multiplied by Spain's share of country o 's exports of product g , following equation (12). The fixed effects included in each regression are listed in the lower part of the table. All variables are defined in Table A.1. Standard errors shown in parentheses are clustered by product and country of origin.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A Additional Tables and Figures

Table A.1: Variable Definitions

Variable	Definition	Source
Firm–bank variables		
$\log(\textit{Credit})$	Logarithm of committed credit granted by bank b to firm f in quarter t .	CIRBE
<i>Termination</i>	Indicator equal to one if the bank–firm relationship is terminated after the policy, conditional on the relationship being active at end-2011.	CIRBE
<i>Long-term credit ratio</i>	Share of loans with residual maturity above one year in total credit granted by bank b to firm f , measured at end-2011.	CIRBE
<i>Collateralization rate</i>	Share of collateralized loans in total credit granted by bank b to firm f , measured at end-2011.	CIRBE
<i>NPL ratio</i>	Ratio of non-performing loans to total credit granted by bank b to firm f , measured at end-2011.	CIRBE
<i>Credit Share</i>	Share of firm f 's total bank debt accounted for by loans from bank b , measured at end-2011.	CIRBE
Bank variables		
<i>Exposure</i>	Ratio of lending to the construction and real estate sectors to total lending to non-financial firms, measured at end-2011.	Supervisory Reports
<i>Size</i>	Logarithm of total assets, measured at end-2011.	Supervisory Reports
<i>Capital</i>	Equity-to-assets ratio, measured at end-2011.	Supervisory Reports
<i>Liquidity</i>	Ratio of liquid assets to total assets, where liquid assets include cash, balances with central banks, and loans and advances to governments and credit institutions, measured at end-2011.	Supervisory Reports
<i>ROA</i>	Return on assets, defined as net income over total assets, measured at end-2011.	Supervisory Reports
<i>NPL ratio</i>	Ratio of non-performing loans to total credit, measured at end-2011.	Supervisory Reports
<i>Rural bank dummy</i>	Indicator equal to one if the bank is a rural savings bank.	Supervisory Reports
<i>% Lending to X or M</i>	Share of lending to exporting and importing firms in total lending to the corporate sector, measured at end-2011.	CIRBE
<i>Local Govt. Credit to Assets</i>	Ratio of credit to local governments to total assets, measured at end-2011.	Credit Register and CIRBE
Firm-level variables		
$\log(\textit{Total Bank Credit})$	Logarithm of total bank credit received by firm f in quarter t .	CIRBE
$\textit{Exposure}_f$	Weighted average of the exposure of the banks lending to firm f , where weights are based on outstanding credit at end-2011.	CIRBE
Trade variables at the firm–market level		
<i>Exports</i>	Exports by firm f of HS2 product g to country c in year t .	Balance of Payments data
<i>Imports</i>	Imports by firm f of HS2 product g from country c in year t .	Balance of Payments data
<i>Entry</i>	Indicator equal to one if exports or imports by firm f in market (g, c) are positive in year t , conditional on being zero in the previous year.	Balance of Payments data

Continued on next page

Table A.1 continued

Variable	Definition	Source
<i>Exit</i>	Indicator equal to one if exports or imports by firm f in market (g, c) are zero in year t , conditional on being positive in the previous year.	Balance of Payments data
<i>No. Countries</i>	Number of countries to which firm f exports, or from which it imports, in year t .	Balance of Payments data
<i>No. Products</i>	Number of products exported or imported by firm f in year t .	Balance of Payments data
Firm balance-sheet variables		
<i>Total Debt</i>	Total debt of firm f in year t , including both bank and non-bank debt.	Mercantile Register
<i>Employment</i>	Average number of employees of firm f in year t .	Mercantile Register
<i>Tangible Assets</i>	Tangible assets of firm f in year t .	Mercantile Register
Bilateral trade-flow variables		
M_{cgot}	Imports of country c from origin country o of product g in year t .	BACI
$Exposure_{go}$	Weighted average of bank exposure to the policy, where weights reflect the relative importance of each bank as a credit supplier to firms importing product g from origin country o .	Balance of Payments data and CIRBE
<i>Agg. Exposure_{go}</i>	Aggregate exposure measure defined as $Exposure_{go}$ interacted with Spain's share of purchases in market (g, o) .	Balance of Payments data, CIRBE, and BACI
X_{got}	Exports of product g by origin country o in year t .	BACI
<i>Comparable countries</i>	Australia, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Japan, Luxembourg, the Netherlands, New Zealand, Portugal, Spain, Switzerland, and the United States.	Authors' classification
Origin-country characteristics		
<i>Close to Spain</i>	Indicator equal to one if the distance between the origin country and Spain is below the 25th percentile of the distance-to-Spain distribution.	CEPII Gravity
<i>Spanish Speaking</i>	Indicator equal to one if the origin country is Spanish-speaking.	CEPII Gravity
<i>FTA</i>	Indicator equal to one if the origin country has a trade agreement with Spain.	CEPII Gravity
<i>Spanish share</i>	Indicator equal to one if Spain's share of world purchases of product g is above the median.	BACI
<i>Price dispersion</i>	Product-level price dispersion, measured as the variance of implicit prices at the HS2 level, using HS6 product-level information across countries.	BACI
<i>Heterogeneous products</i>	Indicator equal to one if product-level price dispersion is above the 25th percentile of its distribution.	BACI
<i>Financial Development</i>	Indicator equal to one if private credit to GDP in the origin country is above the 75th percentile of the corresponding distribution.	Global Financial Development Database
<i>Spanish Banks</i>	Indicator equal to one if Spanish banks operate in the origin country.	BankFocus

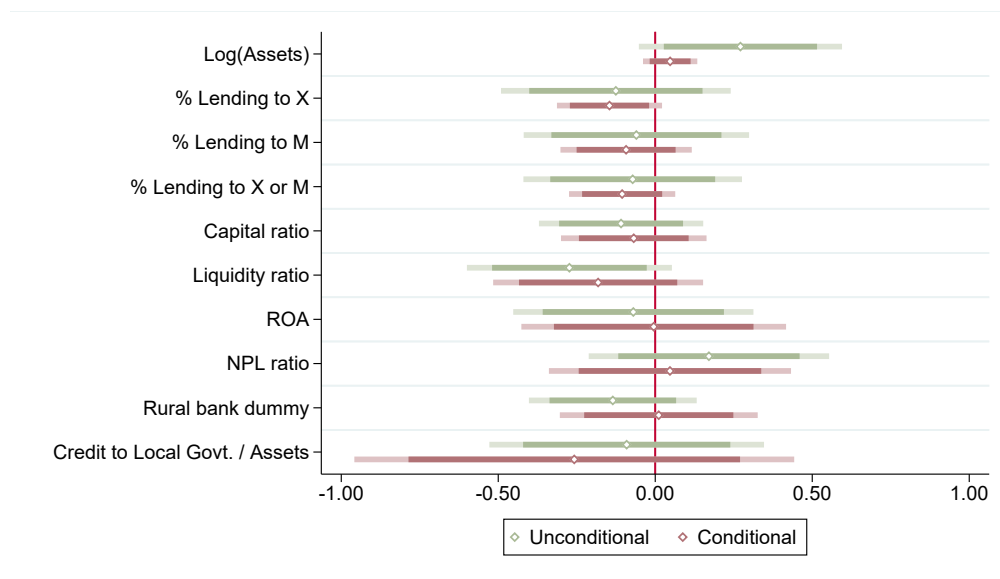


Figure A.1: Bank Covariate Balance

This figure presents coefficient estimates, together with 95% confidence intervals (dark bars) and 99% confidence intervals (light bars), for the effect of a one-standard-deviation increase in bank exposure on different variables. All variables are standardized to have a mean of zero and a standard deviation of one. “Unconditional” estimates compare banks with different levels of exposure without controlling for fixed effects. “Conditional” estimates compare banks within the same size quartile (measured by total assets) and the same trade-exposure quartile (measured by the share of loans to exporters and importers, excluding loans to the construction and real estate sectors). *Rural* is an indicator equal to one for rural savings banks, and *NPL* denotes non-performing loans, defined as outstanding debt with payments overdue by more than 90 days.

Table A.2: Effect of Shift in Provisions on Lending to Importing and Exporting Firms, Bank-Firm Level Analysis Accounting for Bank Specialization

Dep. Variable: Log(Credit)	Importers		Exporters	
	(1)	(2)	(3)	(4)
<i>Exposure</i> $\times$ <i>Policy</i>	-0.031** (0.013)	-0.031** (0.012)	-0.031** (0.012)	-0.032*** (0.011)
$Q_2 \times \text{Policy}$		0.014 (0.011)		0.010 (0.010)
$Q_3 \times \text{Policy}$		-0.002 (0.008)		-0.002 (0.007)
$Q_4 \times \text{Policy}$		-0.022** (0.008)		-0.017** (0.008)
Controls $\times$ time	Yes	Yes	Yes	Yes
Firm-time FE	Yes	Yes	Yes	Yes
Bank-firm FE	Yes	Yes	Yes	Yes
Obs.	962,963	962,963	973,386	973,386
R^2	0.97	0.97	0.97	0.97

This table reports bank–firm-level regressions of log credit based on equation (5). Columns (1)–(2) correspond to importers and columns (3)–(4) to exporters. *Exposure* is defined as bank b ’s pre-reform share of lending to construction and real estate (end-2011). In columns (2) and (4), we augment the specification with interactions between *Policy* and indicators for bank specialization in countries from (or to) which the firm imports (or exports), following Paravisini et al. (2023). *Controls* (measured at end-2011) include bank–firm characteristics (collateralization, share of long-term lending, firm f ’s NPL share at bank b , and bank b ’s share of the firm’s total borrowing) and bank characteristics (size and trade-exposure quartiles). All variables are defined in Table A.1. Firms in construction and real estate are excluded. Standard errors are double-clustered at the bank and firm levels. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Bank Specialization: To ensure that our results reflect the domestic regulatory shock rather than shocks to firms’ trade partners, we test whether bank specialization in specific foreign markets drives our findings. Following Paravisini et al. (2023), we augment specification (5) by interacting the *Policy* dummy with indicators for whether a bank specializes in countries from (or to) which the firm imports (or exports). We measure bank b ’s specialization in country c —for the 50 main export destinations—as: $S_b^c = \frac{\sum_f C_{f,b} \times X_{f,c}}{\sum_j \sum_f C_{f,b} \times X_{f,j}}$, where $C_{f,b}$ is firm f ’s outstanding credit with bank b in 2011, and $X_{f,c}$ denotes firm f ’s exports to country c over 2009–2013. The denominator aggregates this term across all destinations. This measure captures the share of a bank’s lending portfolio associated with firms exporting to country c . Table A.2 shows that the coefficient on *Exposure* $\times$ *Policy* remains stable and statistically significant. Thus, our baseline results are not driven by bank specialization in particular foreign markets, but by the domestic credit supply shock induced by the provisioning reform.

Table A.3: Effect of Shift in Provisions on Lending to Tradable and Non-tradable firms,
Bank-Firm Level Analysis

Dep. Variable: Log(Credit)	<i>All Firms</i> (1)	<i>M + X</i> (2)	<i>All Firms</i> (3)
<i>Exposure</i> $\times$ <i>Policy</i>	-0.019** (0.009)	-0.029** (0.012)	-0.015* (0.009)
<i>Exposure</i> $\times$ <i>Policy</i> $\times$ <i>Tradable</i>			-0.011 (0.008)
Controls $\times$ time	Yes	Yes	Yes
Firm-time FE	Yes	Yes	Yes
Bank-firm FE	Yes	Yes	Yes
Obs.	5,513,983	1,288,007	5,513,983
R^2	0.98	0.97	0.98

This table reports bank–firm-level regressions of log credit at the intensive margin based on equation (5). Columns (1) and (3) include all firms (tradable and non-tradable), while column (2) restricts the sample to firms engaged in international trade. *Exposure* is defined as bank b ’s pre-reform share of lending to construction and real estate (end-2011). *Controls* (measured at end-2011) include bank–firm characteristics (collateralization, long-term lending share, firm f ’s NPL share at bank b , and bank b ’s share of the firm’s total borrowing) and bank characteristics (size and trade-exposure quartiles). All variables are defined in Table A.1. Firms in construction and real estate are excluded. Standard errors are double-clustered at the bank and firm levels. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

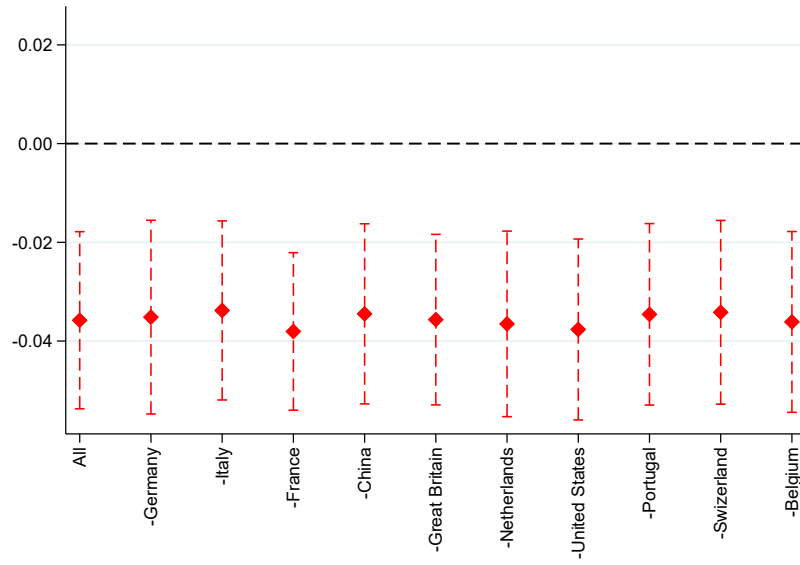
Tradable vs. Non-Tradable firms: As an additional robustness check, we examine whether our results are driven by foreign demand shocks disproportionately affecting tradable firms. To this end, we augment specification (5) by interacting the *Exposure* $\times$ *Policy* term with an indicator for firms engaged in international trade. Table A.3 reports the results. While the baseline effect of *Exposure* $\times$ *Policy* remains negative and statistically significant, the interaction with the tradable indicator is small and statistically insignificant. This indicates that the credit contraction does not differ significantly between tradable and non-tradable firms, suggesting that our findings are unlikely to be driven by shocks to firms’ trade partners and instead reflect the domestic regulatory intervention.

Table A.4: Bank-Firm Relationship Characteristics and Trade Flows

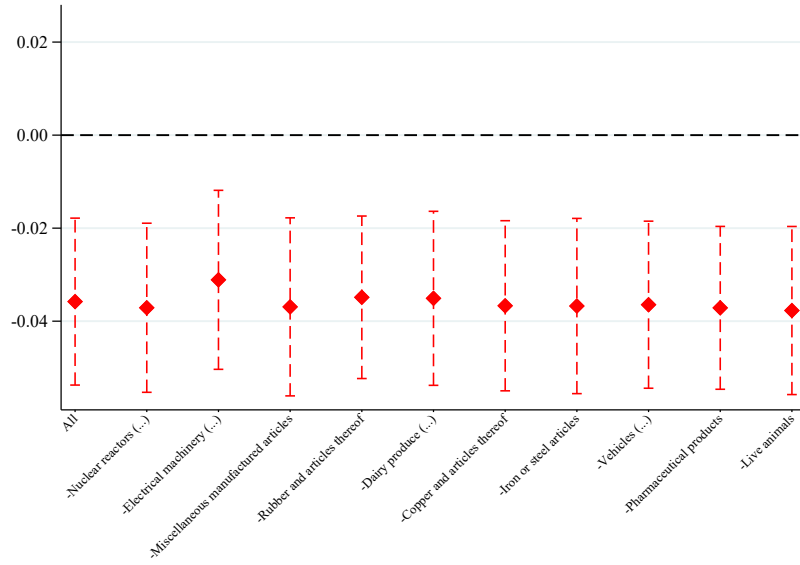
	(1)	(2)	(3)	(4)	(5)
Exposure×Policy	-0.036*** (0.011)	-0.034*** (0.011)	-0.029** (0.011)	-0.063*** (0.010)	-0.050*** (0.010)
× Available funds in CLs		0.010* (0.005)			0.012** (0.005)
× NPL			-0.134*** (0.049)		-0.111** (0.047)
× Share Main Bank				0.036*** (0.007)	0.028*** (0.007)
Product-Country-Firm FE	Yes	Yes	Yes	Yes	Yes
Product-Country-Year FE	Yes	Yes	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes	Yes	Yes
Observations	558,540	558,540	558,540	558,540	558,540
R^2	0.17	0.17	0.17	0.17	0.17

This table presents the heterogeneous effects of the provisioning shift on trade outcomes based on lending relationship characteristics. The dependent variable is the mid-point growth rate of firm f 's exports of product g relative to the 5-year average of firm f 's exports of the same product. We regress this variable on the firm's exposure and a set of fixed effects. *Exposure* is computed as the weighted average of the exposure of banks lending to firm f , using outstanding credit as of the end of 2011 as weights. *Policy* takes a value of one for 2012 and 2013 and zero otherwise. *Available Funds in CLs* represents the share of undrawn to total committed funds available in firm f 's credit lines as of the end of 2011. *Share Main Bank* is the share of firm f 's outstanding credit held with its main bank as of the end of 2011. *NPL* takes a value of one if the firm had at least one non-performing loan as of the end of 2011 and zero otherwise. To facilitate the interpretation of the coefficients, *Available Funds in CLs* and *Share Main Bank* are standardized. The fixed effects included in each regression are specified in the lower part of the table. All variables are defined in Table A.1. Firms in the construction or real estate sectors are excluded. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.



(a) Origin Countries



(b) HS2 Products

Figure A.2: Sensitivity Analysis to Main Origin Countries and Import Products

The figure presents a sensitivity analysis of our baseline results in specification (7). In Panel (a), we present the sensitivity analysis for imports after sequentially excluding each of the top 10 main origin countries for Spanish imports. In Panel (b), we present the sensitivity analysis for imports after sequentially excluding each of the top 10 main imported products for Spanish firms. As a benchmark, the first point in each figure (*All*) coincides with the coefficient in columns 1 and 2 of Table 4, respectively. We drop firms in the construction or real estate sectors. Bands represent the 90% confidence intervals. Standard errors are double-clustered at the main bank and firm levels.

Table A.5: Impact of Provisioning Shift on Firms' Import Flows, Excluding firms highly related to GIIPS

	Mid-point growth	Entry Dummy	Exit Dummy
	(1)	(2)	(3)
<i>Exposure</i> $\times$ <i>Policy</i>	-0.039*** (0.010)	-0.006 (0.004)	0.014*** (0.003)
Product-Country-Year FE	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes
Product-Country-Firm FE	Yes	No	No
Firm FE	No	Yes	Yes
Obs.	175,864	177,588	455,025
R^2	0.19	0.25	0.17

This table presents regression estimates for firms' import flows, replicating the specification from [Table 6](#) but excluding firms highly exposed to GIIP countries. Exposure to GIIP is defined using 2006 data, prior to the global financial crisis, as the share of each firm's imports originating from Greece, Italy, Portugal, and Ireland. Firms in the top quartile of this distribution are excluded. Standard errors are double clustered at the main bank and firm levels and reported in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A.6: Impact of Provisioning Shift on Firms' Import Flows: Excluding Construction-Linked Sectors (IO-Based Robustness)

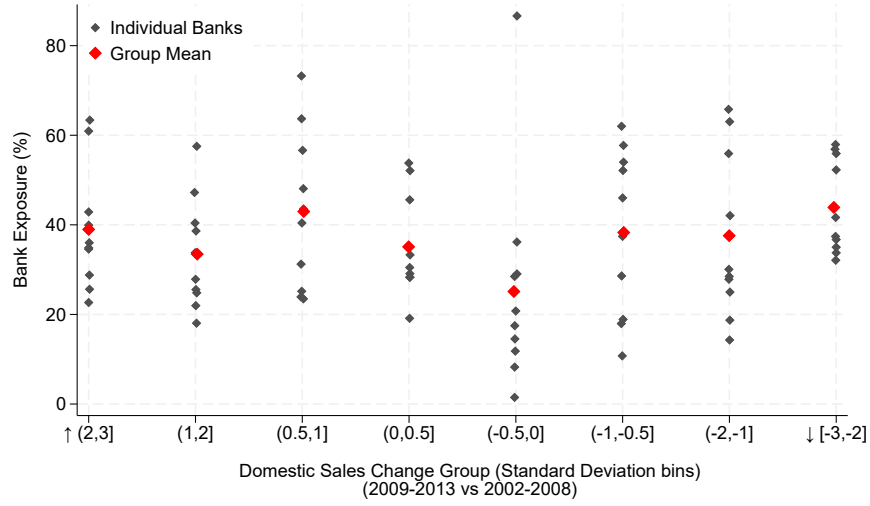
	Mid-point growth	Entry Dummy	Exit Dummy
	(1)	(2)	(3)
Panel A: Excluding upstream dependence (Leontief inverse), 90th percentile			
<i>Exposure</i> $\times$ <i>Policy</i>	-0.033*** (0.012)	-0.003 (0.005)	0.017*** (0.005)
Observations	323,490	124,159	126,038
R^2	0.18	0.20	0.24
Panel B: Excluding downstream dependence (interior coefficient), 90th percentile			
<i>Exposure</i> $\times$ <i>Policy</i>	-0.035*** (0.011)	-0.007** (0.003)	0.014*** (0.004)
Observations	552,365	211,144	219,668
R^2	0.17	0.18	0.23
Product-Country-Year FE	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes
Product-Country-Firm FE	Yes	No	No
Firm FE	No	Yes	Yes

This table presents regression estimates for firms' import flows, replicating the specification from [Table 6](#) while excluding sectors highly connected to construction using Spain's input-output matrix. Panel A excludes sectors with high upstream dependence on construction, measured using the Leontief inverse $L = (I - A)^{-1}$ and the element $L_{i, \text{construction}}$. Panel B excludes sectors with high downstream dependence, measured as the share of intermediate inputs sourced from construction in total inputs ("interior coefficient"). In both panels, sectors above the 90th percentile of the respective dependence measure are excluded. Standard errors are double clustered at the main bank and firm levels and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

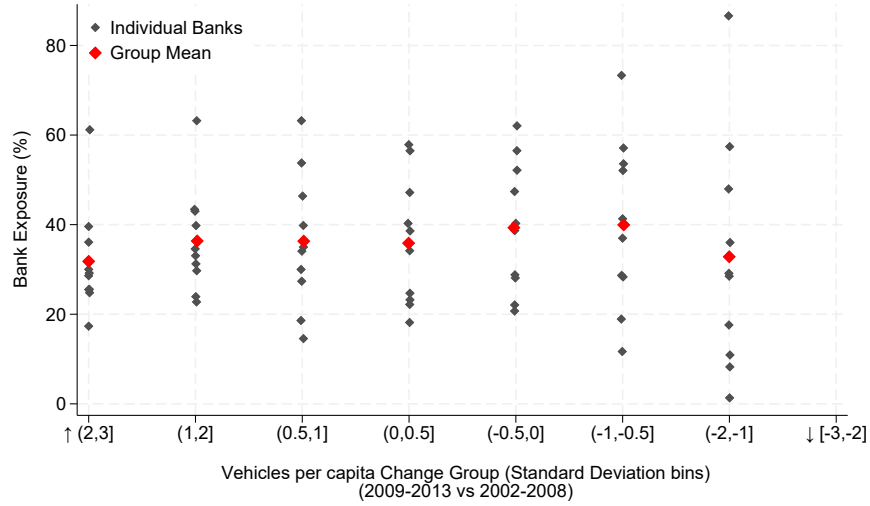
Table A.7: Robustness: Alternative Functional Forms

	Log Specification	PPML Specification
	(1)	(2)
$Exposure_f \times Policy_t$	-0.024*** (0.008)	-0.029*** (0.011)
Firm–Country–Product FE	Yes	Yes
Country–Product–Year FE	Yes	Yes
Province–Industry–Year FE	Yes	Yes
Observations	238,301	549,794
R^2	0.784	–
Pseudo R^2	–	0.734

This table reports robustness checks to alternative functional forms. Columns (1) and (2) estimate the baseline specification using $\log(Z_{f,c,g,t})$ as the dependent variable, thereby focusing on the intensive margin and excluding zero trade flows. Columns (3) and (4) use a Poisson pseudo-maximum likelihood (PPML) estimator, which accommodates zero trade flows and is robust to heteroskedasticity. All regressions include the same set of fixed effects as in Table 6. Standard errors are double-clustered at the main bank and firm levels and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.



(a) Exposure of Top 10 Lenders by Province Grouped by Change in Domestic Sales



(b) Exposure of Top 10 Lenders by Province Grouped by Change in Vehicles per Capita

Figure A.3: Bank exposure across provinces, based on Almunia et. al (2021)

Panel (a) shows, for each group of provinces on the x-axis—classified based on the standardized percentage change in average firm-level domestic sales between the periods 2002–2008 and 2009–2013—the exposure to the regulation (as defined in Section 4.1) for the top 10 banks with the highest share of loans granted to firms operating in each group. In Panel (b), the groups on the x-axis are defined based on the standardized percentage change in the number of vehicles per capita. The classification of provinces across groups is obtained by reverse-engineering the information presented in Figure 3 of Almunia et al. (2021).

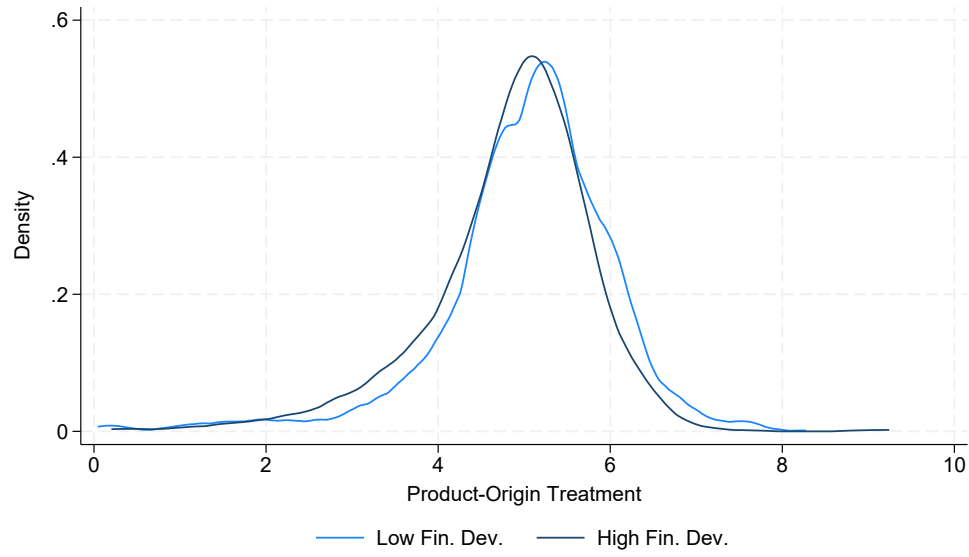
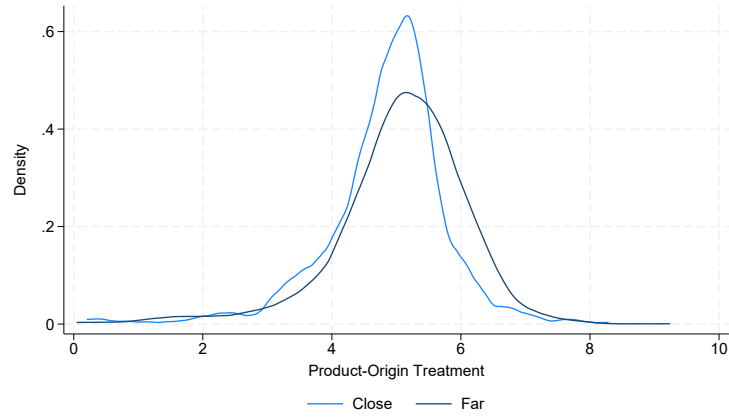
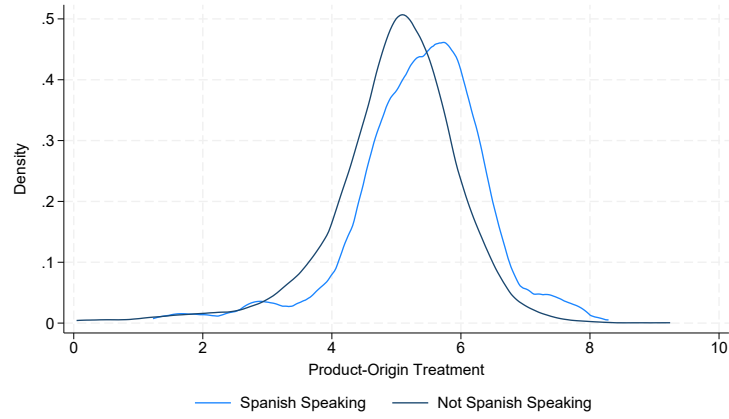


Figure A.4: Product-origin Exposure: Kernel Density Estimates by Financial Development of Origin Country

This figure plots the kernel density of the standardized product-origin treatment, as defined in [Equation 8](#), by the level of financial development in the origin country, where financially developed countries are defined as those in the top quartile of the private credit-to-GDP ratio.



(a) Distance



(b) Language

Figure A.5: Product-origin Exposure: Kernel Density Estimates by Distance and Language

The figure plots the kernel density of the standardized product-origin treatment, as in [Equation 8](#). Panel A splits origin countries into those geographically close to Spain—defined as the bottom quartile of the distance-to-Spain distribution—and those farther away. Panel B divides origin countries into Spanish-speaking and non-Spanish-speaking.

Table A.8: Impact of Provisioning Shift on Exports to Spain by Presence of Spanish Banks

	Mid-point growth Exports to Spain							
	High FD		Low Financial Development					
	Without	With	Without	With	Far		Not Spanish	
					Without	With	Without	With
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Exposure_{go} \times Post_t$	0.004 (0.032)	-0.016 (0.028)	-0.037** (0.018)	-0.032 (0.032)	-0.039* (0.020)	-0.037 (0.037)	-0.034* (0.018)	-0.057* (0.032)
Observations	1,788	915	6,571	1,700	5,366	1,285	6,196	1,035
Fixed Effects								
Exporter $\times$ Product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product charact. $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country charact. $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Spanish share decile $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table presents the regression results for the mid-point growth rate of exports of product g from country o to Spain, relative to its five-year average. Columns (1)-(2) reports estimates for exports to Spain from countries with high financial development, defined as those in the top quartile of the private credit-to-GDP ratio. Columns (3)-(8) report estimates for countries with low financial development, those at the bottom three quartiles. Columns (5)-(6) report estimates for countries that are geographically distant from Spain (top three quartiles of the distance-to-Spain distribution), columns (7)-(8) report estimates for countries that are not Spanish speakers. Columns (1), (3), (5), and (7) report estimates for exporting countries without a Spanish bank presence. Columns (2), (4), (6), and (8) report estimates for exporting countries where Spanish banks are present. The analysis is based on a product-origin level measure of exposure, and incorporates different sets of fixed effects, as specified in equation (10). Product-origin exposure is calculated as the weighted average of the exposures of Spanish banks lending to firms importing product g from country o , following equation (8). The fixed effects included in each regression are listed in the lower part of the table. Product characteristics include a dummy for high or low heterogeneity: i.e, low, if price dispersion of the product is at the bottom quartile of the price dispersion distribution. Exporter country characteristics include: distance to Spain indicator (close to Spain or not), official language (Spanish or non Spanish). Spanish share corresponds to the participation of Spain in total exports of a specific origin $\times$ product pair. All variables are defined in Table A.1. Standard errors shown in parentheses are clustered by product and country of origin.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A.9: Reliance on Supplier Credit and Policy Impact on Trade Flows

	Importers		
	(1)	(2)	(3)
$Exposure \times Policy$	-0.036*** (0.011)	-0.033*** (0.011)	-0.032** (0.013)
$Exposure \times Policy \times Supplier\ Credit$			0.020** (0.008)
Product-Country-Year FE	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes
Product-Country-Firm FE	Yes	Yes	Yes
Imports/Purchases Bins-Year FE	No	No	Yes
Observations	558,540	519,335	519,265
R^2	0.17	0.17	0.18

Notes: This table presents the heterogeneous effects of the provisioning reform on trade outcomes based on firms' reliance on trade credit, specifically importer-received credit in Column (3). *Supplier Credit* is defined as the average share of supplier credit received by importers in total financing (supplier credit plus bank debt) over the same period. This measure is standardized to have mean zero and standard deviation one. Column (1) replicates the main specification from Table 4 using the full sample. Column (2) restricts the sample to firms for which trade credit data are available. Lower-order interaction terms corresponding to the triple interaction are included in the regressions but not reported. Column (3) additionally includes fixed effects at the year-imports-share-bins level. Bins are constructed using quintiles of the firm's average imports-to-purchases ratio over 2010–2011. The dependent variable is the midpoint growth rate of firm f 's imports of product g relative to the 5-year average of firm f 's imports of the same product. *Exposure* is computed as the weighted average of the exposure of banks lending to firm f , using outstanding credit as of the end of 2011 as weights. *Policy* takes a value of one for 2012 and 2013 and zero otherwise. The fixed effects included in each regression are specified in the lower part of the table. Importers operating in the construction or real estate sectors are excluded. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

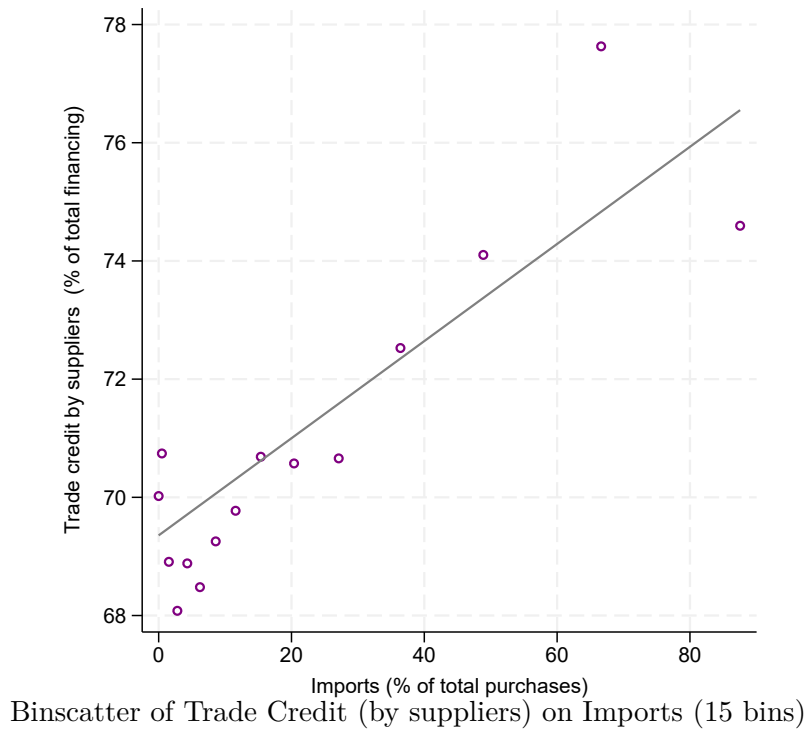
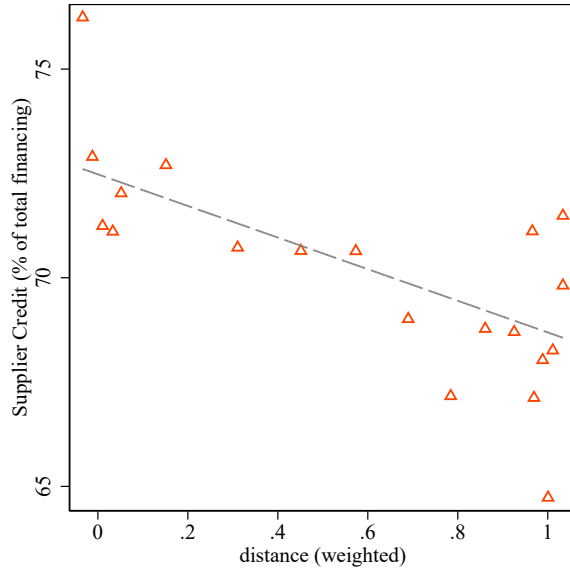
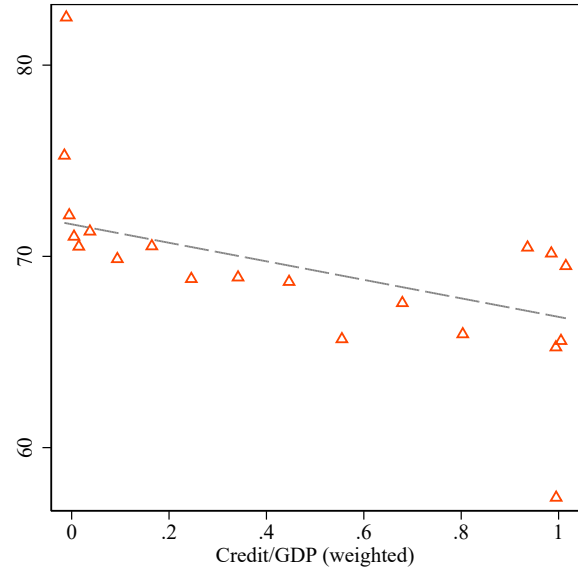


Figure A.6: Trade intensity and use of Trade Credit

This figure presents binned scatterplots illustrating the relationship between firms' trade intensity and their reliance on trade credit. It plots imports as a share of total purchases (x-axis) against trade credit granted by suppliers as a share of total financing (y-axis). Points represent the average values of the X and Y variables within 15 quantile bins of the X-axis variable.



(a) Supplier Credit and Distance



(b) Supplier Credit and Financial Development

Figure A.7: Supplier Credit, Trade Frictions, and Financial Development

This figure presents binned scatterplots illustrating the relationship between supplier credit and the characteristics of firms' import relationships. Panel (a) plots supplier credit against the import-share-weighted fraction of sourcing from geographically distant origins. Panel (b) plots supplier credit against the import-share-weighted fraction of sourcing from financially developed countries, measured using the private-credit-to-GDP ratio. Supplier credit is measured as firm's short-term non-interest-bearing external financing divided by total financing, where total financing is the sum of supplier credit and bank debt. The figures condition on import intensity and restrict the sample to importing firms. Points represent the average values of the X and Y variables within quantile bins of the X-axis variable.

Table A.10: Trade Frictions and Supplier Credit Adjustment

	High FD	Low FD			
	(1)	(2)	(3)	(4)	(5)
<i>Exposure</i>	-0.004 (0.020)	0.231 (0.141)	0.067 (0.085)	0.292* (0.160)	0.217* (0.125)
<i>Exposure</i> $\times$ <i>distant</i>		-0.261* (0.141)		-0.263* (0.141)	
<i>Exposure</i> $\times$ <i>non Spanish</i>			-0.026 (0.083)	-0.062 (0.081)	
<i>Exposure</i> $\times$ <i>distant non Spanish</i>					-0.252** (0.124)
<i>distant</i>		0.867* (0.442)		0.874* (0.442)	
<i>non Spanish lang</i>			0.035 (0.282)	0.159 (0.279)	
<i>distant non Spanish</i>					0.827** (0.389)
Firm Controls	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Main Buyer FE	Yes	Yes	Yes	Yes	Yes
Import/Purchases Bins FE	Yes	Yes	Yes	Yes	Yes
N. Origins quartile FE	Yes	Yes	Yes	Yes	Yes
Observations	5,794	4,639	4,639	4,639	4,639
Clusters	51	50	50	50	50
R^2	0.040	0.033	0.029	0.033	0.033

Notes: The dependent variable is the growth rate of supplier credit at the firm level. *Exposure* is the firm-level exposure to banks affected by the 2012 provisioning reform. Column (1) restricts the sample to firms whose pre-policy import basket is tilted toward origins with above-median financial development, measured by private credit to GDP. Columns (2)–(5) restrict the sample to firms whose pre-policy import basket is tilted toward origins with below-median financial development. *w_far* is the pre-policy import-share-weighted fraction of origins above the median distance to Spain. *w_diffLang* is the weighted fraction of origins that do not share a common language with Spain. *w_far_nonspa* is the weighted fraction of origins that are both distant and non-Spanish-speaking. All specifications include firm controls and absorb sector, postal code, main origin, up/downstream status, and origin-count quantile fixed effects. Standard errors clustered at the banking-group level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

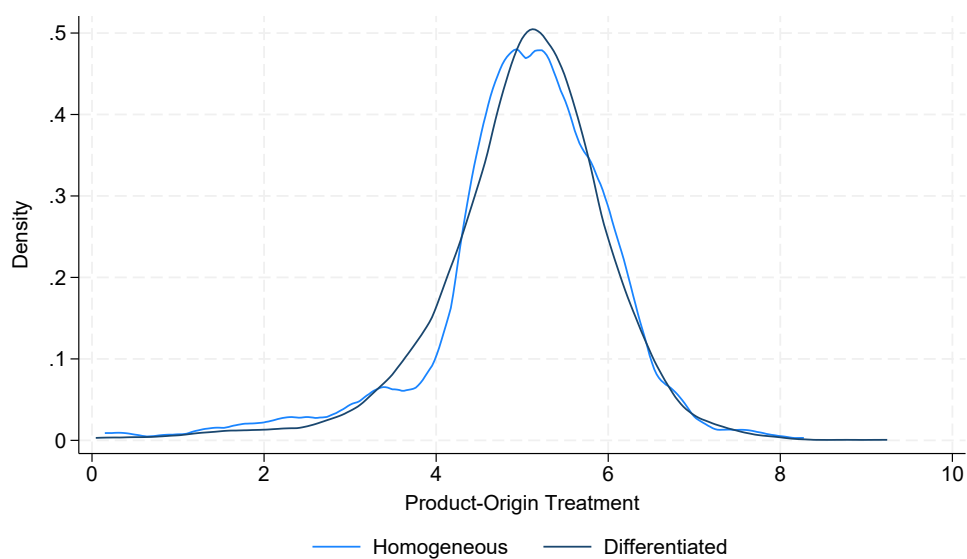


Figure A.8: Product-origin Exposure: Kernel Density Estimates by Product Characteristics

This figure plots the kernel density of the standardized product-origin treatment, as in [Equation 8](#), for heterogeneous products—defined as those in the top three quartiles of the distribution of HS6 price dispersion distribution—and homogeneous products.

Table A.11: Impact of Provisioning Shift on the Total Exports of Spain's Trade Partners and Trade Costs, Low Financially Developed Origins

	Mid-point growth rate of Exports								
	Far from Spain			Not Spanish Speaking			Far and Not Spanish		
	All	Spain	Rest	All	Spain	Rest	All	Spain	Rest
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Agg. Exposure_{go} × Post_t</i>	-0.046* (0.027)	-0.117*** (0.039)	0.009 (0.020)	-0.022 (0.023)	-0.103*** (0.032)	0.025 (0.018)	-0.065* (0.033)	-0.181*** (0.045)	-0.003 (0.025)
Observations	6,651	6,651	6,651	7,231	7,231	7,231	5,611	5,611	5,611
Fixed Effects									
Exporter × Product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product charact. × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country charact. × Year	Yes	Yes	Yes	Yes	Yes	Yes	—	—	—

This table reports regression results for the mid-point growth rate of exports of product g from country o , relative to its five-year average. Estimates consider only low financially developed origins. Columns (1), (4), and (7) present results for total exports to all destinations; columns (2), (5), and (8) for exports to Spain; and columns (3), (6), and (9) for exports to destinations other than Spain. Columns (1)-(3) correspond to exporters that are geographically far from Spain (top three quartiles of the distance-to-Spain distribution). Columns (4)-(6) restrict the sample to exporters from non-Spanish-speaking countries. Columns (7)-(9) restrict the sample to exporters that are both geographically distant and non-Spanish-speaking. The analysis is based on an aggregate measure of market exposure, and incorporates different sets of fixed effects, as specified in equation (13). Aggregate market exposure is calculated as the product-level exposure defined in equation (8) multiplied by Spain's share of country o 's exports of product g , following equation (12). The fixed effects included in each regression are listed in the lower part of the table. Product characteristics include a dummy for high or low heterogeneity: i.e, low, if price dispersion of the product is at the bottom quartile of the price dispersion distribution. Exporter country characteristics include: distance to Spain indicator (close to Spain if it belongs to the bottom quartile of the distance-to-Spain distribution), official language (Spanish or non Spanish). All variables are defined in Table A.1. Standard errors shown in parentheses are clustered by product and country of origin.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B Effects on Spanish Exporting Firms

Table B.1: Domestic credit contraction: intensive and extensive margins, exporting firms

	Intensive Margin		Extensive Margin	
	(1)	(2)	(3)	(4)
<i>Exposure</i> $\times$ <i>Policy</i>	-0.025** (0.011)	-0.031** (0.013)	0.014* (0.008)	0.015 (0.010)
Controls $\times$ time	No	Yes	–	–
Controls	–	–	No	Yes
Firm-time FE	Yes	Yes	–	–
Firm FE	–	–	Yes	Yes
Bank-firm FE	Yes	Yes	–	–
Obs.	962,963	962,963	102,788	102,788
R^2	0.97	0.97	0.52	0.56

This table presents regression results on lending from bank b to firm f at the intensive margin (columns 1–2) and the extensive margin (columns 3–4). In columns 1–2, the dependent variable is the logarithm of credit granted by bank b to firm f in quarter t . In columns 3–4, conditional on the existence of the lending relationship at the end of 2011, the dependent variable is a dummy variable that takes the value of one if the lending relationship was terminated in the aftermath of the policy and zero otherwise. *Exposure* is computed as the ratio of construction and real estate lending to total corporate sector lending by bank b as of the end of 2011. *Controls* are measured as of the end of 2011 and include bank-firm characteristics (the collateralization rate, the ratio of long-term lending, the share of firm f 's NPLs with the bank, and the share of lending coming from bank b out of firm f 's total bank lending) and bank characteristics (dummies for quartiles of size, measured by total assets, and quartiles of exposure to exporter/importer companies, measured as the lending share to exporters and importers relative to total corporate sector lending). All variables are defined in Table A.1. We exclude firms in the construction or real estate sectors. Standard errors are double clustered at the bank and firm levels and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

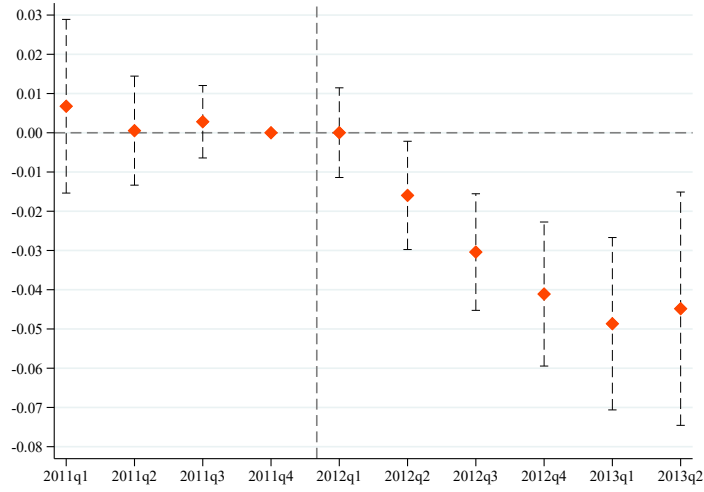


Figure B.1: Impact on Lending to Exporting Firms at the Intensive Margin

This figure plots the event study associated with specification (5), where we replace *Policy* with quarter dummies. The dependent variable is the logarithm of credit. We drop firms in construction or real estate sectors. Each dot represents the estimated coefficient of the interaction of bank exposure and a quarter dummy. We normalize the coefficient of 2011Q4 to zero. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors double clustered at the bank and firm levels.

Table B.2: Impact of Provisioning Shift on Total Bank Debt to Exporting Firms,
Firm Level Analysis

	(1)	(2)
<i>Exposure</i> $\times$ <i>Policy</i>	-0.012** (0.005)	-0.016** (0.007)
Controls $\times$ time	No	Yes
Firm FE	Yes	Yes
Time FE	Yes	No
$\hat{\alpha}_{f,t}$	No	Yes
Obs.	230,099	230,099
R^2	0.98	0.98

This table presents firm-level regression results for firms' total bank debt, estimated using equation (6). The dependent variable is the logarithm of firm f 's total bank debt. *Exposure* is calculated as the weighted average of the exposure of banks lending to firm f , using outstanding credit at the end of 2011 as weights. To account for credit demand, regressions in columns (2) and (4) include estimated firm-quarter fixed effects obtained from specification (5), while regressions in columns (1) and (3) include quarter fixed effects instead. *Controls* are the weighted average of bank-firm characteristics (collateralization rate, ratio of long-term lending, share of firm f 's NPLs with the bank) and bank characteristics (quartiles of size and exposure to exporter/importer companies), measured as of the end of 2011 and weighted by banks' lending to firm f as of 2011. Additionally, we include industry and province indicators as control variables. All variables are defined in Table A.1. Firms in the construction or real estate sectors are excluded. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

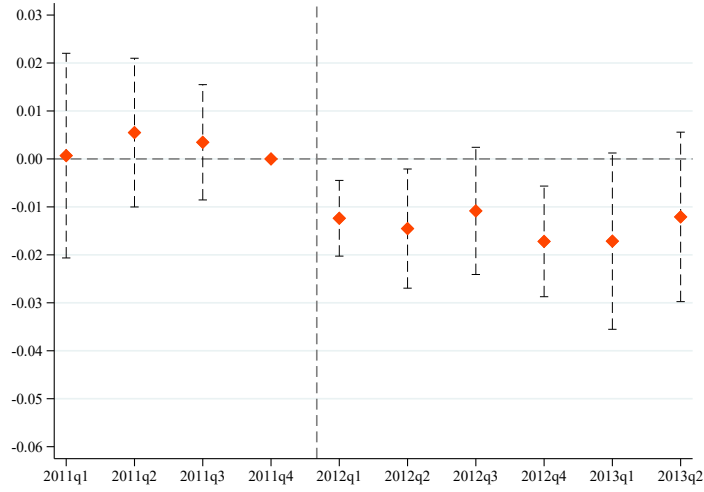


Figure B.2: Impact on Total Bank Debt to Exporting Firms

This figure plots the event study associated with specification (6), where *Policy* is replaced with quarter dummies. The dependent variable is the logarithm of firms' total bank credit. To account for credit demand, all regressions include the estimated firm-quarter fixed effects obtained from our bank-firm specification in (5). Firms in the construction or real estate sectors are excluded. Each dot represents the estimated coefficient of the interaction between a quarter dummy and the firm-level exposure, which is computed as the weighted average of the exposure of banks lending to firm f , using the outstanding credit at the end of 2011 as weights. The coefficient for 2011Q4 is normalized to zero. The dashed lines indicate the 2.5%–97.5% confidence interval, with standard errors double-clustered at the main bank level and firm level.

Table B.3: Effect of Shift in Provisions on annual Firm-level Outcomes, Exporters

	Bank Debt	Total Debt	Total Debt +Equity	Employment	Tangible Assets
	(1)	(2)	(3)	(4)	(5)
<i>Exposure</i>	-0.022 (0.014)	-0.009 (0.012)	0.036 (0.047)	-0.006 (0.004)	-0.008 (0.006)
Firm controls	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Main Buyer FE	Yes	Yes	Yes	Yes	Yes
Obs.	17,194	17,194	17,194	17,194	17,194
R^2	0.01	0.02	0.02	0.02	0.03

This table reports regression results on firm-level outcomes using annual financial statements. The dependent variables are the mid-point growth rates between 2011 and 2012 of total bank debt, total debt, total debt plus equity, employment, and tangible assets. The main explanatory variable, *Exposure*, is computed as the weighted average exposure of banks lending to firm f , using outstanding credit at the end of 2011 as weights. Controls (measured in 2011) include weighted averages of bank characteristics—total assets and the share of lending to importing and exporting firms—as well as firm characteristics (log total assets, cash-to-assets, sales-to-assets, and own funds-to-assets ratios). Regressions include the fixed effects listed in the table, accounting for industry, province, and the firm’s main export destination. All variables are defined in [Table A.1](#). Standard errors are double-clustered at the main bank and firm levels. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.4: Characteristics of Importers and Exporters

Firm characteristics	Importers				Exporters			
	p75	p50	p25	mean	p75	p50	p25	mean
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
log(assets)	9.61	8.46	7.49	8.61	9.66	8.54	7.52	8.65
log(employees)	4.04	2.96	1.94	3.03	4.14	3.17	2.09	3.14
log(age)	3.21	2.83	2.30	2.69	3.26	2.83	2.30	2.70
log(debt)	8.40	6.99	5.56	6.90	8.46	7.12	5.65	6.97
log(bank debt)	7.92	6.61	5.15	6.43	7.99	6.74	5.23	6.50
average collateral	0.11	0	0	0.11	0.14	0	0	0.12
Nr. of banks	5	3	2	3.82	5	3	2	3.88
Credit concentration (HHI_{firm})	0.99	0.52	0.30	0.58	0.98	0.51	0.30	0.58
log(tangible fixed assets)	7.91	6.44	4.67	6.23	8.01	6.62	4.92	6.41
log(LT financial investment)	6.48	4.10	2.00	4.33	6.65	4.29	2.13	4.44
log(LT external funds)	7.65	6.28	4.88	6.24	7.70	6.39	5.00	6.32
log(LT funds from financial institutions)	7.24	5.97	4.59	5.89	7.31	6.09	4.71	5.97

This table reports, in columns 1-4, the 75th, 50th, 25th percentile and mean of importer characteristics, and in columns 5-8, the corresponding values for exporters.

Table B.5: Average difference: exporters only minus importers only

Firm characteristics	Difference between importers and exporters			
	p75	p50	p25	mean
	(1)	(2)	(3)	(4)
log(assets)	0.008	0.002	-0.060**	-0.070***
log(employees)	0.220***	0.290***	0.180***	0.130***
log(age)	0.040***	0.000	0.000	0.024**
log(debt)	0.100***	0.110***	0.015	0.032
log(bank debt)	0.130***	0.110***	0.060	0.070*
average collateral	0.070***	0.000	0.000	0.020***
Nr. of banks	0.000	0.000	—	-0.003
Credit concentration (HHI_{firm})	—	0.004	0.010**	0.005*
log(tangible fixed assets)	0.150***	0.260***	0.350***	0.190***
log(long term financial investment)	0.170**	0.240***	-0.040	0.080
log(long term external funds)	0.090**	0.140***	0.160***	0.100***
log(long term funds from financial institutions)	0.100**	0.120***	0.170***	0.110***

This table reports, in columns 1-3, the difference between the sample of only exporters and only importers, at the 75th, 50th and 25th percentile of their characteristics' distribution. These coefficients are estimated through quantile regressions on a dummy that takes the value of 1 if the firm is classified as exporter only and zero if classified as importer only. Column 4 reports the mean difference in these characteristics for the two groups of firms. The statistical significance of these coefficients is based on the t -test.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.6: Effect of Shift in Provisions on firm's debt for both Importers and Exporters, Importers only and Exporters only

Panel A: Bank-firm level regressions			
Dep. Variable: Log(Credit)	$M \cap X$ (1)	$M \neg X$ (2)	$X \neg M$ (3)
<i>Exposure</i> $\times$ <i>Policy</i>	-0.033** (0.013)	-0.026** (0.012)	-0.028** (0.011)
Controls $\times$ time	Yes	Yes	Yes
Firm-time FE	Yes	Yes	Yes
Bank-firm FE	Yes	Yes	Yes
Obs.	648,342	314,621	325,044
R^2	0.97	0.97	0.97
Panel B: Firm level regressions			
Dep. Variable: Log(Credit)	$M \cap X$ (1)	$M \neg X$ (2)	$X \neg M$ (3)
<i>Exposure</i> $\times$ <i>Policy</i>	-0.022** (0.009)	-0.038*** (0.005)	-0.010* (0.006)
Controls $\times$ time	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
$\hat{\alpha}_{f,t}$	Yes	Yes	Yes
Obs.	141,937	86,796	88,059
R^2	0.98	0.98	0.98

This table presents regression results on lending from bank b to firm f (Panel A) and on total firm's debt (Panel B). Column (1) corresponds to the sample of firms that are simultaneously importers and exporters, column (2) corresponds to the sample of importers that are not exporters, whereas column (3) corresponds to the sample of exporters that are not importers. *Exposure* in Panel A is computed as the ratio of construction and real estate lending to total corporate sector lending by bank b as of the end of 2011. In panel B, it is calculated as the weighted average of the exposure of banks lending to firm f , using outstanding credit at the end of 2011 as weights. In Panel A, *Controls* are measured as of the end of 2011 and include bank-firm characteristics (the collateralization rate, the ratio of long-term lending, the share of firm f 's NPLs with the bank, and the share of lending coming from bank b out of firm f 's total bank lending) and bank characteristics (dummies for quartiles of size, measured by total assets, and quartiles of exposure to importer/exporter companies, measured as the lending share to importers and exporters relative to total corporate sector lending). In Panel B, *Controls* are the weighted average of bank-firm characteristics and bank characteristics, measured as of the end of 2011 and weighted by banks' lending to firm f as of 2011. All variables are defined in [Table A.1](#). We exclude importing and exporting firms in the construction or real estate sectors. Standard errors are double clustered at the bank and firm levels and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.7: Impact of Provisioning Shift on Firms' Export Flows

	Mid-point growth	Entry Dummy	Exit Dummy
	(1)	(2)	(3)
<i>Exposure</i> $\times$ <i>Policy</i>	-0.020** (0.009)	0.000 (0.003)	0.009** (0.004)
Product-Country-Year FE	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes
Product-Country-Firm FE	Yes	No	No
Firm FE	No	Yes	Yes
Obs.	649,050	274,933	229,050
R^2	0.17	0.16	0.25

This table presents regression estimates for firms' export flows at the product-country-year level. In column 1, we compute the mid-point growth rate between firm f 's exports of product g to country c and the 5-year average of exports of the same product to that country. In column 2, we use a binary indicator that, conditional on not participating in the market in the previous year, takes the value of one if the firm enters the market. In column 3, we use a binary indicator that takes the value of one if the firm exits the market, conditional on participation in the same market in the previous year. *Exposure* is computed as the average exposure of banks lending to firm f , weighted by outstanding credit as of the end of 2011. *Policy* is a dummy that takes the value of one after the provisioning increase. Fixed effects are reported in the lower part of the table. All variables are defined in Table A.1. Firms in construction or real estate sectors are excluded. Standard errors are double clustered at the main bank and firm levels and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.8: Impact of Provisioning Shift on Firms' International Trade Flows, Weighted Regressions

	Mid-point growth			
	Unweighted		Weighted	
	<i>M</i>	<i>X</i>	<i>M</i>	<i>X</i>
	(1)	(2)	(3)	(4)
<i>Exposure</i> $\times$ <i>Policy</i>	-0.036*** (0.011)	-0.020** (0.009)	-0.035** (0.014)	-0.018 (0.013)
Product-Country-Year FE	Yes	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes	Yes
Product-Country-Firm FE	Yes	Yes	Yes	Yes
Observations	558,540	649,050	558,540	649,050
R-squared	0.17	0.17	0.27	0.27

This table presents the regression estimates on firm's trade flows at the product-country-year level. The dependent variable is the mid-point growth rate between firm f 's imports (exports) of product g from (to) country c and the 5-year average of firm f 's imports (exports) of the same product from (to) that country, and it is regressed on the firm's exposure and a set of fixed effects. *Exposure* is computed as the average exposure of banks lending to firm f , with weights based on the outstanding credit as of the end of 2011. *Policy* is a dummy variable that takes the value of one after the provisioning increase. The fixed effects included in each regression are noted in the lower part of the table. Columns 1 and 2 present the unweighted results and correspond to columns 1 and 2, respectively, in Table 6. Columns 3 and 4 show the weighted results, using the 5-year average of imported and exported values as weights. Due to the presence of extremely large imported and exported values, weights are winsorized at the 5% level. All variables are defined in Table A.1. Importers and exporters in the construction or real estate sectors are excluded. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.9: Impact of Provisioning Shift on Firms' International Trade Flows
by Cumulative Product-Level Trade Share

<i>Product-Level Trade Share:</i>	Importers			Exporters		
	100%	Bottom 25%	Top 75%	100%	Bottom 25%	Top 75%
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Exposure</i> $\times$ <i>Policy</i>	-0.036*** (0.011)	-0.035*** (0.010)	-0.035* (0.020)	-0.020** (0.009)	-0.022** (0.011)	-0.011 (0.016)
Product-Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Province-Industry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Product-Country-Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	558,540	415,620	135,535	649,050	450,235	188,005
R^2	0.17	0.16	0.28	0.17	0.16	0.26

This table presents the regression estimates on firm's trade flows at the product-country-year level. Columns 1-3 correspond to importers and 4-6 correspond to exporters. Columns 1 and 4 use the full sample of importers and exporters, respectively. Columns 2 and 5 restrict the sample to firms representing the top 75% of total trade value by product in 2011. Columns 3 and 6 restrict the sample to firms representing the bottom 25% of total trade value by product in 2011. *Exposure* is computed as the average exposure of banks lending to firm f , with weights based on the outstanding credit as of the end of 2011. *Policy* is a dummy variable that takes the value of one after the provisioning increase. The fixed effects included in each regression are noted in the lower part of the table. All variables are defined in [Table A.1](#). Importers and exporters in the construction or real estate sectors are excluded. Standard errors are double-clustered at the main bank and firm levels and are reported in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C Appendix to the model

C.1 Derivations and proofs

Notation. Throughout, $\tau_1 = \tau_2 = \tau$ (Assumption 1). Long-term debt costs $(1 + r_0)^2$ per unit in period 2. B is the period-1 borrowing need and $D + B$ is total debt outstanding. We write $MR_i \equiv (1 - 1/\sigma_s)p_i$, $V_i \equiv p_i q_i$, $Q \equiv q_1 + q_2$, $s_i \equiv q_i/Q$, $\tilde{q} \equiv q_1 q_2 / Q$, $\phi_i \equiv \tau / (1 + \tau r_{xi})$, $c_i \equiv V_i \phi_i$, and $\kappa_i \equiv \tau c_i$. The wedge ϕ_i measures the pass-through of the trade-credit rate into the invoice price, $\partial \ln p_i / \partial r_{xi} = -\phi_i$, and $\tilde{q}$ is the harmonic object satisfying $1/q_1 + 1/q_2 = 1/\tilde{q}$, which governs how a difference in marginal revenues translates into redirected quantity. Baseline values carry no bar unless needed for clarity. All statements about the shock are first-order at the baseline.

C.1.1 The importer's problem

The importer in market i has expenditure E_i and Cobb-Douglas preferences across sectors, so that $E_{is} = \alpha_s E_i$ is fixed. Within sector s it maximizes $U_i = \left(\int q_i(\omega)^{(\sigma_s - 1)/\sigma_s} d\omega \right)^{\sigma_s / (\sigma_s - 1)}$ subject to $\int \tilde{p}_i(\omega) q_i(\omega) d\omega \leq E_{is}$. With multiplier μ_i on the budget, the first-order condition for variety ω is $U_i^{1/\sigma_s} q_i(\omega)^{-1/\sigma_s} = \mu_i \tilde{p}_i(\omega)$, so $q_i(\omega) = U_i \mu_i^{-\sigma_s} \tilde{p}_i(\omega)^{-\sigma_s}$. Substituting into the budget, $E_{is} = U_i \mu_i^{-\sigma_s} P_{is}^{1-\sigma_s}$ with $P_{is} \equiv \left(\int \tilde{p}_i(\omega)^{1-\sigma_s} d\omega \right)^{1/(1-\sigma_s)}$, the sectoral index of effective prices. Hence

$$q_i(\omega) = E_{is} P_{is}^{\sigma_s - 1} \tilde{p}_i(\omega)^{-\sigma_s} \equiv A_{is} \tilde{p}_i(\omega)^{-\sigma_s}$$

Origin j is the segment Ω_j of the continuum, of measure M_j , and its firms are identical, so $P_{is}^{1-\sigma_s} = \sum_j M_j \tilde{p}_{ij}^{1-\sigma_s}$, origin j 's share of the index is $M_j \tilde{p}_{ij}^{1-\sigma_s} / P_{is}^{1-\sigma_s}$, and a single firm's share is zero. The exporter is one firm and takes P_{is} , and therefore A_{is} , as given; the origin's flows are $V_{ij} = M_j p_{ij} q_{ij}$, $F_{ij} = M_j f_{ij}$, $\bar{L}_{ij} = M_j \bar{\ell}_{ij}$, so every proportional response and every ratio of responses is the same at the firm and origin level.

The importer pays $p_i q_i$ at delivery and finances a fraction τ of it. The facility $\bar{L}_i$ is a fixed amount attached to the transaction, at a rate below r_{xi} , and is drawn first; the residual $F_i = \tau p_i q_i - \bar{L}_i > 0$ is financed at r_{xi} . The marginal unit is therefore financed entirely with trade credit, at cost τr_{xi} per euro of invoice, so the effective price of the exporter's variety is $\tilde{p}_i = p_i (1 + \tau r_{xi})$ and its demand is $q_i = A_{is} [p_i (1 + \tau r_{xi})]^{-\sigma_s}$. Inverting,

$$p_i = \frac{A_{is}^{1/\sigma_s} q_i^{-1/\sigma_s}}{1 + \tau r_{xi}} \quad V_i = \frac{A_{is}^{1/\sigma_s} q_i^{1-1/\sigma_s}}{1 + \tau r_{xi}} \quad (\text{C.1})$$

We maintain $A_{is} > 0$ and $\sigma_s > 1$ at every quantity, so the variety never leaves the importer's

choice set. Two derivatives are used repeatedly. At given r_{xi} , $\partial V_i/\partial q_i = MR_i$. At given q_i , $\partial \ln p_i/\partial r_{xi} = -\phi_i$, so $\partial V_i/\partial r_{xi} = -V_i\phi_i = -c_i$, and by (1) the interest sensitivity of trade-credit demand is $\partial F_i/\partial r_{xi} = -\tau c_i = -\kappa_i$.

If every imported variety's line is cut, P_{is} rises and A_{is} rises with it, giving every variety a common demand shift $(\sigma_s - 1)d \ln P_{is}$. Propositions 1–4 hold P_{is} fixed; this is exact for a single firm and leaves every relative statement across origins unchanged, since the shift is common.

C.1.2 The exporter's problems

Period 1. Given D_1, D_2 and the rates, the exporter chooses F_1, F_2 to maximize (3), with $B = wD + F - V_1 - V_2$. In the baseline q_i is fixed by capacity and the exporter takes r_{xi} as given, so V_i is fixed and $\partial B/\partial F_i = 1$. Then

$$\frac{\partial \Pi_1}{\partial F_i} = (1 + r_{xi}) - \psi_i F_i - (1 + r_0) - \lambda(D + B)$$

and the first-order condition gives the trade-credit supply curve

$$r_{xi} = r_0 + \lambda(D + B) + \psi_i F_i \quad (\text{C.2})$$

The Hessian in (F_1, F_2) is $-\begin{pmatrix} \lambda + \psi_1 & \lambda \\ \lambda & \lambda + \psi_2 \end{pmatrix}$, with determinant $\psi_1\psi_2 + \lambda(\psi_1 + \psi_2) > 0$, so Π_1 is strictly concave and the period-1 policy is unique; it is interior whenever $r_{xi} > r_0 + \lambda(D + B)$.

Period 0. The exporter chooses D_1, D_2 to maximize $\Pi_1(D_1, D_2)$ at given rates. By the envelope theorem the period-1 choice of F drops out. D_i enters (3) through $(1 + r_0)^2 D$, through the term $\lambda(D + \frac{B}{2})B$ at fixed B , whose derivative is λB , and through B , with $\partial B/\partial D_i = w - \partial V_i/\partial D_i = w - MR_i/(1 + t)$ since $q_i = D_i/(1 + t)$. Collecting terms,

$$\frac{d\Pi_1}{dD_i} = -(1 + r_0)^2 - \lambda B - (1 + r_0 + \lambda(D + B)) \left[w - \frac{MR_i}{1 + t} \right] \quad (\text{C.3})$$

and setting it to zero gives the capacity condition

$$\left(1 - \frac{1}{\sigma_s}\right)p_i = (1 + t) \left[w + \frac{(1 + r_0)^2 + \lambda B}{1 + r_0 + \lambda(D + B)} \right] \quad (\text{C.4})$$

The right-hand side does not depend on i , so $p_1 = p_2$. This holds for any τ_1, τ_2 : the exporter takes r_{xi} as given and does not internalize (1), so neither τ_i nor $\bar{L}_i$ enters (C.3). An exporter that treated $F_i = \tau_i V_i - \bar{L}_i$ as a constraint would have $\partial B / \partial D_i = w - (1 - \tau_i) MR_i / (1 + t)$, the coefficient on MR_i would be destination-specific, and invoice prices would diverge by an amount first-order in $\tau_1 - \tau_2$. The term λB in (C.3) is the crowding-out of trade credit by capacity: an additional unit of long-term debt raises the premium on the short-term debt that finances receivables, and the exporter internalizes it.

Two consequences of (C.4). First,

$$MR_i - (1 + t)w = (1 + t) \frac{(1 + r_0)^2 + \lambda B}{1 + r_0 + \lambda(D + B)} > 0$$

so every unit of capacity earns a strict margin over the wage. An exporter free to leave capacity idle in period 1 would therefore use all of it, and since the margin is bounded away from zero, full capacity remains optimal for shocks small enough that the common invoice price stays above $(1 + t)w\sigma_s/(\sigma_s - 1)$; the first-order responses below are unaffected by whether labour is committed. Second, $\partial B / \partial D_i = -[(1 + r_0)^2 + \lambda B] / [1 + r_0 + \lambda(D + B)] < 0$ at the optimum: at the exporter's own margin, capacity relaxes the period-1 borrowing need, because cash-in-advance receipts per unit exceed the wage bill per unit.

C.1.3 Equilibrium

Given (q_1, q_2) and D , the equilibrium rates solve the supply curves (C.2) together with the clearing conditions $F_i = \tau V_i(q_i, r_{xi}) - \bar{L}_i$, where B itself depends on the rates through V_i . Linearizing this system at fixed quantities gives (C.7)–(C.10) below with $dq_1 = d\varepsilon = 0$, whose determinant is Δ defined in (C.13). Under Assumption 2, $\Delta > 0$, so the rates are locally unique given quantities; capacities then solve (C.4) at those rates. We restrict attention to equilibria with $B > 0$.

Existence and uniqueness. Optimal capacity is finite: as $D_i \rightarrow \infty$, $q_i \rightarrow \infty$ and $MR_i \rightarrow 0$ under CES demand, while (C.3) tends to $-(1 + r_0)^2 - \lambda B - (1 + r_0 + \lambda(D + B))w < 0$, so Π_1 is eventually decreasing in each D_i . The period-1 policy is unique by strict concavity, and the break-even rates are unique given quantities by $\Delta > 0$. The Jacobian of the full system in $(D_1, D_2, r_{x1}, r_{x2})$ is nonsingular at any equilibrium satisfying Assumption 2, so equilibria are locally unique; we do not establish global uniqueness, and the analysis is local to the computed equilibrium. The feedback $\lambda(c_1 + c_2)$ is bounded away from one by Assumption 2(a), which is what keeps the cross slopes finite and the local analysis well posed.

C.1.4 The shock

The period-1 problem with reallocation. With D sunk and $\bar{L}_1$ replaced by $(1-x)\bar{L}_1 = \bar{L}_1 - \varepsilon$, the exporter chooses F_1, F_2 and q_1 , with $q_2 = Q - q_1$, at given rates. The first-order conditions for F_i are unchanged. For q_1 , the quantity enters (3) only through B , with $\partial B / \partial q_1 = -(MR_1 - MR_2)$ along $q_1 + q_2 = Q$, so

$$\frac{\partial \Pi_1}{\partial q_1} = (1 + r_0 + \lambda(D + B))(MR_1 - MR_2)$$

and the condition for shipments is $MR_1 = MR_2$, that is $p_1 = p_2$, for every ε . At the optimum $MR_1 = MR_2$, so the second derivative is

$$\frac{d^2 \Pi_1}{dq_1^2} = (1 + r_0 + \lambda(D + B)) \left[\frac{\partial MR_1}{\partial q_1} + \frac{\partial MR_2}{\partial q_2} \right] < 0$$

since marginal revenue is decreasing in quantity under (C.1), and the shipment choice is a strict maximum. The corner $q_1 = 0$ is never optimal, since $p_1 \rightarrow \infty$ as $q_1 \rightarrow 0$ while p_2 stays bounded.

The cross slopes of the supply curve. Differentiate (C.2) at the baseline. Capacity D and the wage bill wD are fixed by the period-0 commitment, and $q_1 + q_2 = Q$, so $dq_2 = -dq_1$. Writing $d\Omega \equiv \lambda dB$ for the common movement in the shadow cost,

$$dr_{xi} = d\Omega + \psi_i dF_i \quad dB = dF_1 + dF_2 - dV_1 - dV_2 \quad (\text{C.5})$$

By (C.1), $dV_i = MR_i dq_i - c_i dr_{xi}$. At the baseline $MR_1 = MR_2$ and $dq_2 = -dq_1$, so the quantity terms cancel in $dV_1 + dV_2$, and $dB = dF_1 + dF_2 + c_1 dr_{x1} + c_2 dr_{x2}$. Substituting (C.5),

$$d\Omega = \lambda [dF_1 + dF_2 + c_1(d\Omega + \psi_1 dF_1) + c_2(d\Omega + \psi_2 dF_2)]$$

and solving,

$$d\Omega = \tilde{\lambda}_1 dF_1 + \tilde{\lambda}_2 dF_2 \quad \tilde{\lambda}_i \equiv \frac{\lambda(1 + c_i \psi_i)}{1 - \lambda(c_1 + c_2)} \quad (\text{C.6})$$

Under Assumption 2(a), $\tilde{\lambda}_i$ is finite and $\tilde{\lambda}_i > \lambda > 0$; $\tilde{\lambda}_1 = \tilde{\lambda}_2$ iff $c_1 \psi_1 = c_2 \psi_2$; at a given baseline $\partial \tilde{\lambda}_1 / \partial \psi_1 = \lambda c_1 / [1 - \lambda(c_1 + c_2)] > 0$; and $\tilde{\lambda}_i$ is strictly increasing in τ through $c_i = V_i \phi_i$.

Proof. Assumption 2(a) gives $1 - \lambda(c_1 + c_2) > 0$; the rest is inspection of (C.6), using that $\phi_i = \tau / (1 + \tau r_{xi})$ is increasing in τ . $\square$

The cross slope exceeds λ because a euro of receivables raises Ω , which raises both rates, which lowers both invoices by $c_1 + c_2$ per unit of Ω , which raises B , and so on; the loop converges when $\lambda(c_1 + c_2) < 1$. It is destination-specific because a euro extended to destination i also raises r_{xi} through ψ_i , which lowers V_i by $c_i\psi_i$. This amplification is what distinguishes a premium levied on total leverage $D + B$ from one levied on the debt-financed positions alone: in the latter case cash-in-advance receipts do not enter the premium, the loop is absent, and $\tilde{\lambda}_i = \lambda$.

Linearized system. Let $\rho_i \equiv \tilde{\lambda}_i + \psi_i$. Differentiating the five equilibrium conditions at the baseline, where $p_1 = p_2$ and $MR_1 = MR_2 \equiv MR$:

$$dr_{x1} = \rho_1 dF_1 + \tilde{\lambda}_2 dF_2 \quad (\text{C.7})$$

$$dr_{x2} = \tilde{\lambda}_1 dF_1 + \rho_2 dF_2 \quad (\text{C.8})$$

$$dF_1 = d\varepsilon + \tau MR dq_1 - \kappa_1 dr_{x1} \quad (\text{C.9})$$

$$dF_2 = -\tau MR dq_1 - \kappa_2 dr_{x2} \quad (\text{C.10})$$

$$dq_1 = -\sigma_s \tilde{q} (\phi_1 dr_{x1} - \phi_2 dr_{x2}) \quad (\text{C.11})$$

(C.7)–(C.8) are (C.5) with (C.6); (C.9)–(C.10) differentiate the clearing conditions, $dF_i = \tau dV_i + \mathbf{1}_{i=1} d\varepsilon$; (C.11) is $d \ln p_1 = d \ln p_2$, that is $-dq_1/(\sigma_s q_1) - \phi_1 dr_{x1} = dq_1/(\sigma_s q_2) - \phi_2 dr_{x2}$, using $1/q_1 + 1/q_2 = 1/\tilde{q}$.

Solution. Define $m \equiv \sigma_s \tilde{q} MR = (\sigma_s - 1)\tilde{p}\tilde{q}$, and

$$P \equiv \rho_1 \rho_2 - \tilde{\lambda}_1 \tilde{\lambda}_2 = \psi_1 \psi_2 + \tilde{\lambda}_1 \psi_2 + \tilde{\lambda}_2 \psi_1 > 0 \quad N_1 \equiv \rho_1 + \kappa_2 P \quad N_2 \equiv \rho_2 + \kappa_1 P \quad (\text{C.12})$$

$$\Delta \equiv 1 + \rho_1 \kappa_1 + \rho_2 \kappa_2 + \kappa_1 \kappa_2 P \quad \Delta_3 \equiv \Delta + m\tau [(\phi_1 N_1 - \phi_2 \tilde{\lambda}_1) + (\phi_2 N_2 - \phi_1 \tilde{\lambda}_2)] \quad (\text{C.13})$$

Substitute (C.11) into (C.9)–(C.10). Collecting the terms in dr_{x1} and dr_{x2} with $\tilde{\kappa}_1 \equiv \kappa_1 + m\tau\phi_1$, $\tilde{\kappa}_2 \equiv \kappa_2 + m\tau\phi_2$, $a_1 \equiv m\tau\phi_2$, $a_2 \equiv m\tau\phi_1$,

$$dF_1 = d\varepsilon - \tilde{\kappa}_1 dr_{x1} + a_1 dr_{x2} \quad dF_2 = a_2 dr_{x1} - \tilde{\kappa}_2 dr_{x2}$$

The quantity margin raises each destination's own interest sensitivity from κ_i to $\tilde{\kappa}_i$ and creates the cross sensitivities a_1, a_2 : a higher rate in one destination redirects shipments toward the other and raises trade-credit demand there. Substituting these into (C.7)–(C.8)

and collecting,

$$\begin{pmatrix} 1 + \rho_1 \tilde{\kappa}_1 - \tilde{\lambda}_2 a_2 & \tilde{\lambda}_2 \tilde{\kappa}_2 - \rho_1 a_1 \\ \tilde{\lambda}_1 \tilde{\kappa}_1 - \rho_2 a_2 & 1 + \rho_2 \tilde{\kappa}_2 - \tilde{\lambda}_1 a_1 \end{pmatrix} \begin{pmatrix} dr_{x1} \\ dr_{x2} \end{pmatrix} = \begin{pmatrix} \rho_1 \\ \tilde{\lambda}_1 \end{pmatrix} d\varepsilon$$

By Cramer's rule the numerator of dr_{x1} is $\rho_1(1 + \rho_2 \tilde{\kappa}_2 - \tilde{\lambda}_1 a_1) - \tilde{\lambda}_1(\tilde{\lambda}_2 \tilde{\kappa}_2 - \rho_1 a_1)$. The two terms in a_1 are $-\rho_1 \tilde{\lambda}_1 a_1$ and $+\tilde{\lambda}_1 \rho_1 a_1$ and cancel, leaving $\rho_1 + \tilde{\kappa}_2(\rho_1 \rho_2 - \tilde{\lambda}_1 \tilde{\lambda}_2) = \rho_1 + \tilde{\kappa}_2 P$. Symmetrically, the numerator of dr_{x2} is $\tilde{\lambda}_1(1 + \rho_1 \tilde{\kappa}_1 - \tilde{\lambda}_2 a_2) - \rho_1(\tilde{\lambda}_1 \tilde{\kappa}_1 - \rho_2 a_2)$, where the two terms in $\tilde{\kappa}_1$ cancel, leaving $\tilde{\lambda}_1 + a_2 P$. The determinant expands to Δ_3 . Since $\rho_1 + \tilde{\kappa}_2 P = N_1 + m\tau\phi_2 P$,

$$\frac{\partial r_{x1}}{\partial \varepsilon} = \frac{N_1 + m\tau\phi_2 P}{\Delta_3} \quad \frac{\partial r_{x2}}{\partial \varepsilon} = \frac{\tilde{\lambda}_1 + m\tau\phi_1 P}{\Delta_3} \quad \frac{\partial q_1}{\partial \varepsilon} = -\frac{\sigma_s \tilde{q} (\phi_1 N_1 - \phi_2 \tilde{\lambda}_1)}{\Delta_3} \quad (\text{C.14})$$

the last by substituting the rates into (C.11), where $\phi_1 m\tau\phi_2 P - \phi_2 m\tau\phi_1 P = 0$. Substituting into (C.9) and using the identities $\Delta - \kappa_1 N_1 = 1 + \rho_2 \kappa_2$ and $N_2 - \kappa_1 P = \rho_2$ from (C.12)–(C.13),

$$a \equiv \frac{\partial F_1}{\partial \varepsilon} = \frac{1 + \rho_2 \kappa_2 + m\tau(\phi_2 \rho_2 - \phi_1 \tilde{\lambda}_2)}{\Delta_3} \quad (\text{C.15})$$

The value responses are $\partial \ln V_i / \partial \varepsilon = \partial \ln p_i / \partial \varepsilon + \partial \ln q_i / \partial \varepsilon$ with $\partial \ln p_i / \partial \varepsilon = -(\partial q_i / \partial \varepsilon) / (\sigma_s q_i) - \phi_i \partial r_{xi} / \partial \varepsilon$. For destination 2, multiply by Δ_3 and use

$$\frac{m\tau\phi_2}{q_2} = \frac{(\sigma_s - 1)\bar{p}\tilde{q}\tau\phi_2}{q_2} = (\sigma_s - 1)s_1\kappa_2 \quad \text{since } \tilde{q}/q_2 = s_1 \text{ and } \tau\bar{p}q_2\phi_2 = \kappa_2$$

so that the term in P arising from the quantity response, $(\sigma_s - 1)s_1\kappa_2\phi_1 P$, cancels exactly against $-\phi_2 m\tau\phi_1 P$ from the rate response. With $N_1 - \kappa_2 P = \rho_1$,

$$\frac{\partial \ln V_2}{\partial \varepsilon} = \frac{(\sigma_s - 1)\Xi - \phi_2 \tilde{\lambda}_1}{\Delta_3} \quad \Xi \equiv s_1(\phi_1 \rho_1 - \phi_2 \tilde{\lambda}_1) = s_1[\phi_1 \psi_1 + \tilde{\lambda}_1(\phi_1 - \phi_2)] \quad (\text{C.16})$$

The exact cancellation says that the reallocation channel of contagion, through which redirection raises the untouched destination's trade-credit demand and hence its rate, is offset by the quantity gain that produces it; the sign of the third-destination response is therefore governed by the balance-sheet channel alone, weighed against the shocked destination's own pass-through scaled by the elasticity. Finally, two identities follow from the clearing conditions and hold exactly, since $F_i = \tau V_i - \bar{L}_i$ and only $\bar{L}_1$ moves:

$$a = 1 + \tau V_1 \frac{\partial \ln V_1}{\partial \varepsilon} \quad \frac{\partial F_2}{\partial \varepsilon} = \tau V_2 \frac{\partial \ln V_2}{\partial \varepsilon} \quad (\text{C.17})$$

Since $\varepsilon = x\bar{L}_1 = -d\bar{L}_1$, the first identity in elasticity form is

$$\frac{\partial \ln V_1}{\partial \ln \bar{L}_1} = \frac{(1-a)\bar{L}_1}{\tau V_1} = \theta(1-a) \quad \theta \equiv \frac{\bar{L}_1}{\tau V_1} \in (0,1) \quad (\text{C.18})$$

where $\theta < 1$ because $F_1 > 0$. All responses are linear in ε , so the response to the proportional cut x is the response per euro times $\bar{L}_1$; the elasticity form is the one that aggregates across origins in Proposition 3.

C.1.5 The conditions the propositions require

Every sign in Propositions 1–2 follows from $1 - \lambda(c_1 + c_2) > 0$ and three inequalities on baseline objects:

$$(\text{D1}) \quad \phi_1 N_1 > \phi_2 \tilde{\lambda}_1 \quad (\text{D2}) \quad \phi_2 \rho_2 > \phi_1 \tilde{\lambda}_2 \quad (\text{D3}) \quad \phi_1 \rho_1 > \phi_2 \tilde{\lambda}_1 \quad (\text{C.19})$$

Since $N_1 \geq \rho_1$, (D3) implies (D1). (D1) signs redirection; (D2) signs the trade-credit response in the shocked destination; (D3) is $\Xi > 0$. Under (D1)–(D2), $\Delta_3 > \Delta > 0$ by (C.13), since $N_2 \geq \rho_2$. These conditions are exact for the signs they govern: if (D1) fails, $\partial q_1 / \partial \varepsilon > 0$ and shipments move toward the shocked destination; if (D2) fails, $a < 0$ and the invoice in the shocked destination falls by more than the line cut.

Assumption 2(b) is equivalent to (D2)–(D3). Hence Assumptions 1–2 imply (D1)–(D3) and $\Delta_3 > 0$.

Proof. For $i = 1$, (b) reads $\phi_1 \psi_1 > \tilde{\lambda}_1(\phi_2 - \phi_1)$, that is $\phi_1(\tilde{\lambda}_1 + \psi_1) > \phi_2 \tilde{\lambda}_1$, which is (D3). For $i = 2$, (b) reads $\phi_2 \psi_2 > \tilde{\lambda}_2(\phi_1 - \phi_2)$, that is $\phi_2(\tilde{\lambda}_2 + \psi_2) > \phi_1 \tilde{\lambda}_2$, which is (D2). The rest follows from the remarks above. $\square$

Remark 1. Assumption 2(a) is used in its stated form $\lambda(\phi_1 + \phi_2)(V_1 + V_2) < 1$ in the proof of Proposition 1(iii) and in the boundary argument in the proof of Proposition 2(ii), both of which require $\tilde{\lambda}'_1(\phi_2 - \phi_1) < \phi_1$. Part (iv) of Proposition 1 and part (iii) of Proposition 2 require equal wedges instead, and the τ statement in part (iv) additionally uses Assumption 2(c). Every other statement, including finiteness and positivity of $\tilde{\lambda}_i$, holds under the weaker $\lambda(\phi_1 V_1 + \phi_2 V_2) < 1$. The extra strength is used only when the untouched destination carries the larger wedge, $\phi_2 > \phi_1$, where it guarantees that the direct effect of a higher monitoring friction on the shocked destination's pass-through, ϕ_1 , dominates its indirect effect through the cross slope, $\phi_2 \tilde{\lambda}'_1$; the exact condition the proofs use is $\lambda \max_i \phi_i (V_1 + V_2) < 1$. Since $\phi_i \leq \tau$, the stated form is implied by $\lambda(V_1 + V_2) < 1/(2\tau)$. The premium $\lambda(D + B)$ is

in rate units, so with debt of the order of sales and a premium of a few hundred basis points, $\lambda(V_1 + V_2)$ is of the order of a few percent, against a bound of at least one half.

Both parts of (b) hold trivially when $\phi_1 = \phi_2$. Under Assumption 1, $\phi_2 - \phi_1 = \tau^2(r_{x1} - r_{x2})/[(1 + \tau r_{x1})(1 + \tau r_{x2})]$ and, by (C.2), $r_{x1} - r_{x2} = \psi_1 F_1 - \psi_2 F_2$, so (b) fails only when the destination with the higher monitoring premium has it exceed the other's by enough that, multiplied by the cross slope and the wedge, it overturns its own monitoring friction.

C.1.6 Proof of Proposition 1

(i) The numerator of (C.15) is positive by (D2). It is below Δ_3 because

$$\Delta_3 - [1 + \rho_2 \kappa_2 + m\tau(\phi_2 \rho_2 - \phi_1 \tilde{\lambda}_2)] = \rho_1 \kappa_1 + \kappa_1 \kappa_2 P + m\tau[(\phi_1 N_1 - \phi_2 \tilde{\lambda}_1) + \phi_2 \kappa_1 P] > 0$$

using $N_2 - \rho_2 = \kappa_1 P$ and (D1).

(ii) From (C.14), $\partial r_{x1}/\partial \varepsilon > 0$ since $N_1 > 0$, $P > 0$, and $\Delta_3 > 0$; and $\partial q_1/\partial \varepsilon < 0$ by (D1). Both terms of $\partial \ln V_1/\partial \varepsilon = -\phi_1 \partial r_{x1}/\partial \varepsilon + (1 - 1/\sigma_s)(\partial q_1/\partial \varepsilon)/q_1$ are negative. The elasticity is (C.18), and $\theta(1 - a) \in (0, \theta)$ by (i).

(iii) Hold the baseline fixed and vary ψ_1 . The numerator of (C.15) contains ρ_2 , κ_2 , $\tilde{\lambda}_2$, ϕ_1 , ϕ_2 , m , none of which depends on ψ_1 (by (C.6), $\tilde{\lambda}_2$ depends on ψ_2 only). It therefore suffices that Δ_3 is strictly increasing in ψ_1 . By Lemma C.1.4, $\rho_1 = \tilde{\lambda}_1 + \psi_1$ is increasing in ψ_1 with $\rho'_1 = 1 + \tilde{\lambda}'_1$, where $\tilde{\lambda}'_1 \equiv \partial \tilde{\lambda}_1/\partial \psi_1 = \lambda c_1/[1 - \lambda(c_1 + c_2)]$. Differentiating the components of (C.13), using $\partial P/\partial \psi_1 = \rho_2 + \tilde{\lambda}'_1 \psi_2$:

$$\partial \Delta/\partial \psi_1 = \kappa_1 \rho'_1 + \kappa_1 \kappa_2 (\rho_2 + \tilde{\lambda}'_1 \psi_2) > 0$$

$$\partial(\phi_2 N_2 - \phi_1 \tilde{\lambda}_2)/\partial \psi_1 = \phi_2 \kappa_1 (\rho_2 + \tilde{\lambda}'_1 \psi_2) > 0$$

$$\partial(\phi_1 N_1 - \phi_2 \tilde{\lambda}_1)/\partial \psi_1 = \phi_1 [\rho'_1 + \kappa_2 (\rho_2 + \tilde{\lambda}'_1 \psi_2)] - \phi_2 \tilde{\lambda}'_1 = \phi_1 (1 + \rho_2 \kappa_2) + \tilde{\lambda}'_1 [\phi_1 (1 + \psi_2 \kappa_2) - \phi_2]$$

The third is positive if $\phi_1 (1 + \psi_2 \kappa_2) \geq \phi_2$. Otherwise $\phi_2 > \phi_1$, and it suffices that $\phi_1 > \tilde{\lambda}'_1 (\phi_2 - \phi_1)$, that is $\phi_1 [1 - \lambda(c_1 + c_2)] > \lambda c_1 (\phi_2 - \phi_1)$, which rearranges, using $c_1 = V_1 \phi_1$, to $\lambda \phi_2 (V_1 + V_2) < 1$, implied by Assumption 2(a). Hence Δ_3 is strictly increasing in ψ_1 and a strictly decreasing.

(iv), *monotonicity in τ* . With $\phi_1 = \phi_2 \equiv \phi$, write $h \equiv \tau \phi$, so that $\kappa_i = h V_i$, $m\tau \phi = mh$, and $c_i = \phi V_i$. Both $h = \tau^2/(1 + \tau r_x)$ and ϕ are strictly increasing in τ . By (C.15) and (C.13), with $\tilde{\lambda}_2$ cancelling from the numerator and $W \equiv V_1 V_2 + m(V_1 + V_2)$,

$$N \equiv 1 + h n_1 \quad \Delta_3 \equiv 1 + h d_1 + h^2 d_2 \quad n_1 \equiv \rho_2 V_2 + m \psi_2 \quad d_1 \equiv \rho_1 V_1 + \rho_2 V_2 + m(\psi_1 + \psi_2) \quad d_2 \equiv PW$$

and $1/a = \Delta_3/N$. The dependence on τ runs through h directly and through ϕ in the $\tilde{\lambda}_i$, so

$$\frac{d(1/a)}{d\tau} = \frac{\partial(1/a)}{\partial h} h'(\tau) + \left[\frac{\partial(1/a)}{\partial \tilde{\lambda}_1} \tilde{\lambda}'_1 + \frac{\partial(1/a)}{\partial \tilde{\lambda}_2} \tilde{\lambda}'_2 \right] \phi'(\tau)$$

where now $\tilde{\lambda}'_i \equiv \partial \tilde{\lambda}_i / \partial \phi = \tilde{\lambda}_i(e_i + E)$ with $e_i \equiv V_i \psi_i / (1 + \phi V_i \psi_i)$ and $E \equiv \lambda(V_1 + V_2) / [1 - \lambda(c_1 + c_2)]$, both positive. For the first term, at fixed $\tilde{\lambda}_i$,

$$\text{sign} \frac{\partial(\Delta_3/N)}{\partial h} = \text{sign} [(d_1 + 2hd_2)(1 + hn_1) - n_1(1 + hd_1 + h^2d_2)] = \text{sign} [(d_1 - n_1) + 2hd_2 + h^2n_1d_2] > 0$$

since $d_1 - n_1 = \rho_1 V_1 + m\psi_1 > 0$. For the bracket, $\tilde{\lambda}_1$ enters only Δ_3 , through ρ_1 and P , with $\partial \Delta_3 / \partial \tilde{\lambda}_1 = hV_1 + h^2\psi_2 W$, so $\partial(1/a) / \partial \tilde{\lambda}_1 = (hV_1 + h^2\psi_2 W) / N > 0$. For $\tilde{\lambda}_2$, $\partial \Delta_3 / \partial \tilde{\lambda}_2 = hV_2 + h^2\psi_1 W$ and $\partial N / \partial \tilde{\lambda}_2 = hV_2$, so $N^2 \partial(1/a) / \partial \tilde{\lambda}_2 = (hV_2 + h^2\psi_1 W)N - hV_2 \Delta_3 = hV_2(N - \Delta_3) + h^2\psi_1 W N$. Using $\Delta_3 - N = h(\rho_1 V_1 + m\psi_1) + h^2 P W$ and $N = 1 + h(\rho_2 V_2 + m\psi_2)$, the terms of order h^2 collect to $h^2[\psi_1 W - V_2(\rho_1 V_1 + m\psi_1)] = h^2 V_1(m\psi_1 - V_2 \tilde{\lambda}_1)$, and the terms of order h^3 to $h^3 W[\psi_1(\rho_2 V_2 + m\psi_2) - V_2 P] = h^3 W \psi_2(m\psi_1 - V_2 \tilde{\lambda}_1)$, so

$$\frac{\partial(1/a)}{\partial \tilde{\lambda}_2} = \frac{h^2(m\psi_1 - V_2 \tilde{\lambda}_1)(V_1 + h\psi_2 W)}{N^2}$$

Since $hV_1 + h^2\psi_2 W = h(V_1 + h\psi_2 W)$, the bracket equals

$$\frac{h(V_1 + h\psi_2 W)}{N^2} [N \tilde{\lambda}'_1 - hV_2 \tilde{\lambda}_1 \tilde{\lambda}'_2 + hm\psi_1 \tilde{\lambda}'_2]$$

The last term inside is positive. For the first two, using $N = 1 + h(\tilde{\lambda}_2 + \psi_2)V_2 + hm\psi_2$ and $\tilde{\lambda}_2 \tilde{\lambda}'_1 - \tilde{\lambda}_1 \tilde{\lambda}'_2 = \tilde{\lambda}_1 \tilde{\lambda}_2 (e_1 - e_2)$,

$$N \tilde{\lambda}'_1 - hV_2 \tilde{\lambda}_1 \tilde{\lambda}'_2 = \tilde{\lambda}'_1 [1 + h\psi_2(V_2 + m)] + hV_2 \tilde{\lambda}_1 \tilde{\lambda}_2 (e_1 - e_2)$$

If $e_1 \geq e_2$ this is positive. If $e_2 > e_1$, use $\tilde{\lambda}'_1 \geq \tilde{\lambda}_1 E$ and $e_2 - e_1 \leq e_2 \leq V_2 \psi_2$: it suffices that $E[1 + h\psi_2(V_2 + m)] \geq hV_2^2 \psi_2 \tilde{\lambda}_2$, that is, after substituting E and $\tilde{\lambda}_2$ and cancelling $\lambda/[1 - \lambda(c_1 + c_2)]$, that $(V_1 + V_2)[1 + h\psi_2(V_2 + m)] \geq hV_2^2 \psi_2 (1 + \phi V_2 \psi_2)$. Since $(V_1 + V_2)h\psi_2 V_2 \geq hV_2^2 \psi_2$, it suffices that $V_1 + V_2 \geq h\phi V_2^3 \psi_2^2$, which holds because $h\phi = \tau \phi^2 \leq 1$ and $\psi_2 V_2 \leq 1$ by Assumption 2(c). Hence every term in $d(1/a)/d\tau$ is positive, and a is strictly decreasing in τ .

(iv), *monotonicity in λ* . With $\phi_1 = \phi_2 \equiv \phi$, the numerator of (C.15) is $N = 1 + \rho_2 \kappa_2 + m\tau \phi \psi_2$, since $\tilde{\lambda}_2$ cancels, and the bracket in Δ_3 is $\phi[\psi_1 + \psi_2 + (\kappa_1 + \kappa_2)P]$. By (C.6),

$d \ln \tilde{\lambda}_i / d\lambda = 1/[\lambda(1 - \lambda(c_1 + c_2))] \equiv g$ for both i , so $\tilde{\lambda}'_i = g\tilde{\lambda}_i$ and $P' = g(\tilde{\lambda}_1\psi_2 + \tilde{\lambda}_2\psi_1)$. Writing $N_0 \equiv 1 + \rho_2\kappa_2$, the sign of $\partial(N/\Delta_3)/\partial\lambda$ is that of:

$$\begin{aligned} & [\kappa_2\tilde{\lambda}'_2\Delta - N_0\Delta'] + m\tau\phi g \left\{ -2\kappa_1\tilde{\lambda}_1\psi_2 - \kappa_1\tilde{\lambda}_2\psi_1 - \kappa_2\tilde{\lambda}_1\psi_2 - (\kappa_1 + \kappa_2)\kappa_2\tilde{\lambda}_1\psi_2^2 - \psi_2\kappa_1\kappa_2P \right\} \\ & \quad - (m\tau\phi)^2\psi_2(\kappa_1 + \kappa_2)P' \end{aligned}$$

For the first bracket, write $N_0/\Delta = 1/(u - \tilde{\lambda}_1\tilde{\lambda}_2\kappa_1\kappa_2/N_0)$ with $u = 1 + \rho_1\kappa_1$. The derivative of the reciprocal is $\kappa_1(1 + \psi_2\kappa_2)N_0^{-2}[\tilde{\lambda}'_1N_0 - \kappa_2\tilde{\lambda}_1\tilde{\lambda}'_2]$, and with the common elasticity $\tilde{\lambda}'_1N_0 - \kappa_2\tilde{\lambda}_1\tilde{\lambda}'_2 = \tilde{\lambda}'_1(N_0 - \kappa_2\tilde{\lambda}_2) = \tilde{\lambda}'_1(1 + \psi_2\kappa_2) > 0$, so the bracket equals $-\kappa_1\tilde{\lambda}'_1(1 + \psi_2\kappa_2)^2 < 0$. In the brace, the term $\kappa_2\tilde{\lambda}_2\psi_1$ contributed by N' cancels against its counterpart in $N_0(\kappa_1 + \kappa_2)P'$, and every surviving term is negative. The last term is negative. Hence a is strictly decreasing in λ . $\square$

Remark 2 (When the replaced share vanishes). As $\psi_1 \rightarrow \infty$, the numerator of (C.15) is unchanged while Δ_3 grows without bound by the proof of (iii), so $a \rightarrow 0$. As $\tau \rightarrow 1$, every object in (C.15) stays finite and the numerator stays positive by (D2), so a is bounded away from zero. As $\sigma_s \rightarrow \infty$, $m \rightarrow \infty$ and $a \rightarrow (\phi_2\rho_2 - \phi_1\tilde{\lambda}_2)/[(\phi_1N_1 - \phi_2\tilde{\lambda}_1) + (\phi_2N_2 - \phi_1\tilde{\lambda}_2)] > 0$ by (D1)–(D2). Under Assumption 2(a) the $\tilde{\lambda}_i$ are finite, so a is bounded away from zero for any admissible λ . Hence an exporter that replaces none of the lost credit is one facing a large monitoring friction; financing needs, substitutability, and the leverage premium cannot drive the replaced share to zero.

C.1.7 Proof of Proposition 2

(i) Redirection is $-\partial q_1/\partial\varepsilon > 0$ by (D1). The identity is (C.17). By (C.16) and $\Delta_3 > 0$, the sign of $\partial \ln V_2/\partial\varepsilon$ is that of $(\sigma_s - 1)\Xi - \phi_2\tilde{\lambda}_1$, which is the displayed inequality. For the thresholds, define for ψ_1 satisfying (D3)

$$\bar{\sigma}(\psi_1) \equiv 1 + \frac{\phi_2\tilde{\lambda}_1}{s_1[\phi_1\psi_1 + \tilde{\lambda}_1(\phi_1 - \phi_2)]} \quad (\text{C.20})$$

so that $\partial \ln V_2/\partial\varepsilon > 0$ iff $\sigma_s > \bar{\sigma}(\psi_1)$; for ψ_1 violating (D3), $\Xi \leq 0$ and the response is negative for every σ_s . Part (ii) shows $\bar{\sigma}$ is continuous and strictly decreasing in ψ_1 on the set where (D3) holds, which is an interval (ψ_0, ∞) with $\bar{\sigma} \rightarrow \infty$ as $\psi_1 \downarrow \psi_0$. Hence for each σ_s the set of ψ_1 at which the response is positive is an interval $(\bar{\psi}(\sigma_s), \infty)$, with $\bar{\psi}(\sigma_s)$ the unique solution of $\bar{\sigma}(\bar{\psi}) = \sigma_s$ when σ_s lies in the range of $\bar{\sigma}$, and $\bar{\psi}(\sigma_s) = +\infty$ otherwise. In the limit $\lambda \rightarrow 0$, $\tilde{\lambda}_1 \rightarrow 0$, $\bar{\psi} \rightarrow 0$ and $\bar{\sigma} \rightarrow 1$.

(ii) *Monotonicity in σ_s .* Only m depends on σ_s in (C.16), and $m = (\sigma_s - 1)\bar{p}\tilde{q}$. Writing $K_3 \equiv (\phi_1 N_1 - \phi_2 \tilde{\lambda}_1) + (\phi_2 N_2 - \phi_1 \tilde{\lambda}_2) > 0$,

$$\frac{\partial \ln V_2}{\partial \varepsilon} = \frac{(\sigma_s - 1)\Xi - \phi_2 \tilde{\lambda}_1}{\Delta + (\sigma_s - 1)\bar{p}\tilde{q}\tau K_3}$$

is a ratio of two affine functions of $\sigma_s - 1$; its derivative has the sign of $\Xi\Delta + \phi_2 \tilde{\lambda}_1 \bar{p}\tilde{q}\tau K_3 > 0$ under (D3).

Monotonicity of $\bar{\sigma}$ in ψ_1 . Hold the baseline fixed. Let $g(\psi_1) \equiv \tilde{\lambda}_1/[\phi_1\psi_1 + \tilde{\lambda}_1(\phi_1 - \phi_2)]$, so $\bar{\sigma} = 1 + \phi_2 g/s_1$. Then

$$g' = \frac{\tilde{\lambda}'_1[\phi_1\psi_1 + \tilde{\lambda}_1(\phi_1 - \phi_2)] - \tilde{\lambda}_1[\phi_1 + \tilde{\lambda}'_1(\phi_1 - \phi_2)]}{[\cdot]^2} = \frac{\phi_1(\tilde{\lambda}'_1\psi_1 - \tilde{\lambda}_1)}{[\cdot]^2} = -\frac{\phi_1\lambda}{[1 - \lambda(c_1 + c_2)][\cdot]^2} < 0$$

using (C.6), since $\tilde{\lambda}'_1\psi_1 - \tilde{\lambda}_1 = \lambda[c_1\psi_1 - (1 + c_1\psi_1)]/[1 - \lambda(c_1 + c_2)]$. So $\bar{\sigma}$ is strictly decreasing wherever (D3) holds. The set where (D3) holds is an up-set in ψ_1 : its boundary is $\phi_1\psi_1 = \tilde{\lambda}_1(\psi_1)(\phi_2 - \phi_1)$, and the left side grows at rate ϕ_1 while the right grows at rate $\tilde{\lambda}'_1(\phi_2 - \phi_1) < \phi_1$ by Assumption 2(a); when $\phi_2 \leq \phi_1$, (D3) holds for all $\psi_1 > 0$.

Monotonicity of $\bar{\psi}$ in σ_s . $\bar{\psi}$ is the inverse of a strictly decreasing continuous function on the range of $\bar{\sigma}$, hence strictly decreasing there, and equal to $+\infty$ below that range.

(iii) With $\phi_1 = \phi_2$, the wedge cancels from the reallocation condition, which becomes $(\sigma_s - 1)s_1\psi_1 > \tilde{\lambda}_1$, so $\bar{\sigma} = 1 + \tilde{\lambda}_1/(s_1\psi_1)$. At a given baseline, τ enters only through $\tilde{\lambda}_1$, which is strictly increasing in τ by Lemma C.1.4. $\square$

Remark 3 (The limit of large frictions). Because $\tilde{\lambda}_1$ grows with ψ_1 , $\bar{\sigma}$ does not fall to one as $\psi_1 \rightarrow \infty$. From (C.20) and (C.6),

$$\lim_{\psi_1 \rightarrow \infty} \bar{\sigma}(\psi_1) = 1 + \frac{\lambda c_1 \phi_2}{s_1[\phi_1(1 - \lambda(c_1 + c_2)) + \lambda c_1(\phi_1 - \phi_2)]} \equiv \sigma_\infty > 1$$

whose denominator is positive because (D3) holds in the limit. Hence for $\sigma_s \leq \sigma_\infty$ no monitoring friction is large enough and $\bar{\psi}(\sigma_s) = +\infty$. The reason is the feedback: a higher monitoring premium in the shocked destination raises its rate, lowers its invoice, raises the borrowing need, and so raises the premium that spills into the untouched destination. Without the feedback, $c_1 = 0$, $\sigma_\infty = 1$, and every good reallocates at a high enough friction. Under equal wedges, $c_1/s_1 = \phi V_1 Q/q_1 = \phi \bar{p}Q = \phi(V_1 + V_2)$, so $\sigma_\infty = 1 + \lambda\phi(V_1 + V_2)/[1 - \lambda\phi(V_1 + V_2)]$.

Remark 4 (Magnitude in ψ_1). Proposition 2(ii) signs the thresholds, not the magnitude of

the third-destination response in ψ_1 . At a given baseline, (C.16) is a ratio of two functions of ψ_1 that are affine in ρ_1 ; writing it as $f(\psi_1)$ and $\partial_{\psi_1}\Delta_3$ for the derivative of (C.13) computed in the proof of Proposition 1(iii),

$$\text{sign } f' = \text{sign} \left[(\sigma_s - 1)s_1(\phi_1 + \tilde{\lambda}'_1(\phi_1 - \phi_2)) - \phi_2\tilde{\lambda}'_1 - \partial_{\psi_1}\Delta_3 \cdot f \right]$$

so the response is strictly increasing in ψ_1 whenever it is negative, and when positive whenever it lies below $[(\sigma_s - 1)s_1(\phi_1 + \tilde{\lambda}'_1(\phi_1 - \phi_2)) - \phi_2\tilde{\lambda}'_1]/\partial_{\psi_1}\Delta_3$. We do not use this in the text.

C.1.8 Proof of Proposition 3

The shocked importer sources from origins j , each a segment of the continuum whose representative firm is the exporter of Section 2 with parameters $(\tau_j, \psi_{1j}, \psi_{2j}, \lambda_j)$. Since each firm is of measure zero, the origins' responses are independent, and each is given by Propositions 1–2 with the appropriate parameters. Write a_j for origin j 's replaced share and $\theta_j \equiv \bar{L}_{1j}/(\tau_j V_{1j})$. Every facility is cut by the same proportion, $\varepsilon_j = x\bar{L}_{1j}$, so by (C.18)

$$d \ln V_{1j} = -(1 - a_j) \theta_j x \quad (\text{C.21})$$

(i) $dF = \sum_j dF_{1j} = \sum_j a_j \varepsilon_j$, and $a_j > 0$ for every j by Proposition 1(i).

(ii) With $F = \sum_j \tau_j V_{1j} - \bar{L}$, $\bar{\tau} \equiv \sum_j v_j \tau_j$, and $S = \bar{\tau} - \bar{L}/V$,

$$V dS = dF - S dV = \varepsilon + \sum_j (\tau_j - S) dV_{1j} = \varepsilon + \frac{\bar{L}}{V} dV + \sum_j (\tau_j - \bar{\tau}) dV_{1j}$$

Since $dV_{1j} = V_{1j} d \ln V_{1j} = V v_j d \ln V_{1j}$ and $\sum_j v_j (\tau_j - \bar{\tau}) = 0$,

$$\sum_j (\tau_j - \bar{\tau}) dV_{1j} = V \sum_j v_j (\tau_j - \bar{\tau}) (d \ln V_{1j} - d \ln V) = V \text{Cov}_v(\tau_j, d \ln V_{1j})$$

which is the decomposition. Nothing uses the demand system, so it holds exactly, and for any distribution of the cuts ε_j since only their sum enters. For the sign of the bracket, $\bar{L} d \ln V = (\bar{L}/V) \sum_j V_{1j} d \ln V_{1j}$ and $\bar{L}/V \leq \tau_j$ for every j under $\bar{L} \leq \min_j \tau_j V$; if all $d \ln V_{1j} \leq 0$, then $\bar{L} |d \ln V| \leq \sum_j \tau_j V_{1j} |d \ln V_{1j}| = \varepsilon - dF < \varepsilon$ by (i), using $dF = \sum_j (\varepsilon_j + \tau_j V_{1j} d \ln V_{1j})$; if some $d \ln V_{1k} > 0$, drop those terms and apply the same bound to the rest, using $dF_{1j} = \varepsilon_j + \tau_j V_{1j} d \ln V_{1j} > 0$ for each remaining j .

(iii) If $\tau_j = \tau$ for all j , then $\tau_j - \bar{\tau} = 0$ and the covariance vanishes. If origins differ only in τ_j , with common θ and a common cut x , then by (C.21) $d \ln V_{1j} = -(1 - a_j)\theta x$, which

is strictly decreasing in τ_j by Proposition 1(iv), so τ_j and $d \ln V_{1j}$ are strictly comonotone in opposite directions and the covariance is negative. In general the covariance is negative if and only if origins with higher τ_j have lower $d \ln V_{1j}$ in the import-share-weighted sense, which is the statement.

(iv) Rearranging (ii). The bracket requires the importer's total line cut ε , its pre-shock line $\bar{L}$, and its import growth $d \ln V$; the left side requires V and dS . None requires origin-level data. $\square$

Remark 5 (The extensive margin). The level of trade credit received by the importer falls only if an origin is dropped. Let the importer bear a fixed cost of maintaining a sourcing relationship, and let it drop origin k when its surplus from access, which under CES is increasing in the origin's spending share, falls below that cost; by Proposition 1 the share falls first for the origin whose effective price rises most. Since surplus is continuous in the shock and strictly above the fixed cost at the baseline for every origin retained, no origin is dropped for shocks small enough, so the extensive margin is inactive at first order and absent from Propositions 1–3. For larger shocks, after the drop $F_{1k} = V_{1k} = 0$, the freed line serves the remaining origins, and they absorb the product's spending at the new index. With primes for post-drop values, $F' - F = \sum_{j \neq k} \tau_j (V'_{1j} - V_{1j}) - \tau_k V_{1k} + \varepsilon$. Since spending does not rise when a variety is lost, $\sum_{j \neq k} (V'_{1j} - V_{1j}) \leq V_{1k}$, so $F' < F$ whenever $(\tau_k - \max_{j \neq k} \tau_j) V_{1k} > \varepsilon$: the gap in financing needs times the dropped origin's sales exceeds the line cut. On the exporter's side the same event is a withdrawal of the capacity that served the pair. When financing needs are common, $F' - F = \tau(\sum_{j \neq k} (V'_{1j} - V_{1j}) - V_{1k}) + \varepsilon$, negative only if product spending falls by more than ε/τ ; like the intensive margin, the extensive margin reduces aggregate credit mainly through the difference in financing needs.

C.1.9 Proof of Proposition 4

Let $d \ln E_{1s} = -\delta$ with both facilities unchanged. We work directly with the three log changes $\pi \equiv d \ln p$ (common to both destinations, since $p_1 = p_2$ holds after the shock by the argument of Appendix C.1.4), $x_1 \equiv d \ln V_1$, and $x_2 \equiv d \ln V_2$. Write $\gamma \equiv \lambda(1 - \tau)(V_1 + V_2)$ and $\eta_i \equiv \tau \psi_i V_i$. Four exact relations hold at the baseline.

First, clearing with the facilities fixed gives $dF_i = \tau dV_i = \tau V_i x_i$ for both i . Second, since Q is fixed and the price is common, $dV_1 + dV_2 = Q dp$, that is

$$\pi = s_1 x_1 + s_2 x_2 \tag{C.22}$$

using $V_i/(V_1 + V_2) = q_i/Q = s_i$. Third, the borrowing need moves by $dB = dF_1 + dF_2 -$

$dV_1 - dV_2 = -(1 - \tau)(V_1 + V_2)\pi$, so $d\Omega = \lambda dB = -\gamma\pi$, and by (C.5)

$$dr_{xi} = -\gamma\pi + \psi_i dF_i = -\gamma\pi + \eta_i x_i \quad (\text{C.23})$$

Fourth, from $q_i = A_{is}\tilde{p}_i^{-\sigma_s}$ with $d\ln \tilde{p}_i = \pi + \phi_i dr_{xi}$ and $d\ln A_{1s} = -\delta$, $d\ln A_{2s} = 0$,

$$x_1 = \pi + d\ln q_1 = -\delta + (1 - \sigma_s)\pi - \sigma_s \phi_1 dr_{x1} \quad x_2 = (1 - \sigma_s)\pi - \sigma_s \phi_2 dr_{x2}$$

Substituting (C.23) and collecting, with $\beta_i \equiv 1 + \sigma_s \phi_i \eta_i > 1$ and $\alpha_i \equiv (\sigma_s - 1) - \sigma_s \phi_i \gamma$,

$$\beta_1 x_1 = -\delta - \alpha_1 \pi \quad \beta_2 x_2 = -\alpha_2 \pi \quad (\text{C.24})$$

Substituting (C.24) into (C.22), $\pi[1 + s_1 \alpha_1 / \beta_1 + s_2 \alpha_2 / \beta_2] = -s_1 \delta / \beta_1$, so with

$$\mathcal{D} \equiv \beta_1 \beta_2 + s_1 \alpha_1 \beta_2 + s_2 \alpha_2 \beta_1 \quad (\text{C.25})$$

the solution is

$$\pi = -\frac{s_1 \beta_2}{\mathcal{D}} \delta \quad x_1 = -\frac{\beta_2 + s_2 \alpha_2}{\mathcal{D}} \delta \quad x_2 = \frac{s_1 \alpha_2}{\mathcal{D}} \delta \quad (\text{C.26})$$

where x_2 is (C.24) with π substituted, and x_1 follows from $\beta_1 x_1 = -\delta - \alpha_1 \pi = -\delta(\mathcal{D} - s_1 \alpha_1 \beta_2) / \mathcal{D} = -\delta \beta_1 (\beta_2 + s_2 \alpha_2) / \mathcal{D}$.

Two facts about the pieces. By Assumption 2(a), $\gamma \phi_i = \lambda(1 - \tau) \phi_i (V_1 + V_2) < \lambda \phi_i (V_1 + V_2) < 1$. Hence, using $s_1 + s_2 = 1$,

$$\beta_2 + s_2 \alpha_2 = 1 + \sigma_s \phi_2 \eta_2 + s_2 (\sigma_s - 1) - \sigma_s s_2 \phi_2 \gamma > 1 + s_2 (\sigma_s - 1) - \sigma_s s_2 + \sigma_s \phi_2 \eta_2 = s_1 + \sigma_s \phi_2 \eta_2 > 0$$

and, writing $\mathcal{D} = \beta_1 \beta_2 + (\sigma_s - 1)(s_1 \beta_2 + s_2 \beta_1) - \sigma_s \gamma (s_1 \phi_1 \beta_2 + s_2 \phi_2 \beta_1)$ and bounding the last term by $\sigma_s (s_1 \beta_2 + s_2 \beta_1)$,

$$\mathcal{D} > \beta_1 \beta_2 - s_1 \beta_2 - s_2 \beta_1 = \sigma_s \phi_1 \eta_1 s_1 + \sigma_s \phi_2 \eta_2 s_2 + \sigma_s^2 \phi_1 \phi_2 \eta_1 \eta_2 > 0$$

(i) By (C.26) and the two facts, $x_1 < 0$, so $dF_1 = \tau V_1 x_1 < 0$. Also $\pi < 0$. For the share, $V_1 dS = dF_1 - S dV_1 = (\tau - S) dV_1 = (\bar{L}_1 / V_1) dV_1 < 0$.

(ii) At a given baseline, ψ_1 enters (C.26) only through η_1 , which appears in β_1 and hence in $\mathcal{D}$ but not in the numerator of x_1 . From (C.25), $\partial \mathcal{D} / \partial \beta_1 = \beta_2 + s_2 \alpha_2 > 0$, and $\partial \beta_1 / \partial \psi_1 = \sigma_s \phi_1 \tau V_1 > 0$, so $\mathcal{D}$ is strictly increasing in ψ_1 and $|x_1|$ strictly decreasing.

(iii) By (C.26), $x_2 > 0$ iff $\alpha_2 > 0$, that is $\sigma_s - 1 > \sigma_s \lambda(1 - \tau) \phi_2 (V_1 + V_2)$, or $\sigma_s > \bar{\sigma}^d \equiv 1 / [1 - \lambda(1 - \tau) \phi_2 (V_1 + V_2)]$, whose denominator is positive by the first fact. The threshold

contains no ψ_i . Under $\phi_1 = \phi_2 \equiv \phi$, write $y \equiv \lambda\phi(V_1 + V_2) < 1$; by Remark 3, $\bar{\sigma}(\psi_1)$ is strictly decreasing in ψ_1 with limit $\sigma_\infty = 1 + y/(1 - y)$, while $\bar{\sigma}^d = 1 + (1 - \tau)y/[1 - (1 - \tau)y]$. Since $z \mapsto z/(1 - z)$ is increasing on $[0, 1)$ and $(1 - \tau)y < y$, $\bar{\sigma}^d < \sigma_\infty < \bar{\sigma}(\psi_1)$ for every ψ_1 . $\square$

The contrast with the credit shock is in the numerators. In x_2 , the positive term $\sigma_s - 1$ in α_2 is the direct redirection, present with $\psi_1 = 0$; the credit shock has $(\sigma_s - 1)\Xi$ in its place in (C.16), which vanishes as $\psi_1 \rightarrow 0$. The contagion term carries $1 - \tau$ because a fall in expenditure raises the borrowing need only through the cash it stops, whereas a cut in the facility lands on the exporter's receivables in full.

C.2 Large shocks

Let the exporter choose production $Y \leq D$ in period 1 at wage w per unit, with $(1 + t)Q = Y$. The marginal value of a unit produced and shipped to destination i is $(1 + r_0 + \lambda(D + B))[MR_i - (1 + t)w]/(1 + t)$, positive at the baseline by (C.4). As ε grows, the common invoice price $p_1 = p_2$ falls, and production is cut below capacity once p reaches the reservation price $\bar{p}_w \equiv (1 + t)w\sigma_s/(\sigma_s - 1)$. Beyond that point the price is pinned at $\bar{p}_w$, both quantities fall to hold it there, and the shock is absorbed by output rather than by trade credit and reallocation; in particular V_2 falls with the common quantity cut, and the reallocation-versus-contagion trade-off of Proposition 2 no longer applies. The threshold shock $\bar{\varepsilon}$ solves $p(\bar{\varepsilon}) = \bar{p}_w$ along the first-order responses, $\bar{\varepsilon} \approx [\ln p(0) - \ln \bar{p}_w]/|\partial \ln p / \partial \varepsilon|$, and is strictly positive since $p(0) > \bar{p}_w$. We state this as a sketch; the full characterization of the constrained regime is left aside.

D Without committed capacity

This appendix shows that the balance-sheet friction alone transmits the shock across destinations only as contagion, so that reallocation in Proposition 2 rests on committed capacity. Let the exporter produce freely in period 1 at marginal cost $c \equiv w(1 + t)$ per unit delivered, with no ceiling, and let there be no long-term debt, so that the borrowing need is $B = cQ + F - V_1 - V_2$ and the premium is λB . Everything else is as in Section 2.

Period-1 problem. The exporter chooses q_1, q_2, F_1, F_2 at given rates. The conditions for F_i are unchanged and give $r_{xi} = r_0 + \lambda B + \psi_i F_i$. For quantities, q_i enters only through B ,

with $\partial B/\partial q_i = c - MR_i$, so

$$\frac{\partial \Pi_1}{\partial q_i} = (1 + r_0 + \lambda B)(MR_i - c) = 0 \quad \implies \quad MR_i = c \quad \text{for each } i$$

Each market clears on its own. The invoice price is the constant markup $p_i = c\sigma_s/(\sigma_s - 1)$, and from (C.1), $q_i = A_{is}[(1 - 1/\sigma_s)/c]^{\sigma_s}(1 + \tau r_{xi})^{-\sigma_s}$, so quantity in each destination depends on that destination's rate only,

$$d \ln V_i = d \ln q_i = -\sigma_s \phi_i dr_{xi}$$

Linearization. Since $p_i - c = p_i/\sigma_s$, the borrowing need responds by $dB = dF_i - (p_i - c)dq_i = dF_i + V_i \phi_i dr_{xi}$ per destination, the same feedback as in Appendix C.1.4, so $d\Omega = \tilde{\lambda}_1 dF_1 + \tilde{\lambda}_2 dF_2$ with $\tilde{\lambda}_i$ as in (C.6). Clearing gives $dF_1 = d\varepsilon - \sigma_s \kappa_1 dr_{x1}$ and $dF_2 = -\sigma_s \kappa_2 dr_{x2}$. The system is

$$\begin{aligned} dr_{x1} &= \rho_1 dF_1 + \tilde{\lambda}_2 dF_2 & dF_1 &= d\varepsilon - \sigma_s \kappa_1 dr_{x1} \\ dr_{x2} &= \tilde{\lambda}_1 dF_1 + \rho_2 dF_2 & dF_2 &= -\sigma_s \kappa_2 dr_{x2} \end{aligned}$$

which is the system of Appendix C.1.4 without the quantity margin, (C.11) absent, and with κ_i replaced by $\sigma_s \kappa_i$. Its solution is

$$\frac{\partial r_{x1}}{\partial \varepsilon} = \frac{\rho_1 + \sigma_s \kappa_2 P}{\Delta^\circ} \quad \frac{\partial r_{x2}}{\partial \varepsilon} = \frac{\tilde{\lambda}_1}{\Delta^\circ} \quad \Delta^\circ \equiv (1 + \rho_1 \sigma_s \kappa_1)(1 + \rho_2 \sigma_s \kappa_2) - \tilde{\lambda}_1 \tilde{\lambda}_2 \sigma_s^2 \kappa_1 \kappa_2 > 0$$

Proposition 5 (No committed capacity). *Without committed capacity: (i) Proposition 1 holds in full, with κ_i replaced by $\sigma_s \kappa_i$ and $m = 0$ in every expression; (ii) the spillover to the untouched destination is*

$$\frac{\partial \ln V_2}{\partial \varepsilon} = -\frac{\sigma_s \phi_2 \tilde{\lambda}_1}{\Delta^\circ} < 0 \quad \frac{\partial F_2}{\partial \varepsilon} = -\frac{\sigma_s^2 \kappa_2 \tilde{\lambda}_1}{\Delta^\circ} < 0$$

for every σ_s and every ψ_1 .

Proof. (i) The proofs in Appendix C.1.6 of parts (i)–(iii) and of the λ statement in (iv) use only the 2×2 block, which is unchanged up to the rescaling of κ_i ; the proof of the τ statement in (iv) uses the structure $N = 1 + hn_1$, $\Delta_3 = 1 + hd_1 + h^2 d_2$ with $m = 0$, which the rescaling $h \rightarrow \sigma_s h$ preserves. (ii) $\partial r_{x2}/\partial \varepsilon = \tilde{\lambda}_1/\Delta^\circ > 0$ since $\Delta^\circ > 0$ by the same argument as for Δ , and $d \ln V_2 = -\sigma_s \phi_2 dr_{x2}$, $dF_2 = -\sigma_s \kappa_2 dr_{x2}$, with $\tilde{\lambda}_1 > 0$ by Lemma C.1.4. $\square$

The leverage premium does one thing to the untouched destination: it raises the rate charged there, which lowers the effective price and the quantity sold. There is no term that could make V_2 rise, because q_2 is pinned by $MR_2 = c$ and responds to nothing in destination 1 except through r_{x2} . In the model of Section 2, the positive term in (C.16) comes from the quantity margin, $dq_2 = -dq_1 > 0$, which exists only because the exporter's shipments are constrained to sum to Q . More generally, reallocation requires the production side to link the two markets: under an increasing marginal cost $C'(Q)$, the quantity conditions become $MR_i = C'(Q)$ for both i , so that a fall in q_1 lowers C' and raises q_2 . Committed capacity is the limit $C'' \rightarrow \infty$, in which the link is a vertical supply curve and the redirection is one for one.